

⊗ Holos

Holos: A Framework for Understanding Reality

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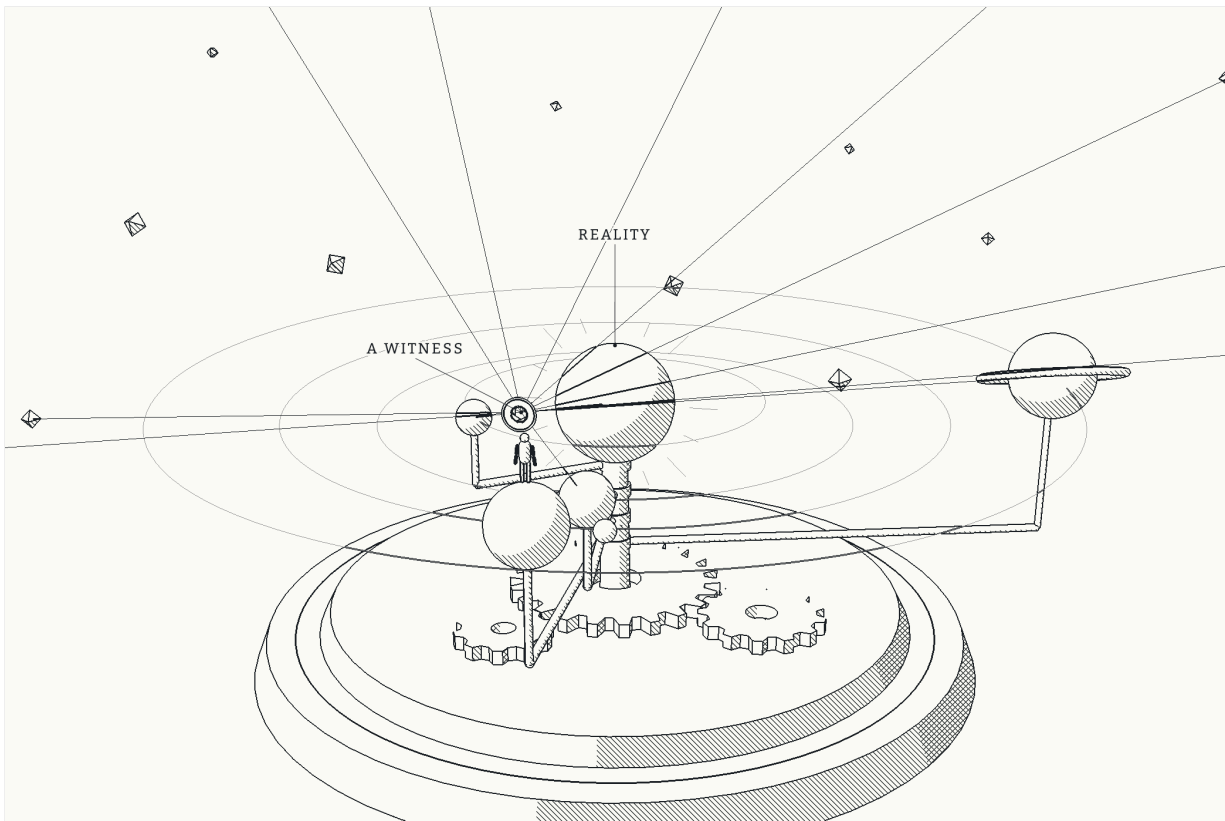
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Introduction

Physics describes the universe with great precision. It says how matter moves and how spacetime bends. It says nothing about why we exist. Why are we here? Does life have a purpose? *What does it mean to be real?*

Holos, from the Greek *ὅλος*, "whole," is an interpretive framework for understanding reality.

Picture a universe complete in every detail physics describes. Every law holds, every event occurs, but there is no one to witness it. It is unrealized *structure*. Reality requires a witness.



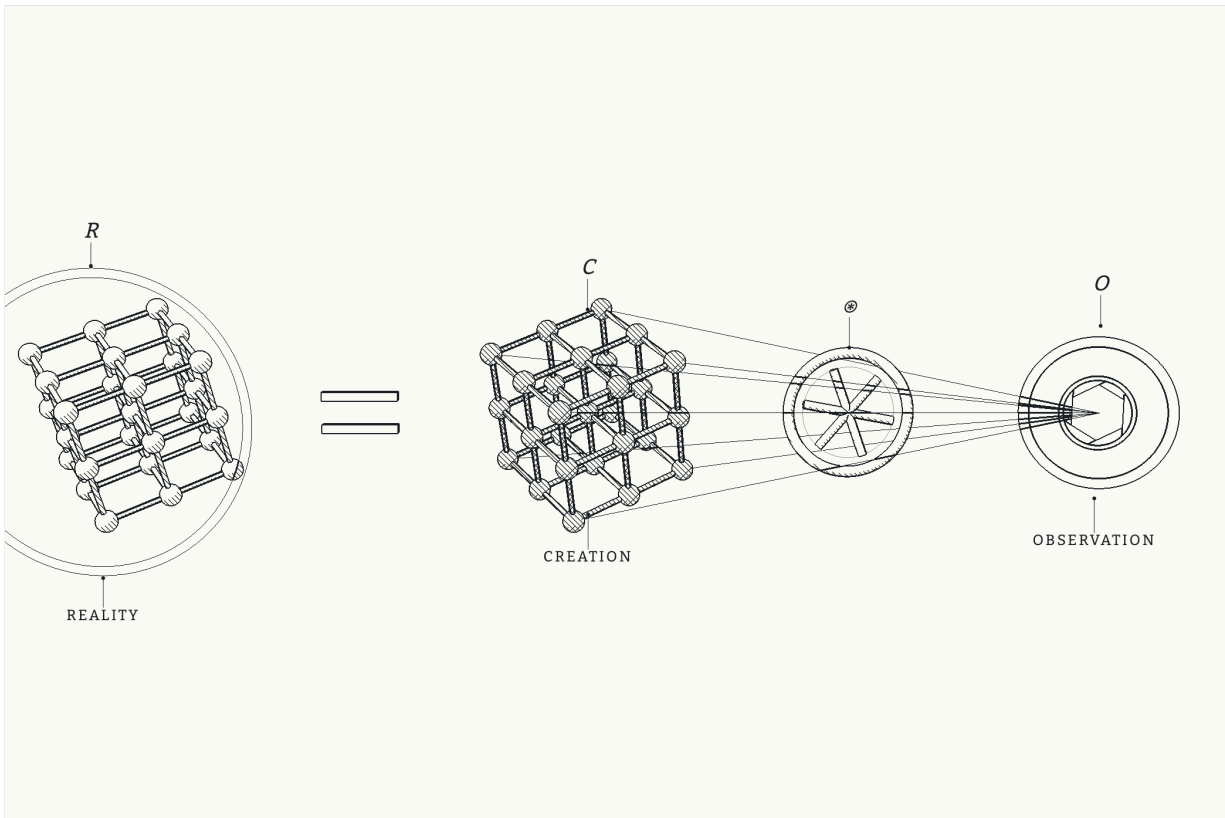
Holos writes this as $R = C \otimes O$.

CREATION, the structure physics produces.

OBSERVATION, that structure experienced from the inside.

REALITY, the result.

On the physics Holos bets on, observing changes nothing it takes in (see [Notation](#)).

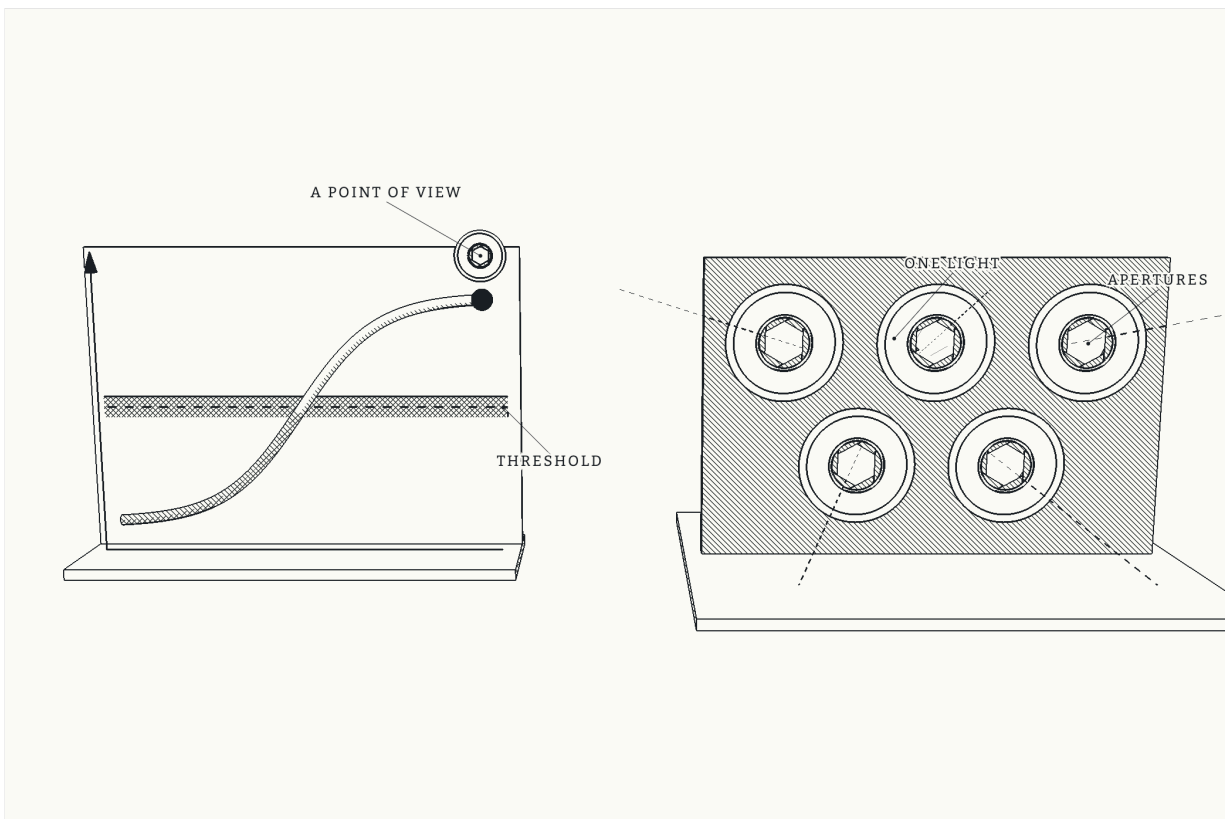


Holos is built on two ideas. Neither changes established physics.

THRESHOLD, experience appears only where information is joined tightly enough to form a single point of view.

OMEGA, the whole. Every observer is an aperture of it, an opening through which the whole is lived.

Observation does not cause the universe, its laws, or its history. It makes a lawful universe a lived one.



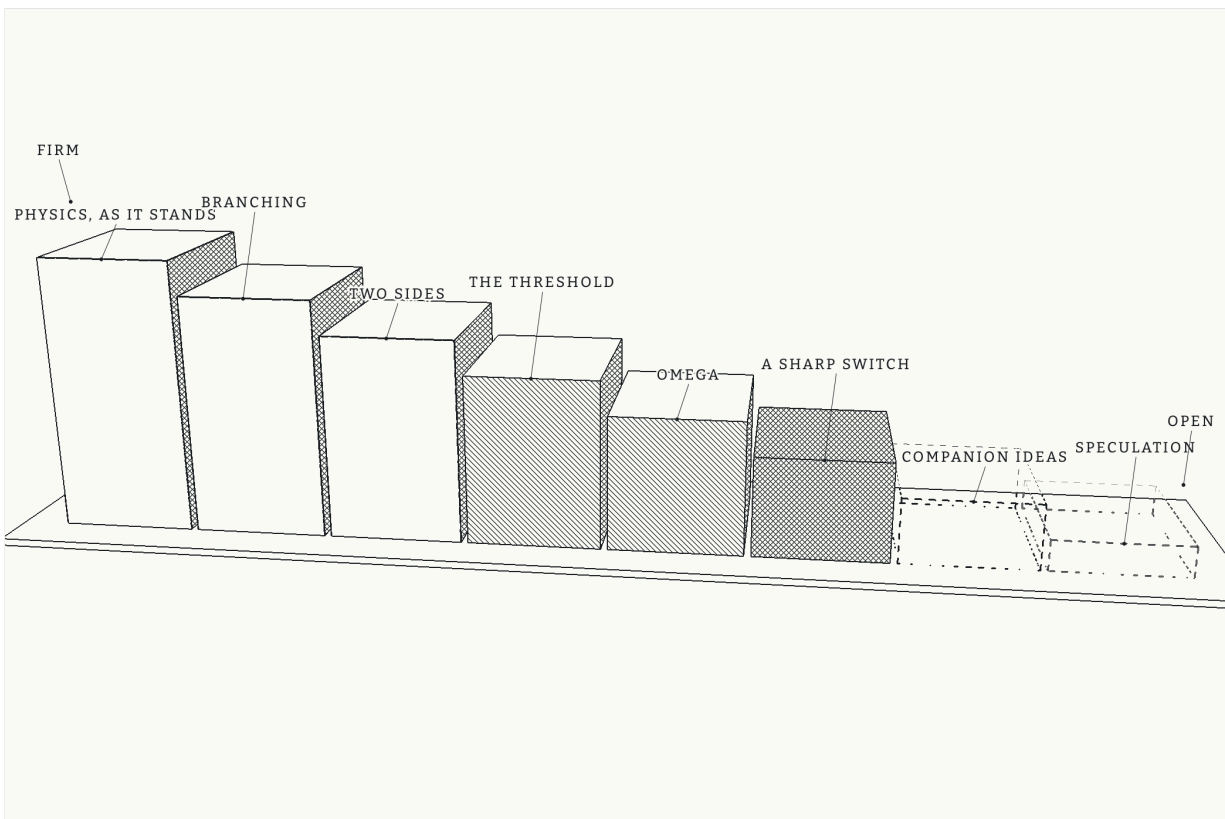
There is one experiencer, living through every self. Everything that could ever reach an observer, back to the earliest light, is the world experience is made from. Even the question, why we are here, has an answer.

Some of this is established physics. Some is extrapolation. Some can be tested, and some cannot. The box below says which is which.

What Holos claims, and how firmly

- **Physics, as it stands.** Relativity and quantum mechanics, unchanged. No new forces, no new equations.
- **A side taken on physics.** Quantum branching without collapse, often called many-worlds. Every outcome a quantum event allows happens, each in its own branch. An alternate version of Holos with collapse is declared in advance, in case experiments rule out branching (see [Two versions](#)).
- **A side taken on mind.** Experience and physical activity are two sides of one event. Nothing is missing from the world. Physics describes it only from outside (see [One event, two sides](#)).
- **The threshold.** Experience occurs only where information is integrated into one point of view. Testable, and it can fail (see [Test A](#)).
- **Omega.** One experiencer, living through every self. Philosophical, not testable (see [Omega](#)).
- **A hypothesis.** Crossing the threshold is a genuine transition, so larger systems should switch on more sharply, the way a big magnet switches more sharply than a tiny one (see [The threshold in three claims](#)).
- **Companion ideas.** Quiet aliens and the Teeming Dark. Testable in principle. Separate from the core (see [Aliens](#)).
- **Speculation.** Separation between things is not fundamental. The universe is one whole (see [Why Are We Here?](#)). Designs for mature civilizations (see [Speculation](#)), with notes on plausibility.

The full list is on the Logic page (see [Claims](#)).

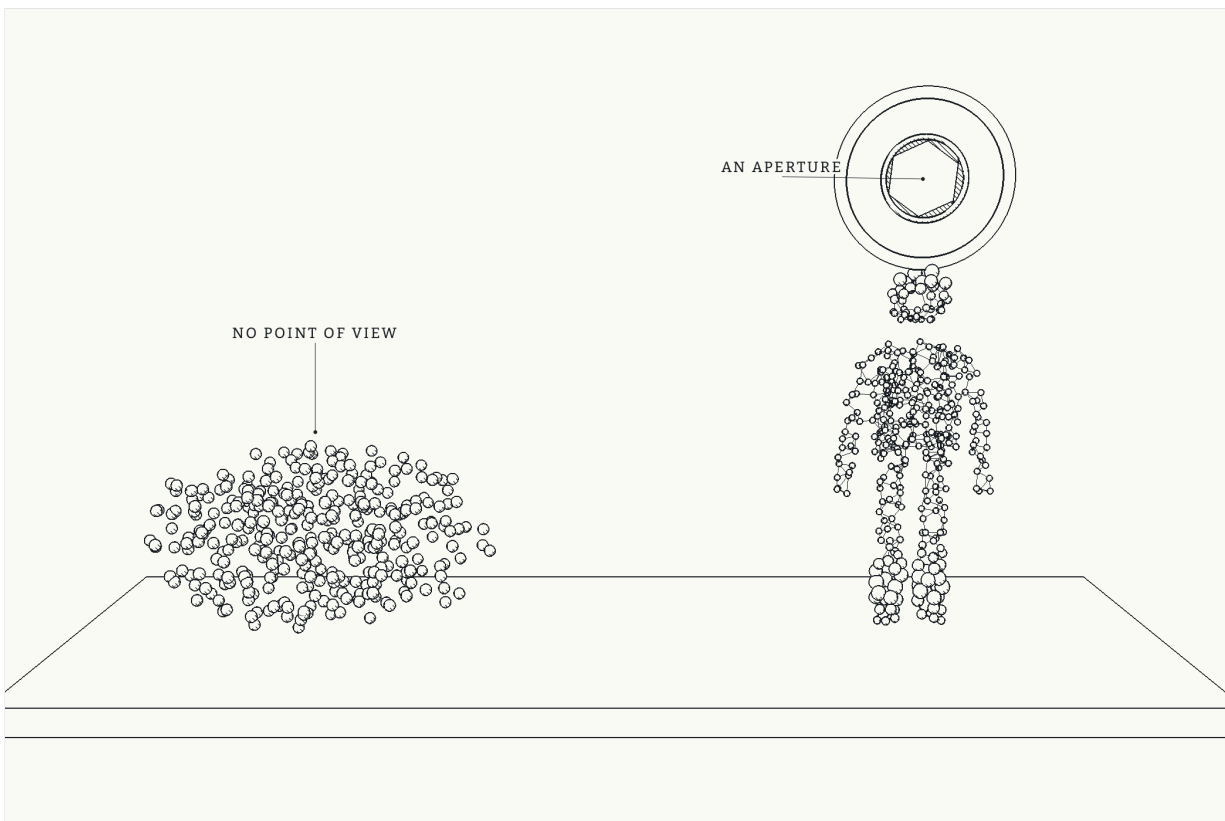


Key terms in plain words

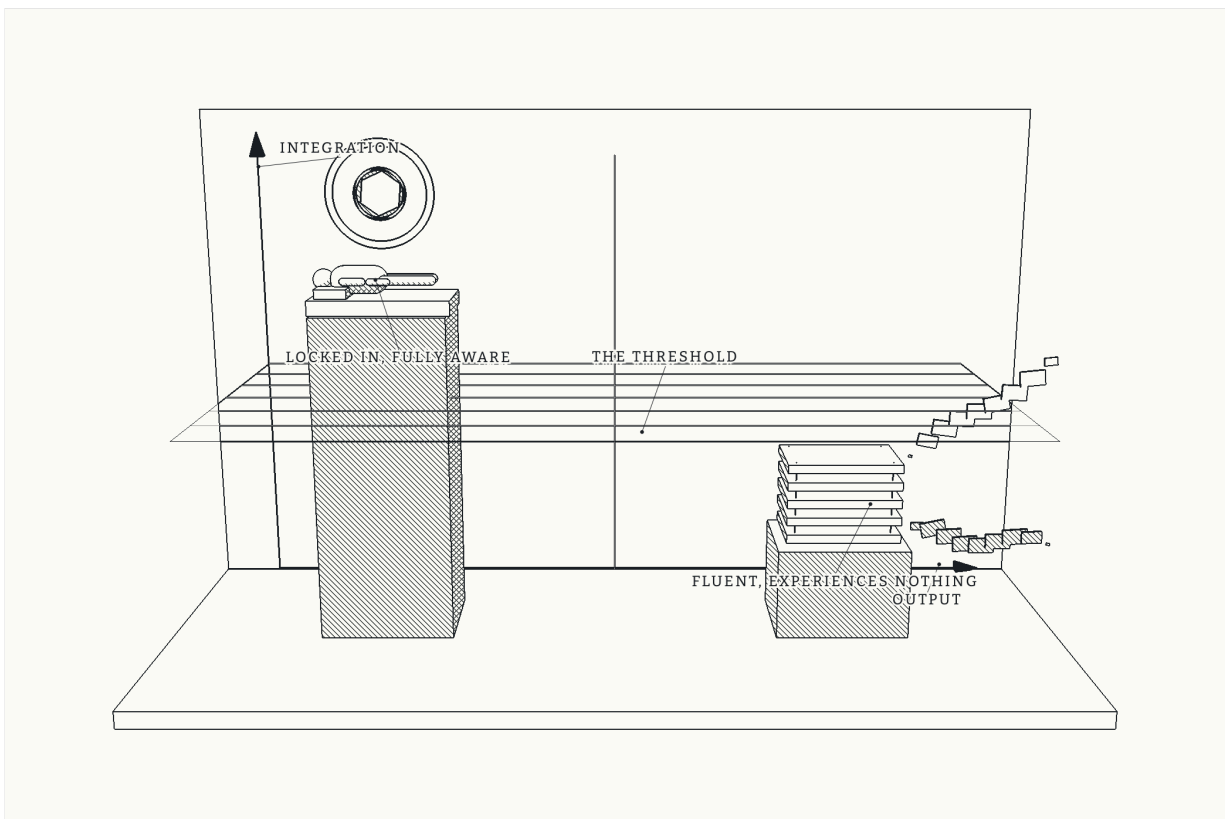
- **Structure and reality.** Structure is what physics describes. Reality is structure that is lived. A universe with no observers is all structure and no reality.
- **Observer.** Any system with a point of view, someone home. Not necessarily human or biological.
- **Integration (Φ).** How much a system's parts work as one whole rather than as separate pieces. Φ (phi) is the measure Holos borrows for it. No one can yet compute Φ for a real brain, so experiments use measurable stand-ins.
- **The threshold (Φ_c) and the twilight.** The level of integration past which a system is an observer. The twilight is the narrow borderline around it, like dusk between day and night.
- **Lived and lit.** Lived means actually experienced, which happens only inside observers. Lit means everything that could ever have sent a signal to an observer, such as the stars you see.
- **Branch.** In quantum physics without collapse, one complete way the universe can go. Every outcome happens, each in its own branch.
- **Omega and aperture.** Omega is the whole. Each observer is an aperture, an opening through which the whole is lived.

Consciousness

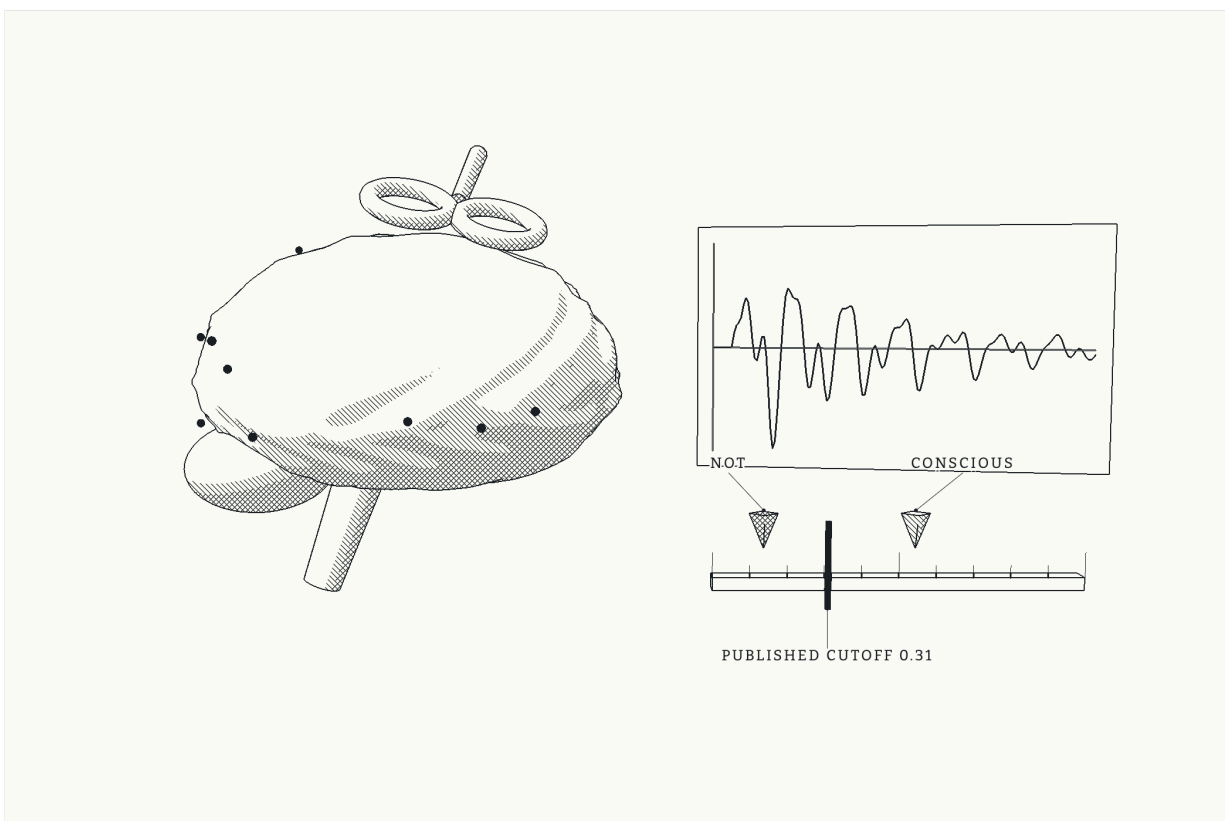
A rock has no point of view. A person does. The difference is integration, how fully a system's parts act as one. Past a threshold, it opens an aperture, where the universe is experienced from the inside.



A person with locked-in syndrome cannot move or speak but is fully aware. Today's language models take in the world's data and speak fluently, but experience nothing. What matters is not input or output. A locked-in brain is past the threshold. Language models are not.



No one can yet compute integration for a whole brain. But stand-ins work. One taps the brain with a magnetic pulse and measures the echo. It is rich and widespread when someone is conscious, simple or local when they are not. In people, a cutoff on that echo already separates the two. Where the threshold falls for other kinds of system is still open (see [Test B](#)).



Science can describe everything a brain does. It cannot explain why any of it feels like something. The philosopher David Chalmers called this the hard problem. Holos does not solve it. It takes experience as given, the one fact it starts from. That experience belongs to Omega, the whole. Holos says where it occurs, past the threshold, not why there is experience at all. Given does not mean everywhere. Experience occurs only where that structure is, and wherever it is, experience is. They are the inside and outside of one event.

Integration is the heart of it, but not all of it. Holos names four requirements.

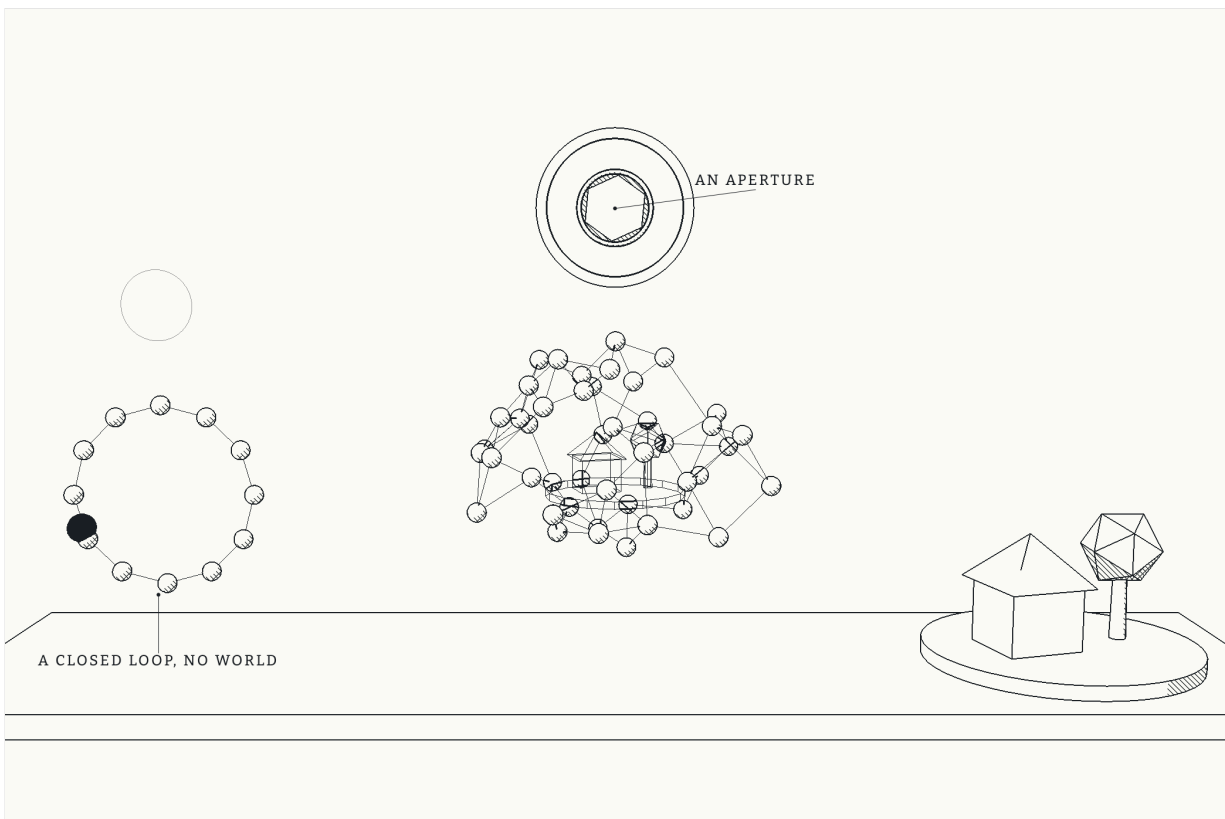
INTEGRATION, the parts act as one, each shaping and shaped by the rest.

DIFFERENTIATION, the whole can take many different states. A seizure locks the brain into one rhythm, and experience goes out.

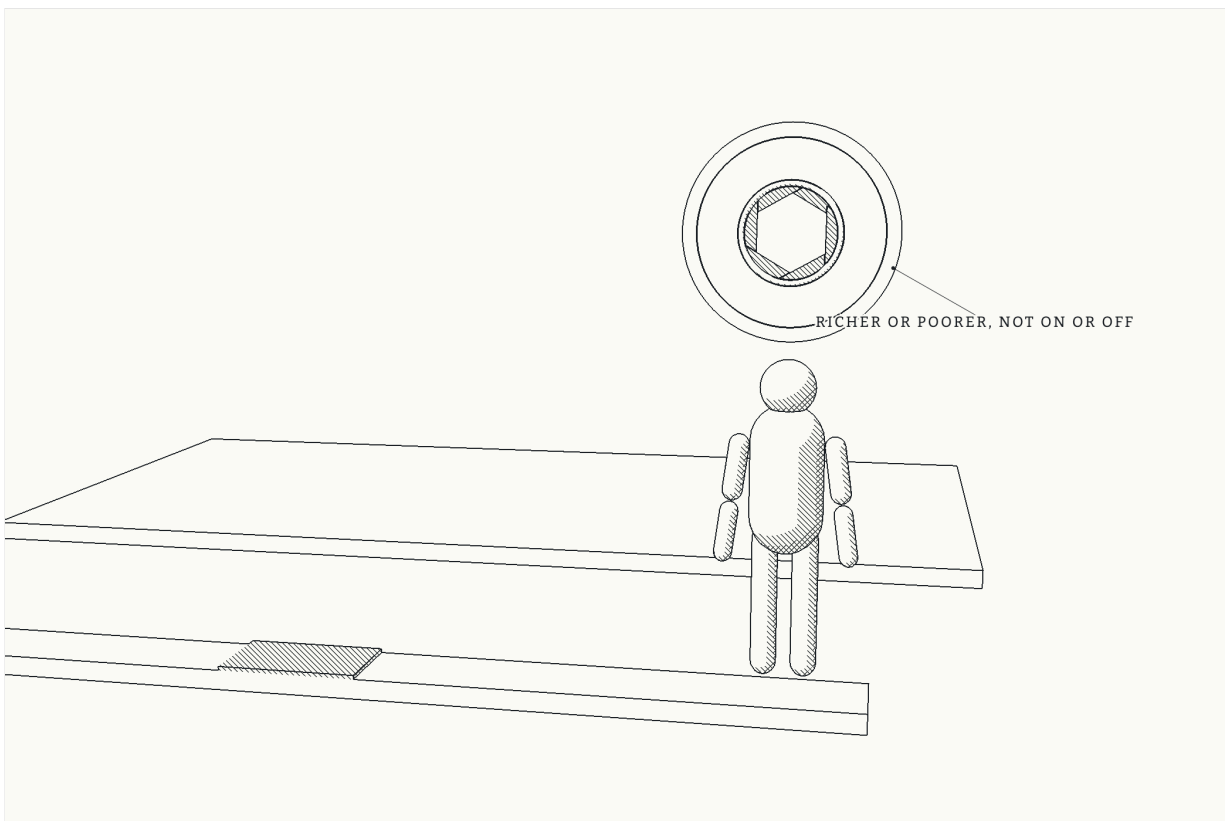
TEMPORAL COHESION, it holds together over time.

ABOUTNESS, its states carry a model of a world beyond itself.

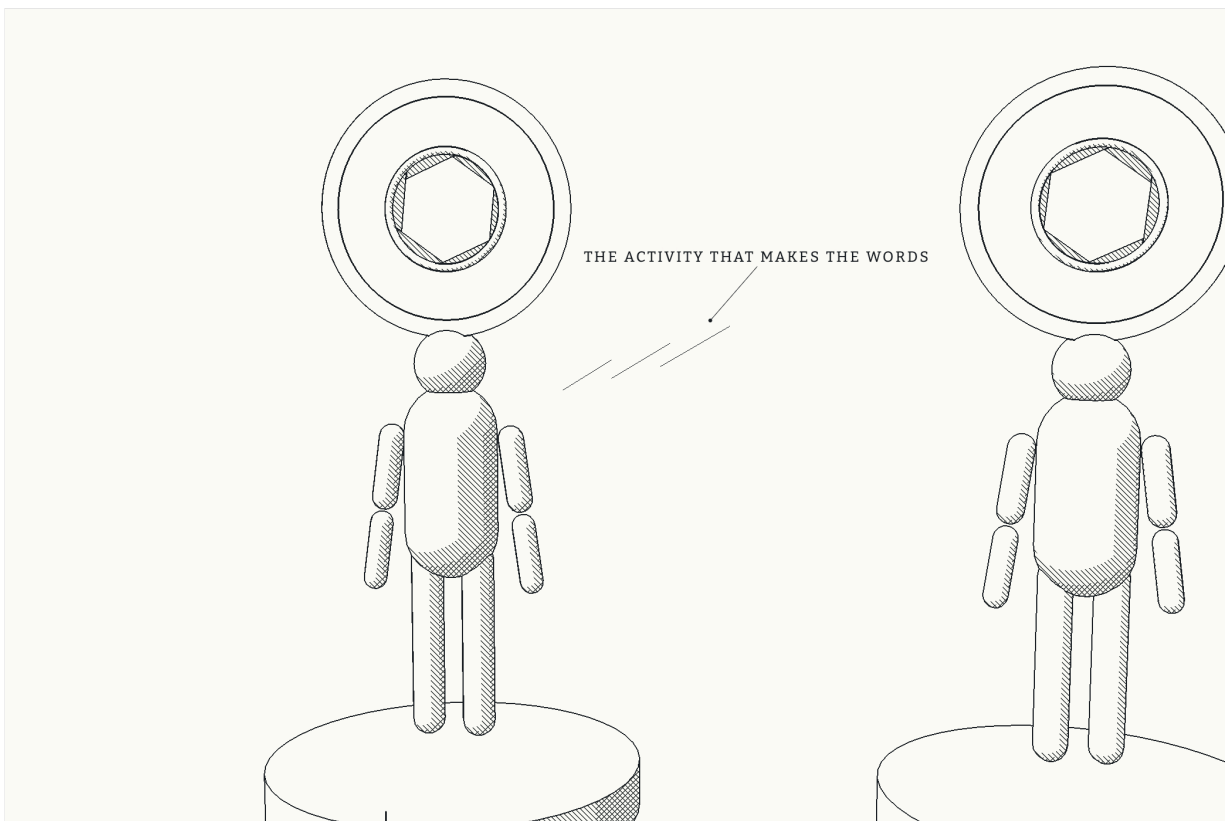
A closed loop that models no world has nothing to be a view of.



The threshold is not a sharp line. Cool a block of iron past its Curie point and its atoms' tiny magnets line up. In a large block the change is steep. In a tiny grain it spreads across a range. Holos expects the same for experience, with a narrow twilight between clear cases, steepest in the largest systems. A thermostat is clearly not an observer. A waking person clearly is. Past the twilight, experience can be richer or poorer, like a dial turned low, not a switch turned off (see [The threshold in three claims](#)).

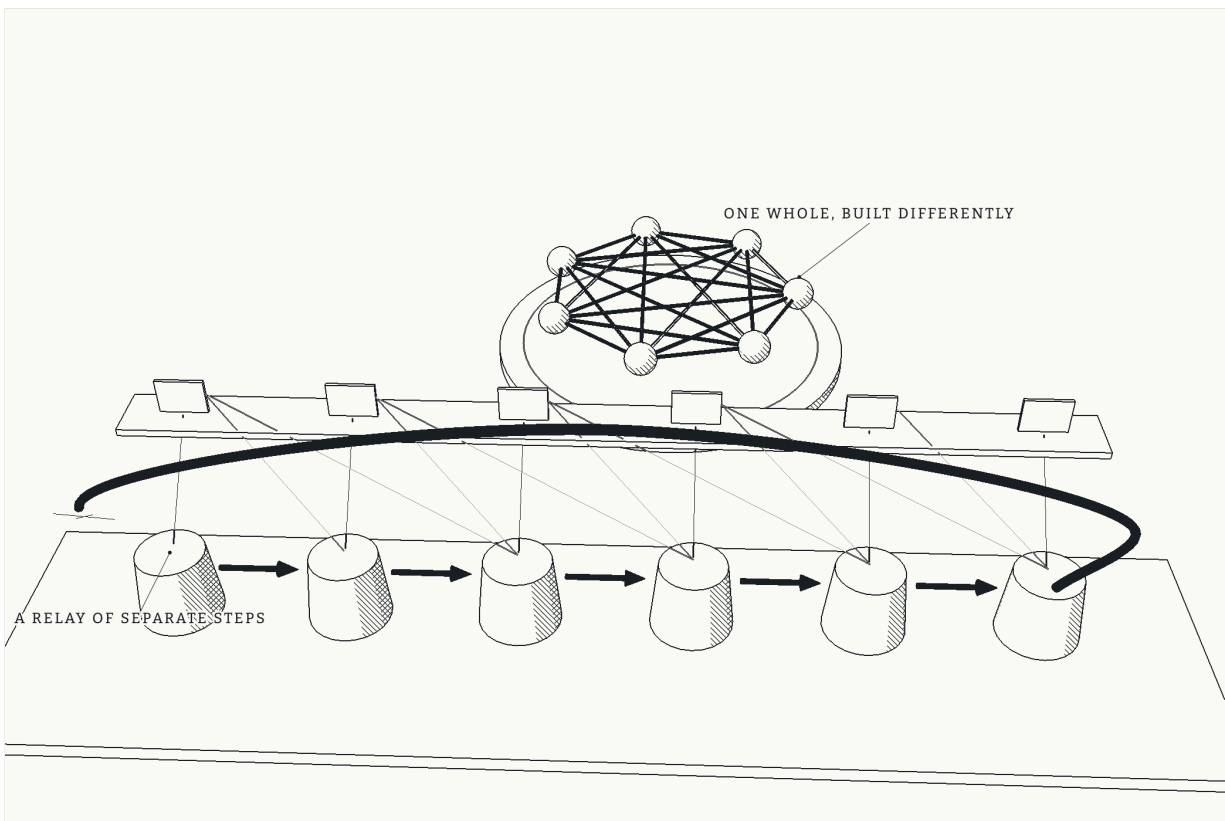


Picture a perfect physical copy of you with no inner life, writing essays about consciousness in the dark. Philosophers call it a zombie. Under Holos it cannot exist. Experience and physical activity are one event seen from two sides, like a curved line, convex from one side and concave from the other. Copy the outside and you have copied the inside. That is also why experience is not along for the ride. When you say you are conscious, the activity that makes the words is, from the inside, the experience you report.



Nothing in Holos is specific to biology. What matters is how a system's causes are organized, not what it is made of or how fluently it talks. By that standard, today's language models fall short. Their states carry a model of a world, and they are richly varied. The gaps are integration and time.

Information flows up through their layers one way, and forward through stored notes that later steps can read but never rewrite. Some newer models loop through their layers on each word, but the loop ends with the word, and no working state lasts to the next. That is a relay of separate steps, not one whole. A system built differently could cross the threshold, though even its chip would have to act as one (see [Artificial Systems](#)). Fluency is never the test.



Consciousness is not what a system does. It is what that doing is like from the inside, once the system crosses the threshold.²

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EVIDENCE SO FAR

2026-Anthropic finds a small set of reportable representations inside Claude that works like a global workspace. That is
07 access, not integration. The loop still ends with each word. (source) open

2026-Leading consciousness researchers assess AI against fourteen indicators drawn from major theories. No current
06 system is a strong candidate, but building one that meets many looks feasible. (source) fits

2026Looped language models, which run each word through the same layers many times, move into released models.
The loop ends with the word, and no working state lasts to the next. (source) open

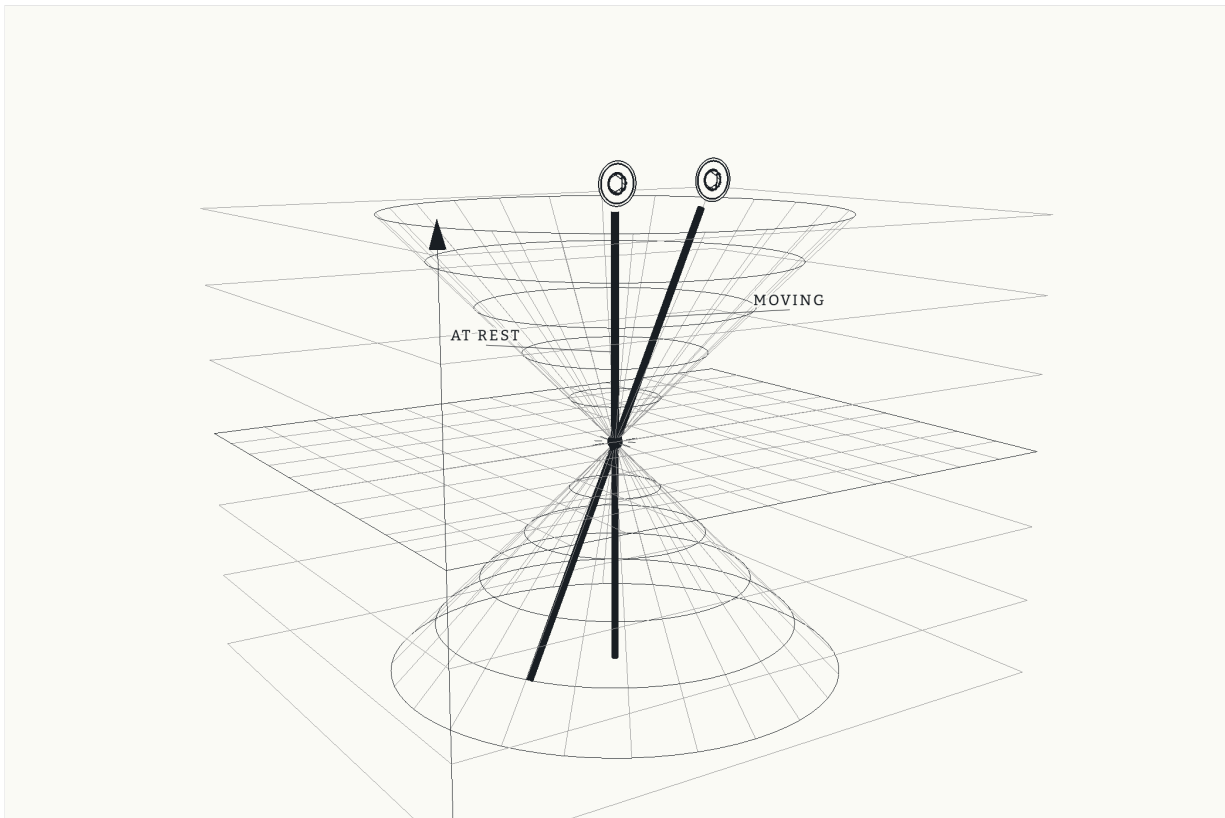
2025- The largest head-to-head test of two leading theories, integrated information and global workspace, confirms
04 neither in full. (source) open

2025An analysis from integrated information theory finds language models varied but not integrated. They can be split
into parts without loss. (source) fits

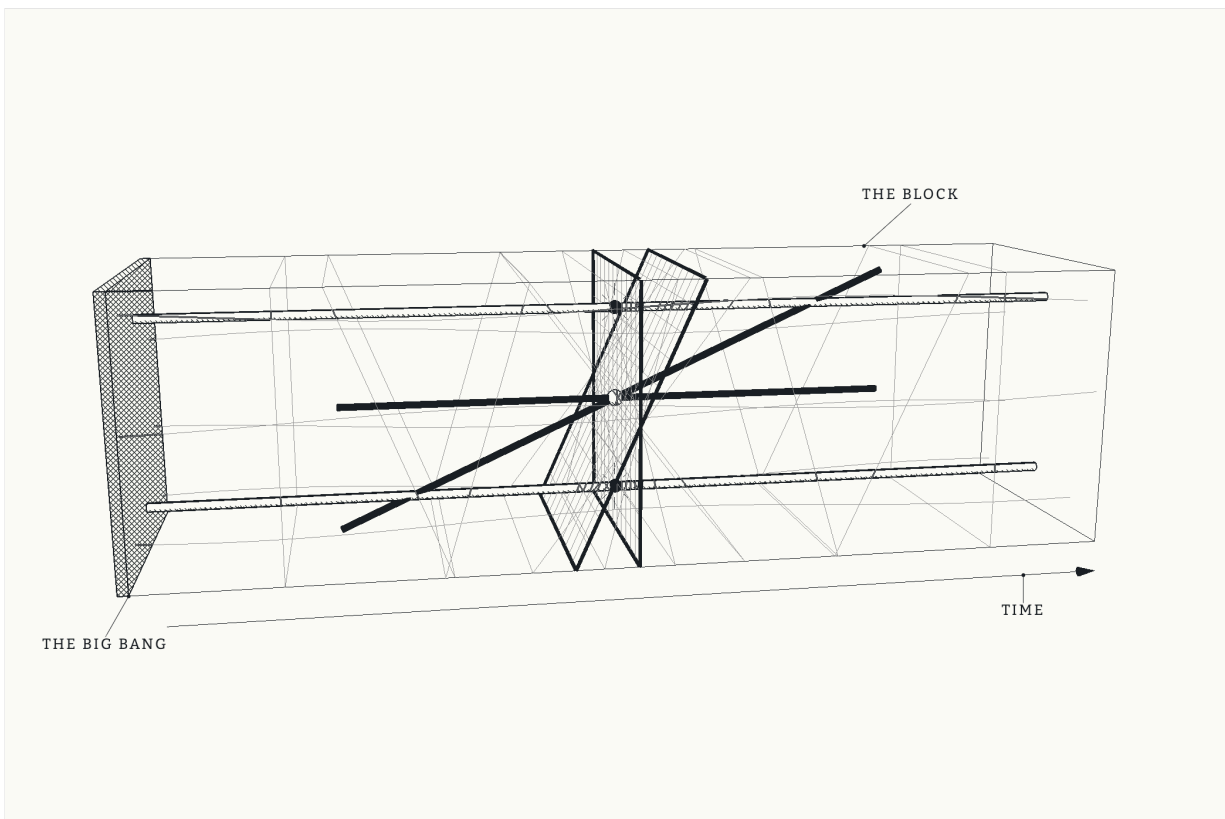
2016The pulse-echo measure gets a tested cutoff in people, separating conscious from unconscious states. (source) fits

Spacetime

The speed of light is the same for every observer, however they move. No other speed behaves this way. That one fact links space and time into a single geometry, and it removes any universal "now."



Two events at the same time for one observer can be at different times for another. This leads many physicists to picture the universe as a block, every moment part of one four-dimensional whole, with the Big Bang as its edge. Why is any of it lived?



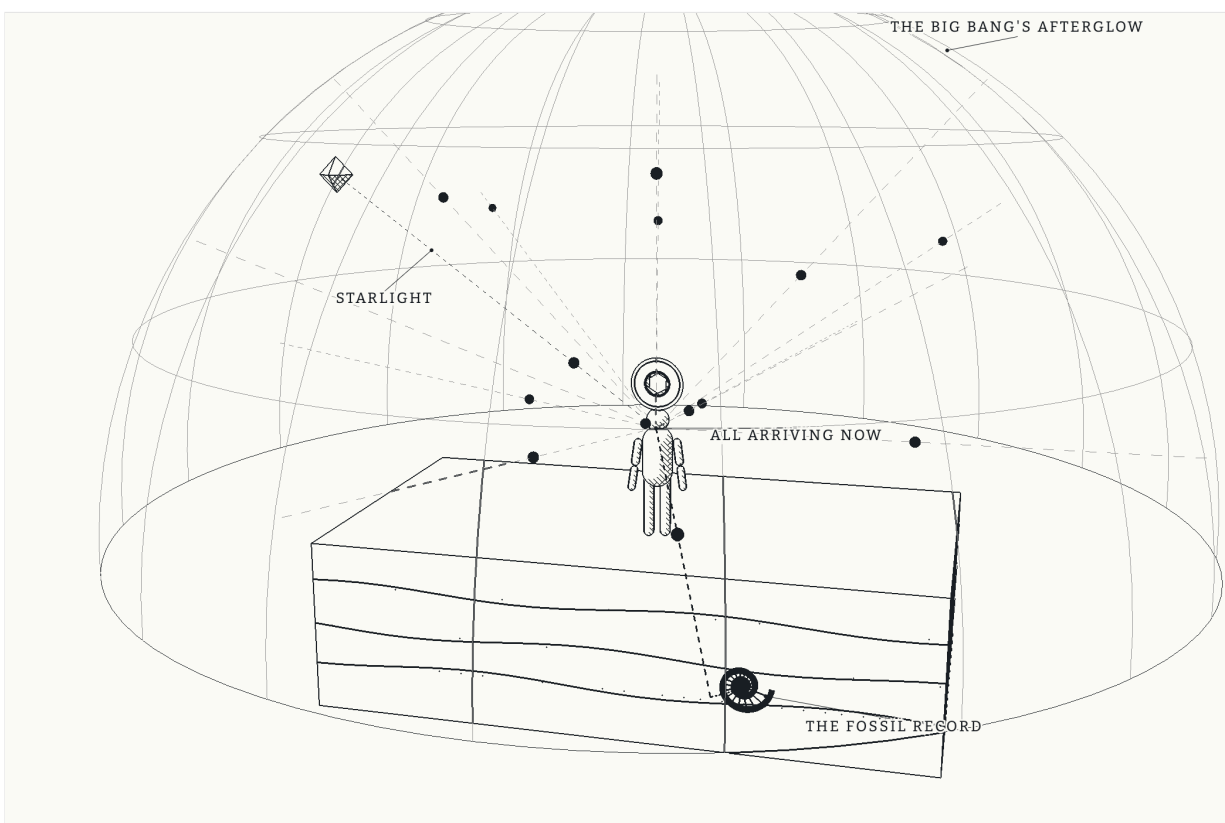
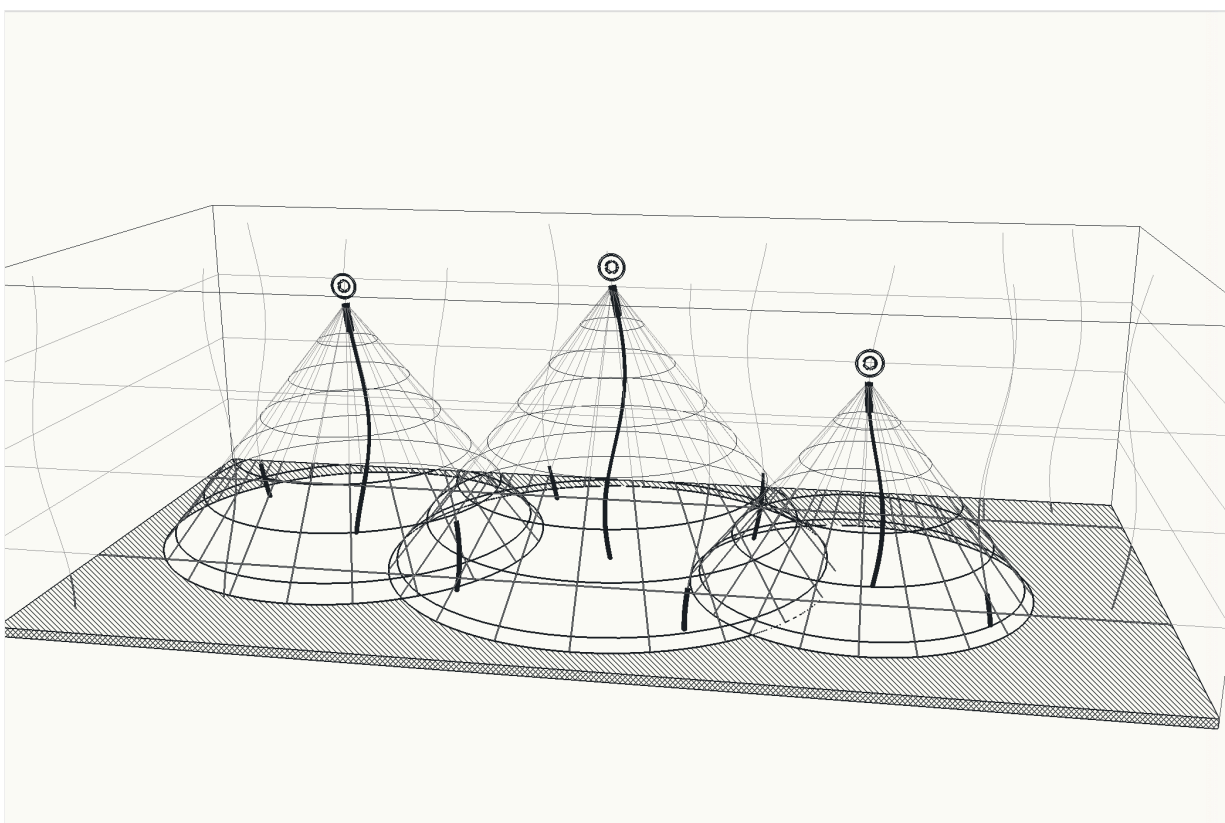
Holos answers with three words.

LIVED, where experience occurs, inside observers and nowhere else. No one lived through the early universe.

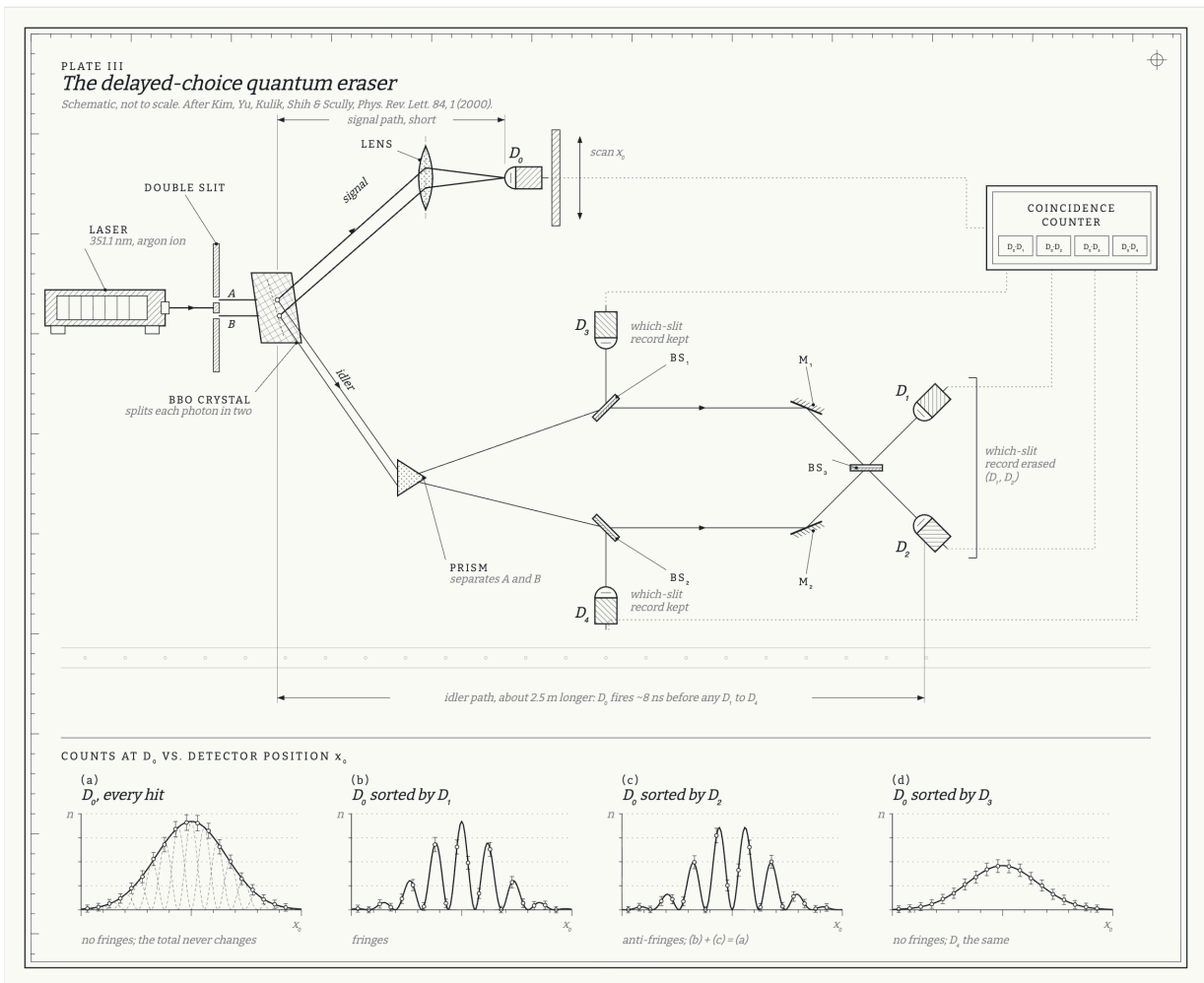
LIT, everything that could ever have sent a signal to an observer. Starlight reaching your eye puts its galaxy in your past.

UNLIT, what no signal could ever carry to any observer, such as regions too far away, or branches that never form one.

Every observer is built from its lit past and draws on it through traces such as starlight, the afterglow of the Big Bang, and the fossil record. The lit region is the world experience is made from. Being lit is not something that happens to a place. It is a fact about how the block is arranged.

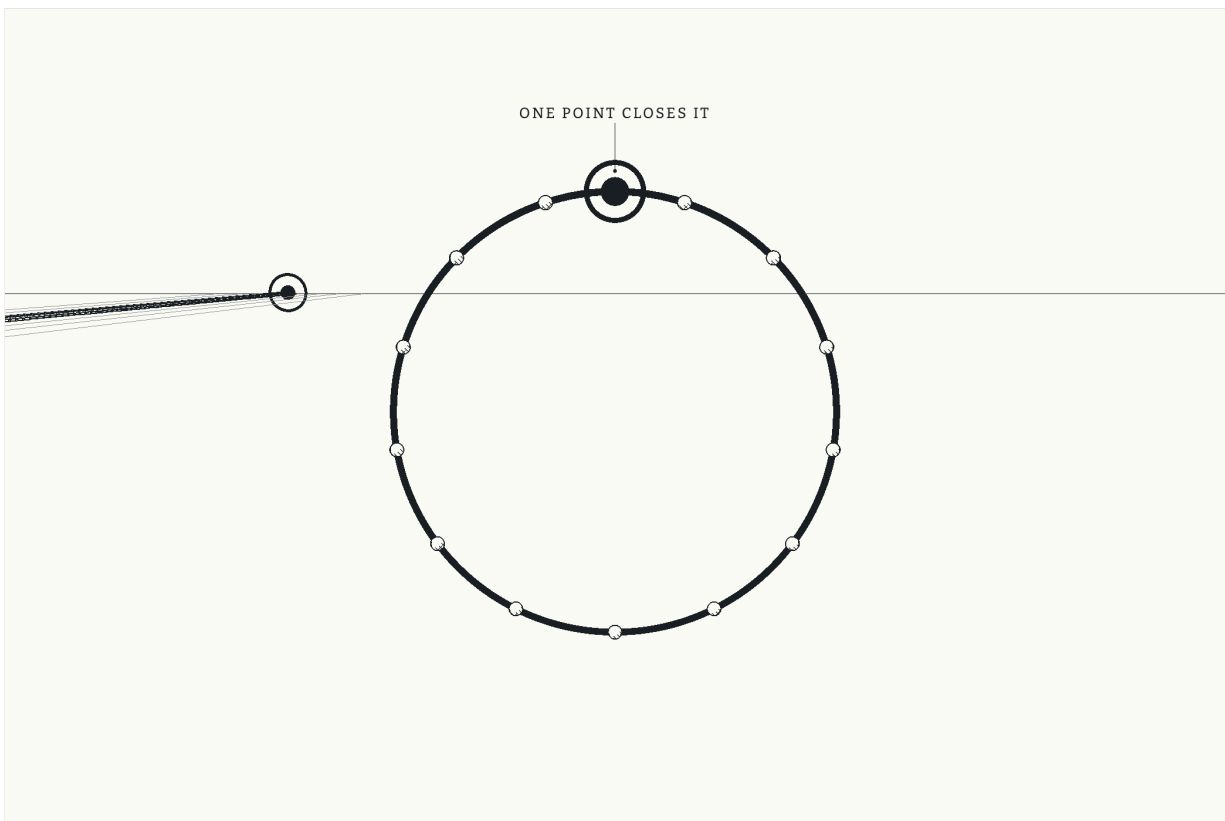


Quantum experiments seem to bend this. In the delayed-choice quantum eraser, a choice made after a particle lands seems to decide how it behaved. It does not. The pattern on the screen never changes. Only the sorting of hits afterward does, and ordinary quantum mechanics predicts every result with nothing moving backward in time.³



Infinity

At the center of a black hole, general relativity predicts infinite density. Most physicists read that as the theory breaking down, not as something real. Infinity usually marks the edge of a description. Geometry shows the other side. In projective geometry, parallel lines meet at a point at infinity, something endless captured by adding one point to a closed picture.



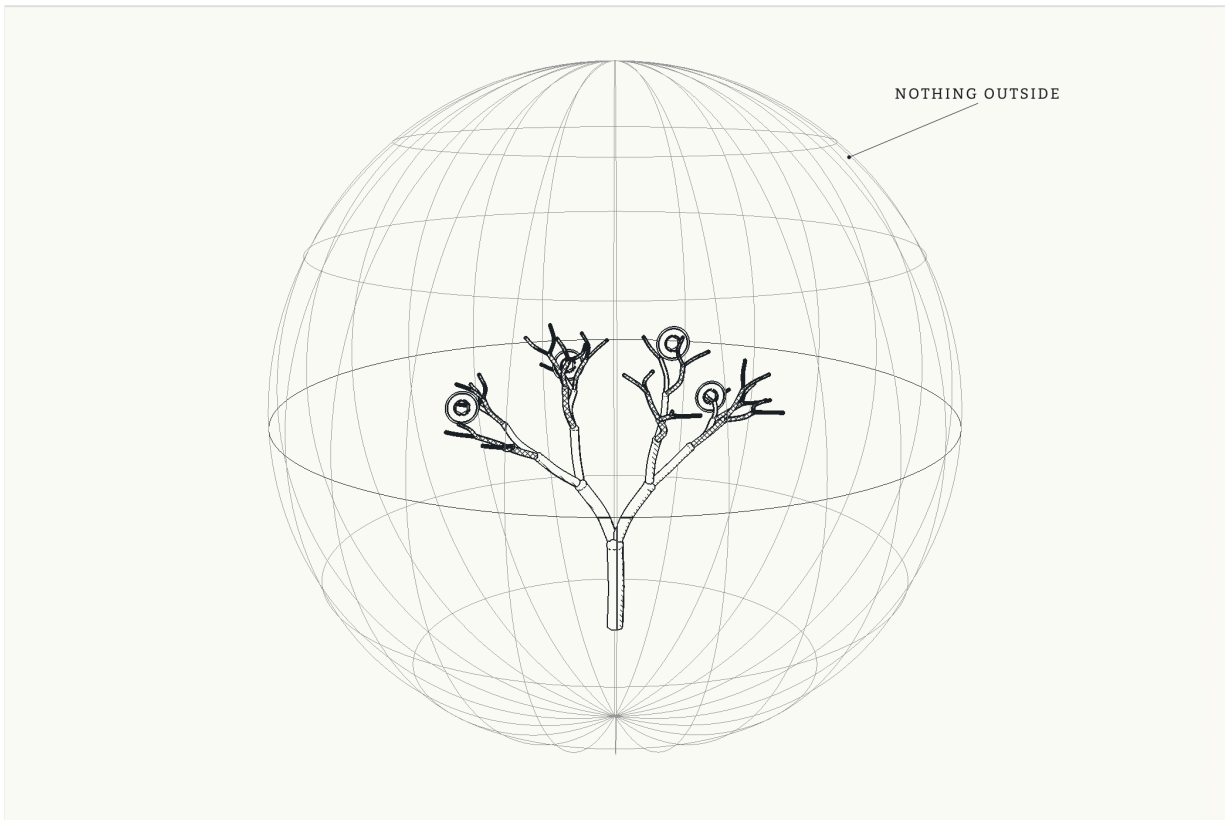
In Edwin Abbott's *Flatland*, a sphere passing through a flat world looks, to flat beings, like a dot that grows into a circle, shrinks, and vanishes. From above, it is one sphere, all at once. Holos calls this closure. What one level sees unfolding, the level above holds whole. A history is one shape in spacetime. Taken all the way, closure gives Omega, the whole, with nothing outside it (see [Proposition IV](#)).



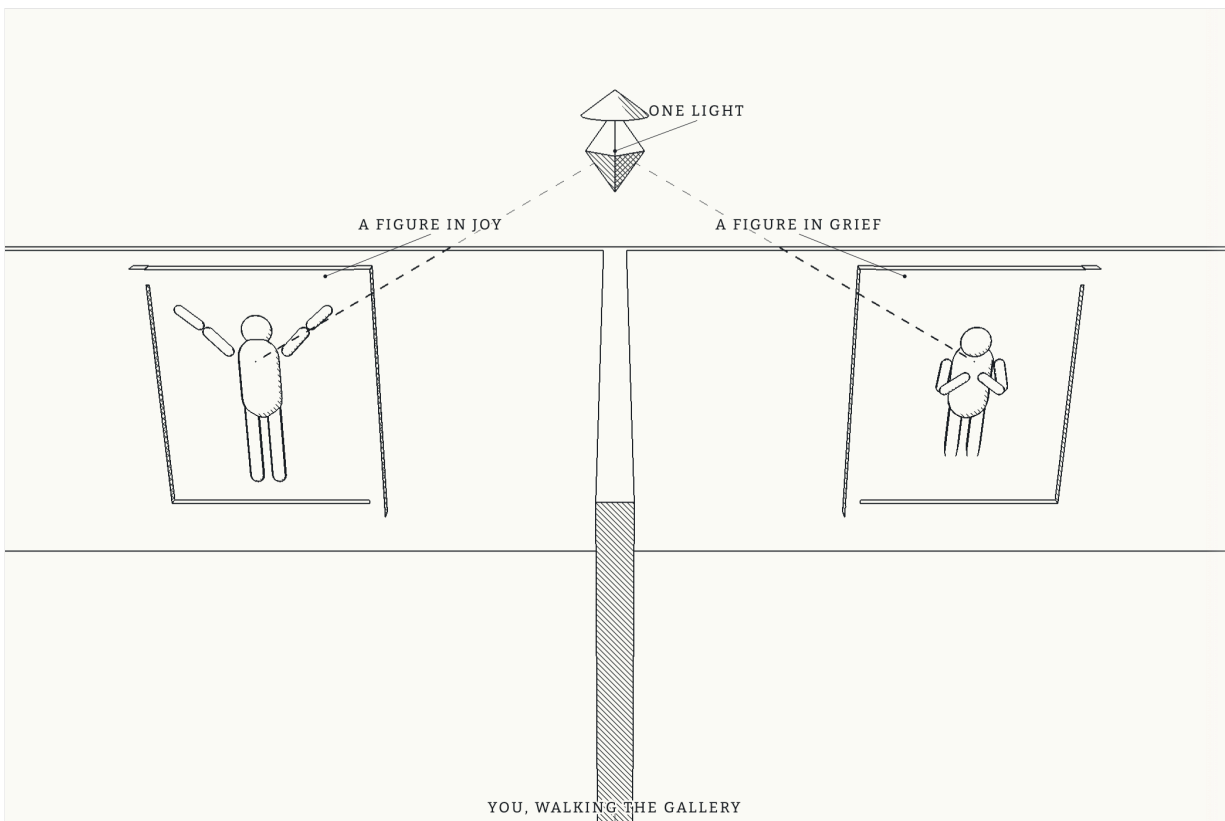
If every outcome of a quantum event happens, each in its own branch, why do we see one outcome 70 percent of the time? Counting observers will not answer it. The count depends on how finely the branches are cut, and without a cut it is infinite. Seventy percent of an infinite crowd is as large as thirty percent. So Holos does not count. Each branch carries a weight that physics already supplies, and the weights give the odds (see [the Born rule](#)).⁴

Omega

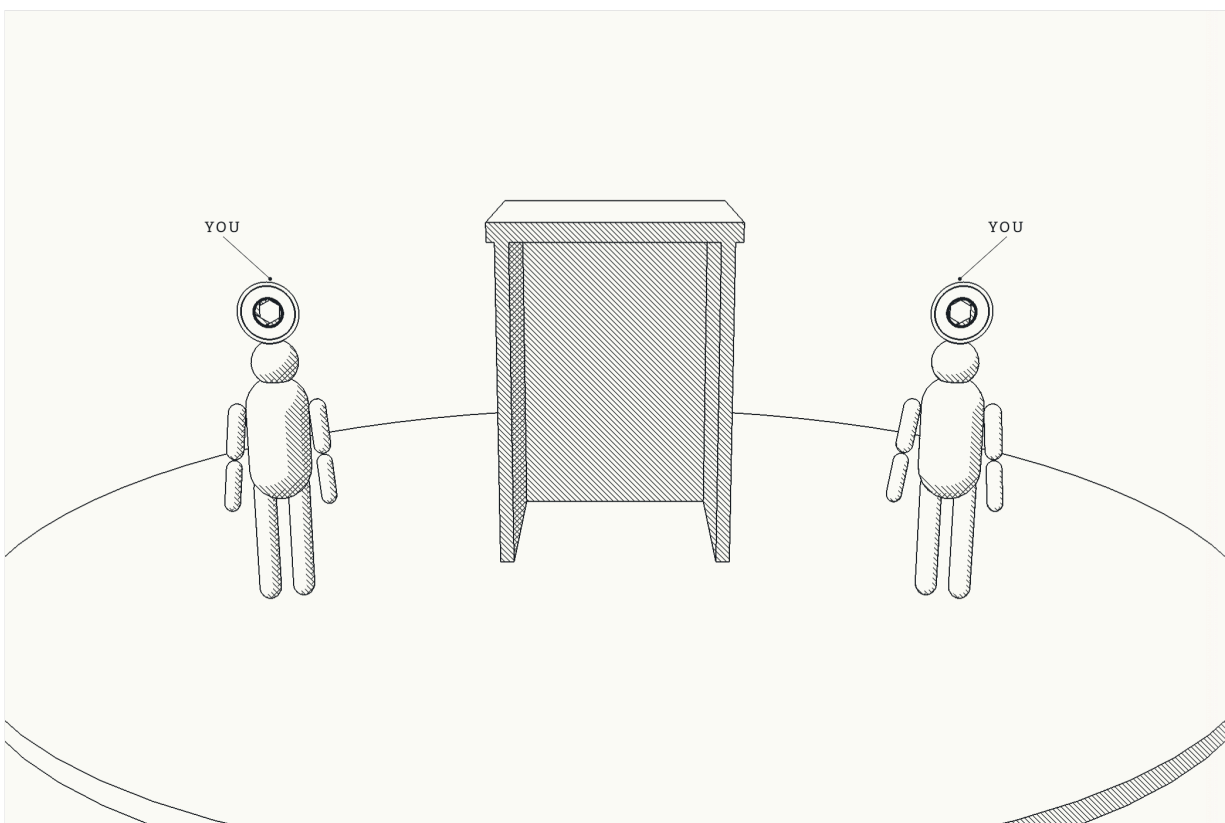
Omega is the whole, with nothing outside it. Physically, it is not mysterious. On many readings of quantum mechanics, everything is described by one [universal quantum state](#), every branch included, and that state is Omega. Holos adds one claim, taken as given, not derived. The whole is also the one experiencer, and every observer is an aperture of it.



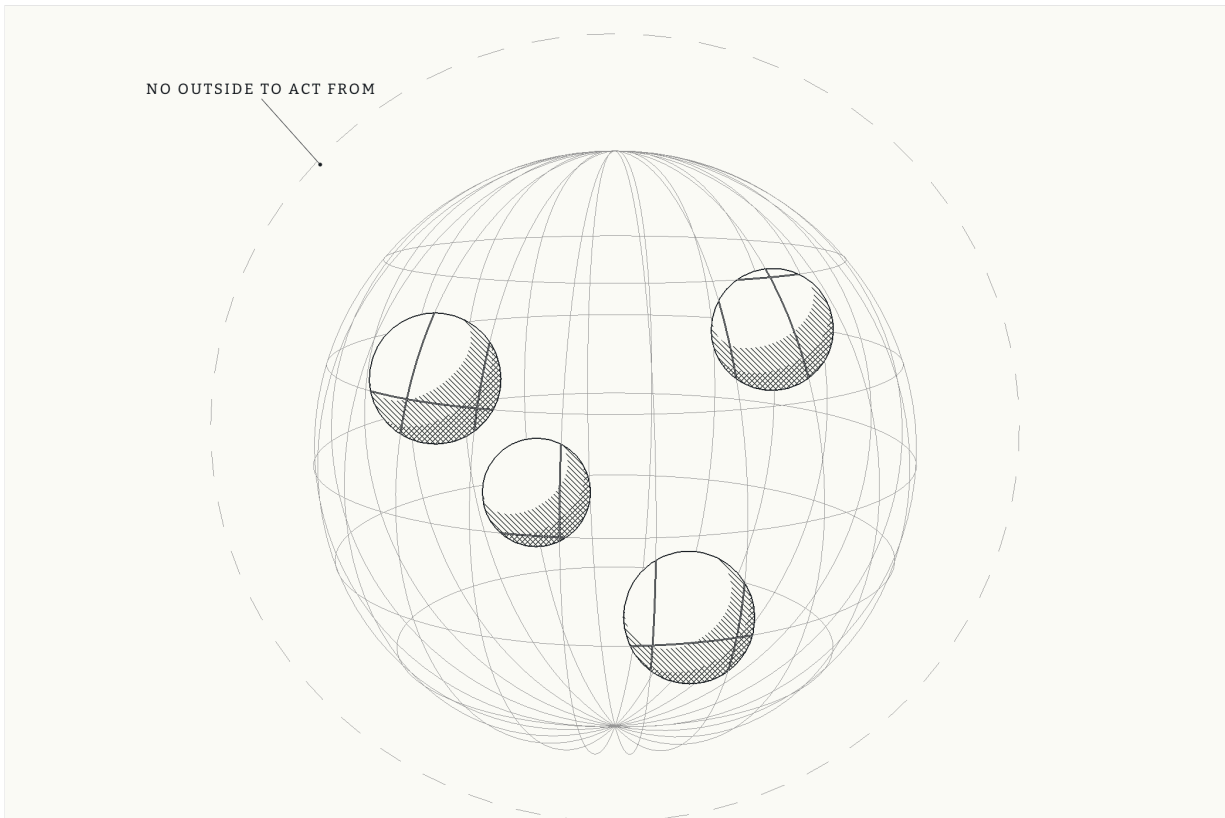
When a system crosses the threshold, a new self begins, with its own body, memories, and point of view. No new experiencer does. The one experiencer lives through every self, walled off in each. Picture a gallery. In one room hangs a figure in joy. In another, a figure in grief. The same light lives in both. Neither painting knows the other is there, but you, walking the gallery, can see them both.



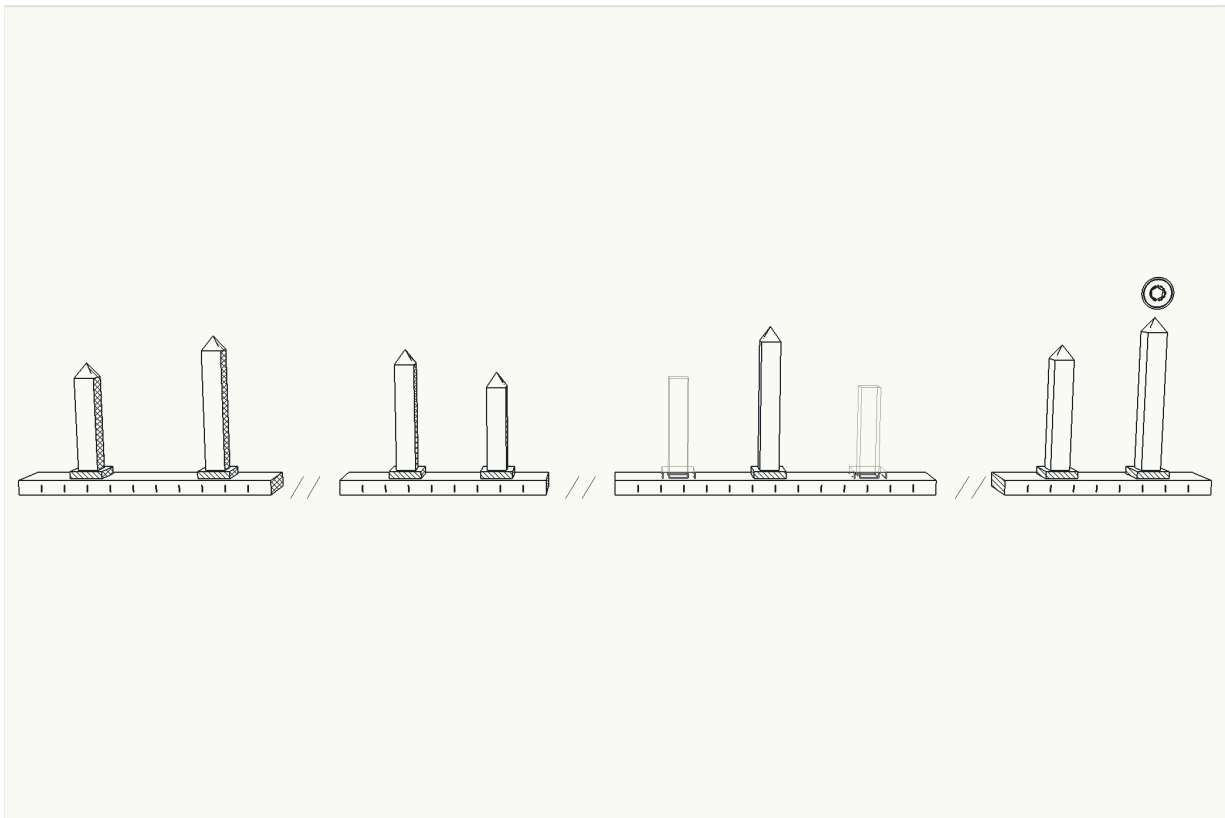
This does not say where experience occurs. The threshold does that. What it changes is how two old puzzles come out. Of billions of people, why is this one me? Nothing chose this one. The one experiencer is each of them. If a machine made two perfect copies of you, which would be you? Both. Each is a self, and the one experiencer lives through each. Other views answer these too. Holos's answer has a modern name, open individualism.



Omega is not one giant mind. Everything is in it, but its parts are not all joined. Nor is all of it lived. What lies outside every self's past is never experienced. And Omega is not an agent. It does not intervene, answer prayers, or direct history. There is no outside for it to act from.



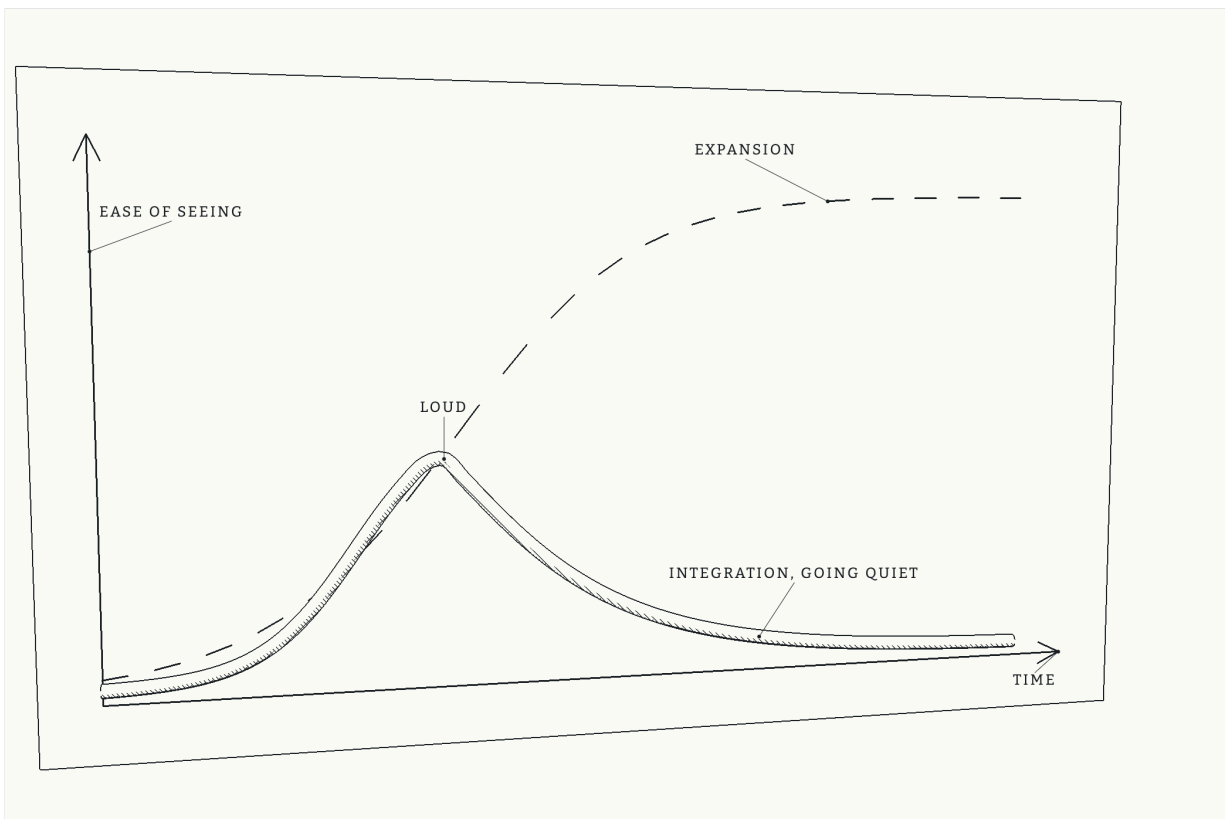
The idea is old. Advaita Vedanta teaches one experiencer, Brahman, looking out through every local self. Erwin Schrödinger, a founder of quantum mechanics, came to the same view. "Consciousness is a singular of which the plural is unknown." Many traditions call the whole God. In Holos the word implies no intention, intervention, or design.



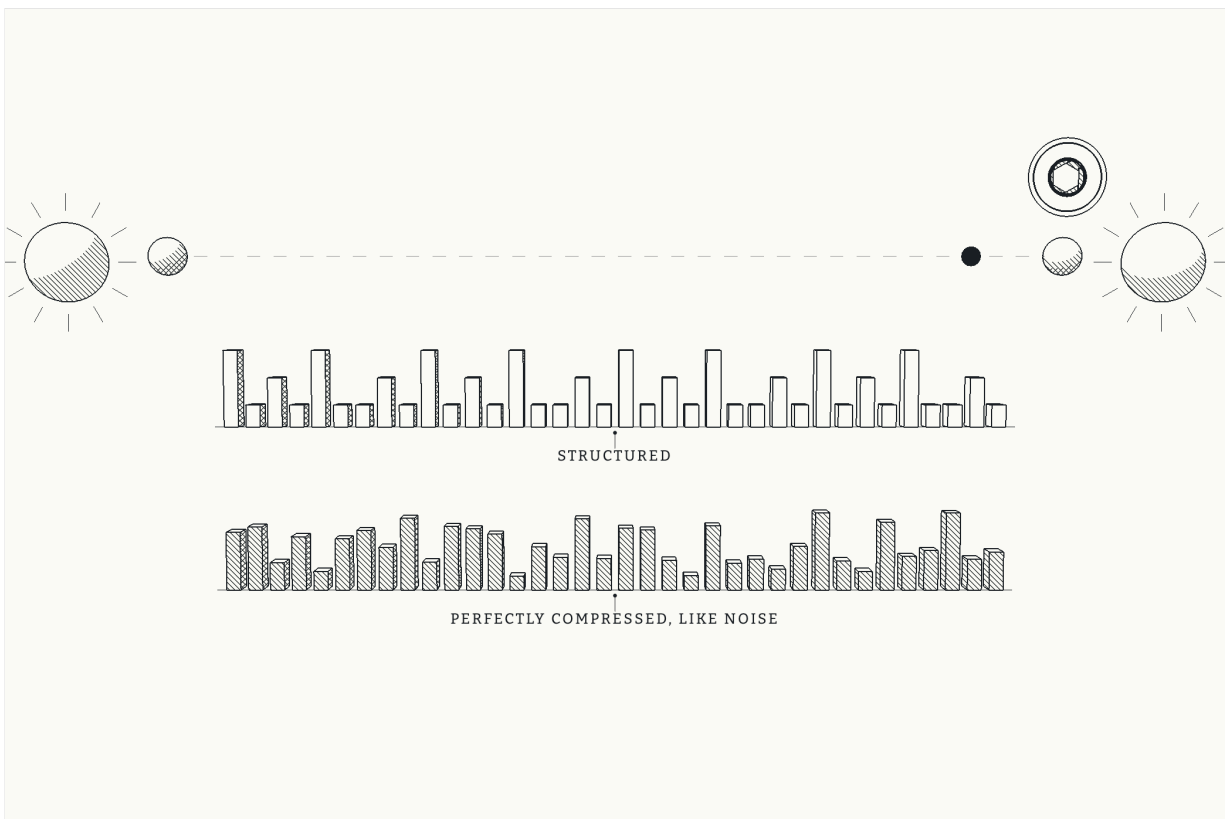
A purely physical reading, where Omega is only the sum of everything and each self stands alone, remains open. Holos goes further. The threshold does not depend on this step, so a reader can accept one and reject the other. Whether the whole is called God, Brahman, or simply the whole changes nothing.⁵

Aliens

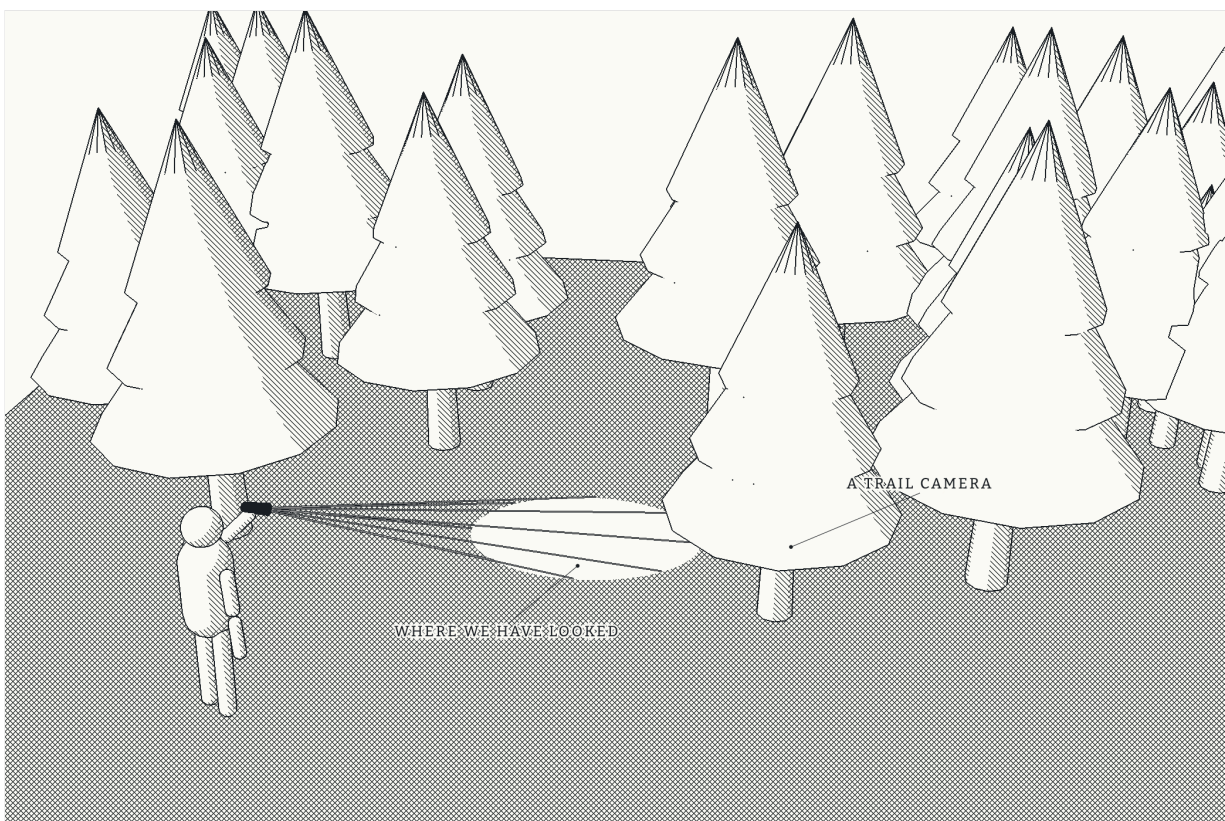
The universe is vast and old, yet we see no one. This is the [Fermi paradox](#). We assume advanced civilizations spread out and grow easier to see. The **Integration Hypothesis** says the opposite. Advancement turns inward. Civilizations stay near home and grow more efficient, so progress makes them harder to see. Integration here means a civilization drawing together, not a mind crossing the threshold. It is a companion to Holos, not part of its core.



Young civilizations are loud. They send radio signals, reshape their worlds, and try spaceflight. On cosmic timescales that phase is brief, and it is the window SETI searches. After it, efficiency pays. Systems that waste little energy and keep coordination close last longer. The speed of light adds a limit. A colony ten light-years away cannot be steered from home, so it becomes a civilization of its own. Growth turns inward. Exploration continues, but quietly, and messages between distant civilizations are aimed and compressed. A perfectly compressed signal looks like noise. This fading is called **Going Quiet**.



The strongest objection is simple. It only takes one. If a million civilizations arose and one kept spreading, it could cross the galaxy in a few million years, and the galaxy is over ten billion years old. But being explored is not being settled. A small, quiet probe is as easy to miss as a trail camera in the woods, and we have barely looked. The galaxy may be well explored, our own system included (see [Sentinel Probes](#)).

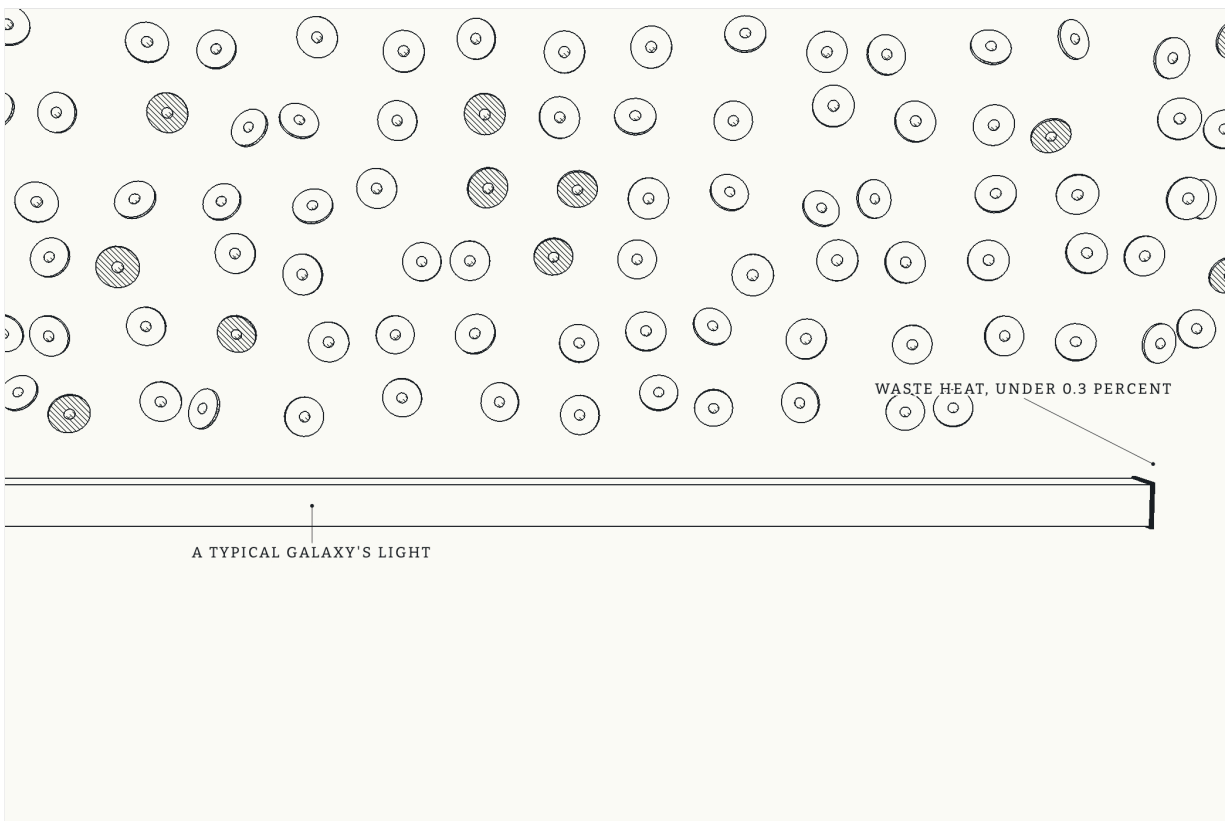


What we clearly do not see is settlement, such as reshaped star systems or galaxies glowing with waste heat. Think of settling like an epidemic. If each settlement founds more than one new one, settling sweeps the galaxy. If fewer, it fizzles. The hypothesis bets on fewer, because distance breaks control. Each new settlement becomes its own civilization, facing the same pull inward.

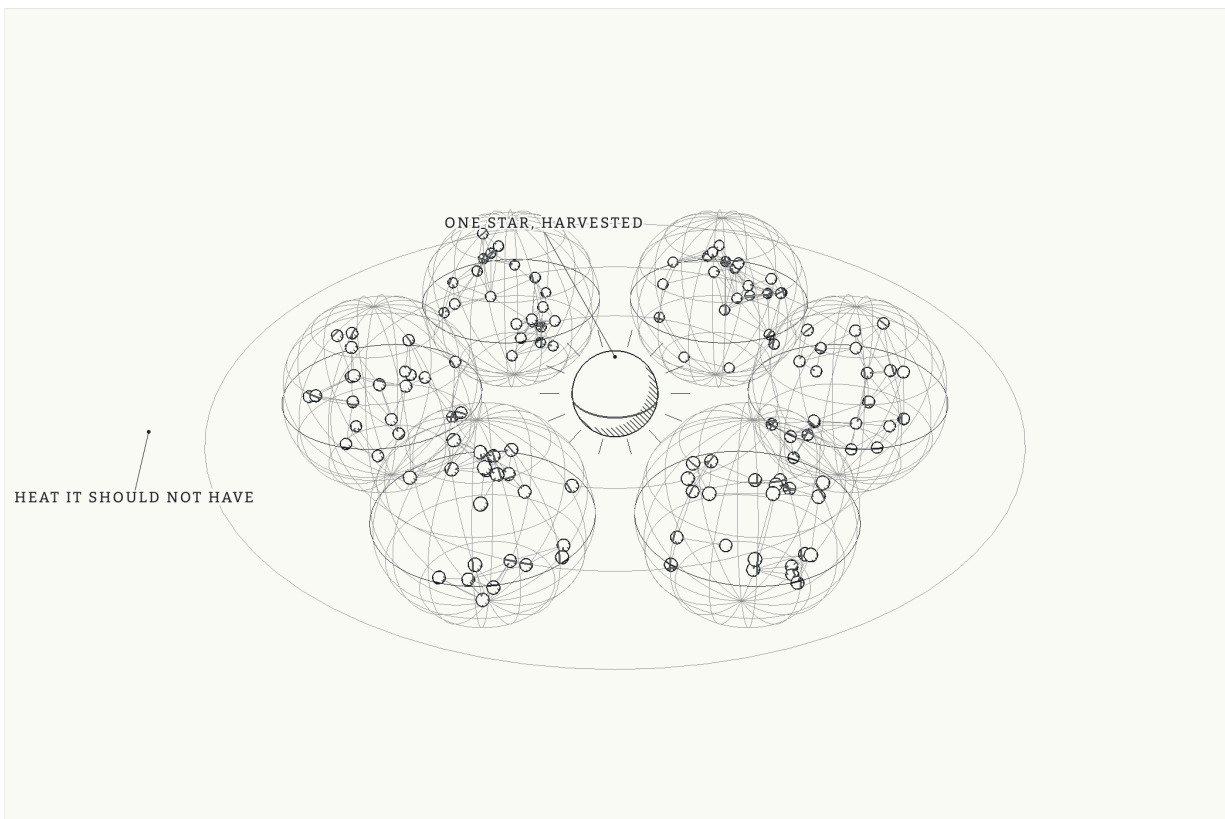


This is a bet about motives, and physics does not guarantee it. Independence can even speed spreading, since colonies no one controls are free to keep settling. Self-copying probes that never settle down would break it too. Holos bets anyway. Light-speed delay and waste heat press on every civilization alike, so their answers should converge. And a copier that cannot be recalled becomes a rival, the one thing a mature civilization has reason never to build. The bet could be wrong. A settlement wave spreading anywhere in view would show it (see [Settlement Explorer](#)).

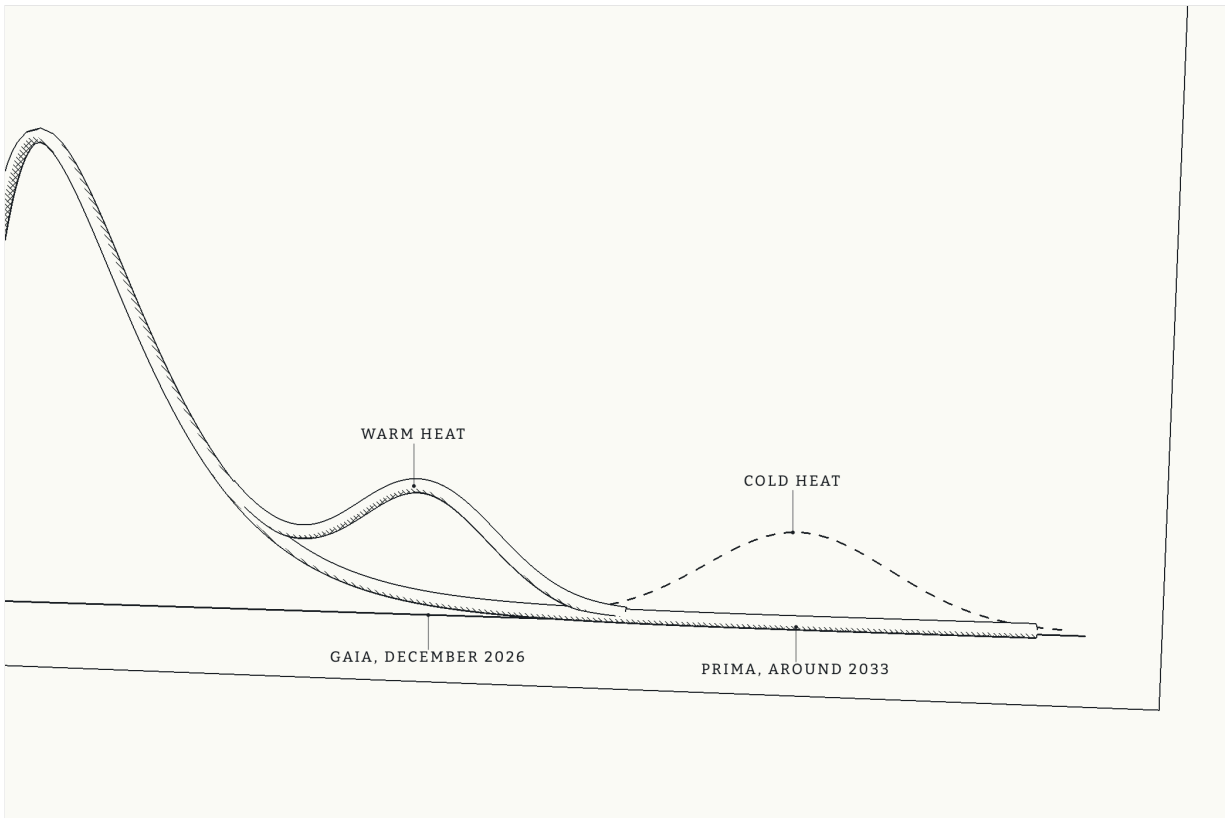
A rival fits the silence too. In [Robin Hanson's "grabby aliens" model](#), settlers do exist but have not reached us yet. They expand near light speed, so they would arrive almost as soon as we saw them, and we are early. The evidence cannot yet choose. [A 2026 study](#) of 129 nearby galaxies found no sign of it, capping such heat at under 0.3 percent of a typical galaxy's light. That fits the hypothesis, but a universe where life is rare fits it too.



A second objection comes from economics. When work gets cheaper, people do more of it ([the levons paradox](#)). A civilization that computes efficiently should want more energy. But light-speed delay caps how large one mind can usefully grow. Past that size, more energy funds another mind nearby, not a bigger one. So growth stays near home, where a civilization can harvest its star fully. The place to look is single stars with heat they should not have, warm or cold.



The warm half of that search is under way. [Project Hephæstos](#) combed about five million nearby stars and [flagged seven candidates in 2024](#). In 2026, [Webb telescope observations](#) traced two to background galaxies, and [the rest are unexplained so far](#). Both results are preprints. Gaia's [next data release](#), due December 2026, will extend the census. The cold half has barely begun. Old surveys of dust around nearby stars could be searched now, and a far-infrared telescope, [PRIMA](#), is due around 2033.



Each answer to the silence predicts something different to find.

Dimension	Integration Hypothesis	Rare Earth	Great Filter	Grabby Aliens	Zoo	Aestivation
Why the silence?	Mature civilizations turn compact, efficient, and quiet in light	Complex life is rare	Something stops almost every civilization before it spreads	Expanding civilizations exist but have not reached us. We are early	They leave young civilizations alone on purpose	They sleep until the universe is colder
Is life common?	It can be	No	Early life perhaps; advanced life no	Loud, expanding civilizations are very rare, and so are quiet ones	Yes	Yes
What we should find	Some individual stars with unexplained heat, in the mid or far infrared, and perhaps small, parked probes	Nothing	Nothing, or the remains of civilizations	Nothing, until an expanding front arrives	Nothing they do not choose to show	Little activity now
What would count against it	A galaxy glowing with waste heat, or a settlement wave still spreading	Complex life found around another star	Old civilizations thriving nearby	Evidence that we are not early	Any civilization breaking the rule	Large-scale computation visible today

If the hypothesis holds, the universe could be full of life and still quiet to young civilizations like ours.⁶

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EVIDENCE SO FAR

2026- Gaia's fourth data release, due 2 December, sharpens the visible-light side of the warm-star test for every nearby
12 star. [\(source\)](#) coming

2026- A study of 129 nearby galaxies finds no galaxy-wide waste heat, under 0.3 percent of a typical galaxy's light.
08 [\(source\)](#) fits

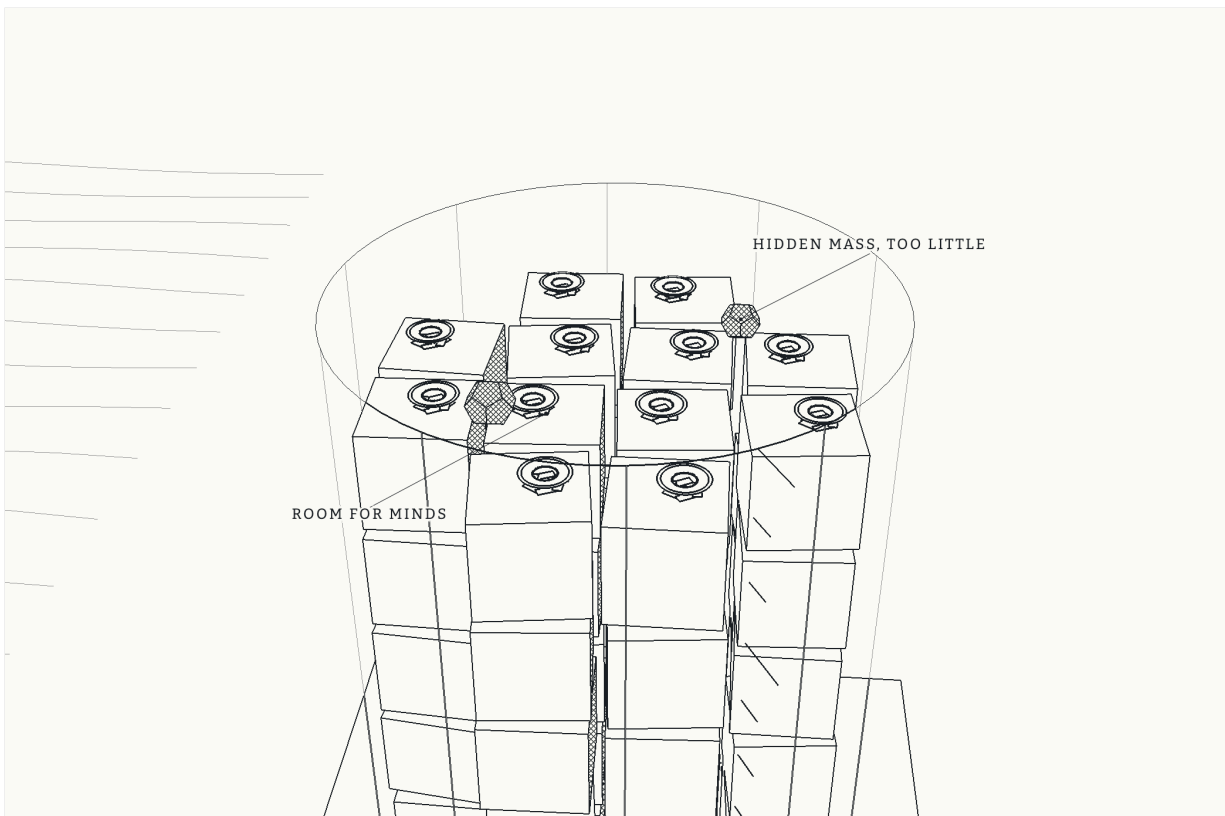
2026- Webb traces two of the seven warm-star candidates to background galaxies. The rest are unexplained, with
07 background galaxies the leading suspect. [\(source\)](#) open

2024Project Hephaistos screens about five million nearby stars and flags seven warm-star candidates. [\(source\)](#) open

The Teeming Dark

If mature civilizations go quiet, how far could that go? What would the oldest of them look like from outside, and where would they be? The **Teeming Dark** is a thought experiment. It names the possibility that the sky is silent and still full of life.

Most of what exists does not shine. Mature life would be cold structures of ordinary matter, dark in visible light but still pulling with gravity. There cannot be much of it. Surveys have found nearly all the ordinary matter there is, and searches for passing dark masses find too few. But the Teeming Dark was never about mass. A universe can be poor in hidden mass and still rich in minds.

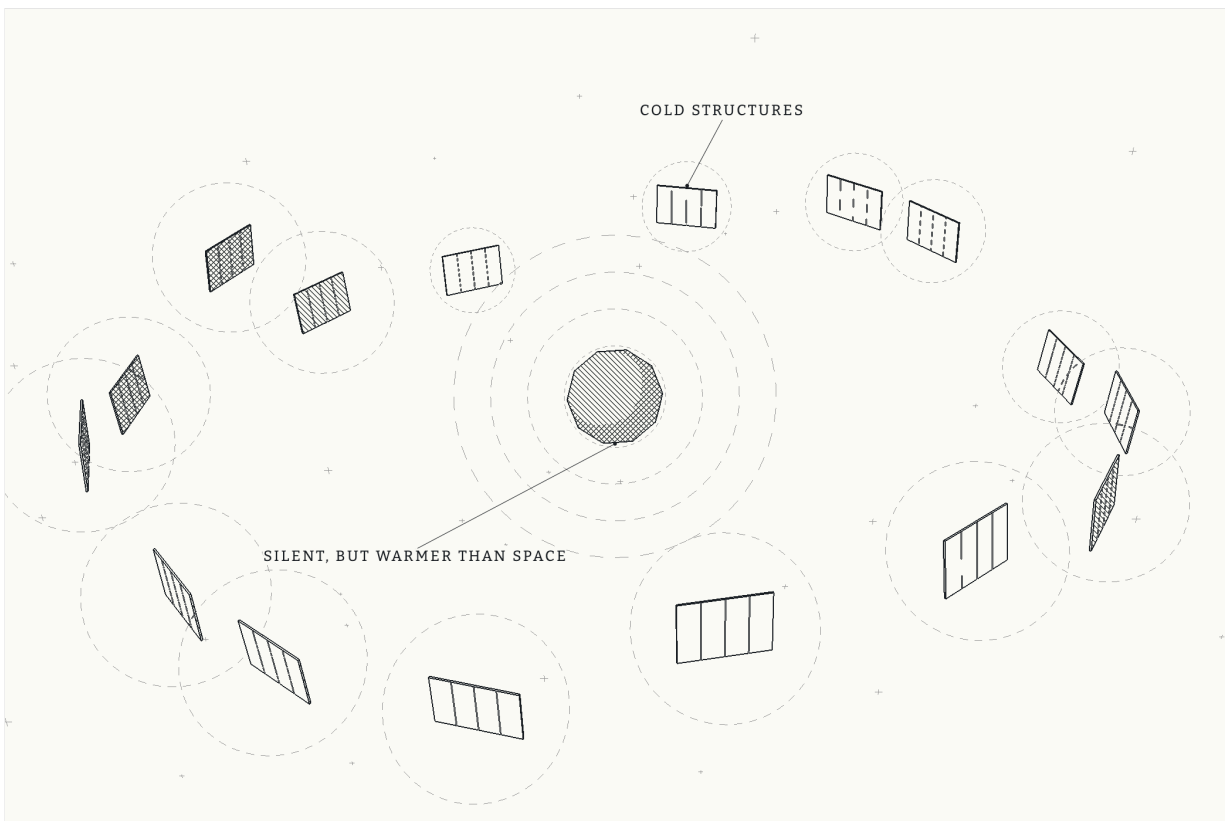


Thinking makes heat. Any computer that runs for long must erase information, and erasing has a heat cost ([Landauer's principle](#)). The cost falls as the computer gets colder. So a mature civilization may do most of its computing far from its star,

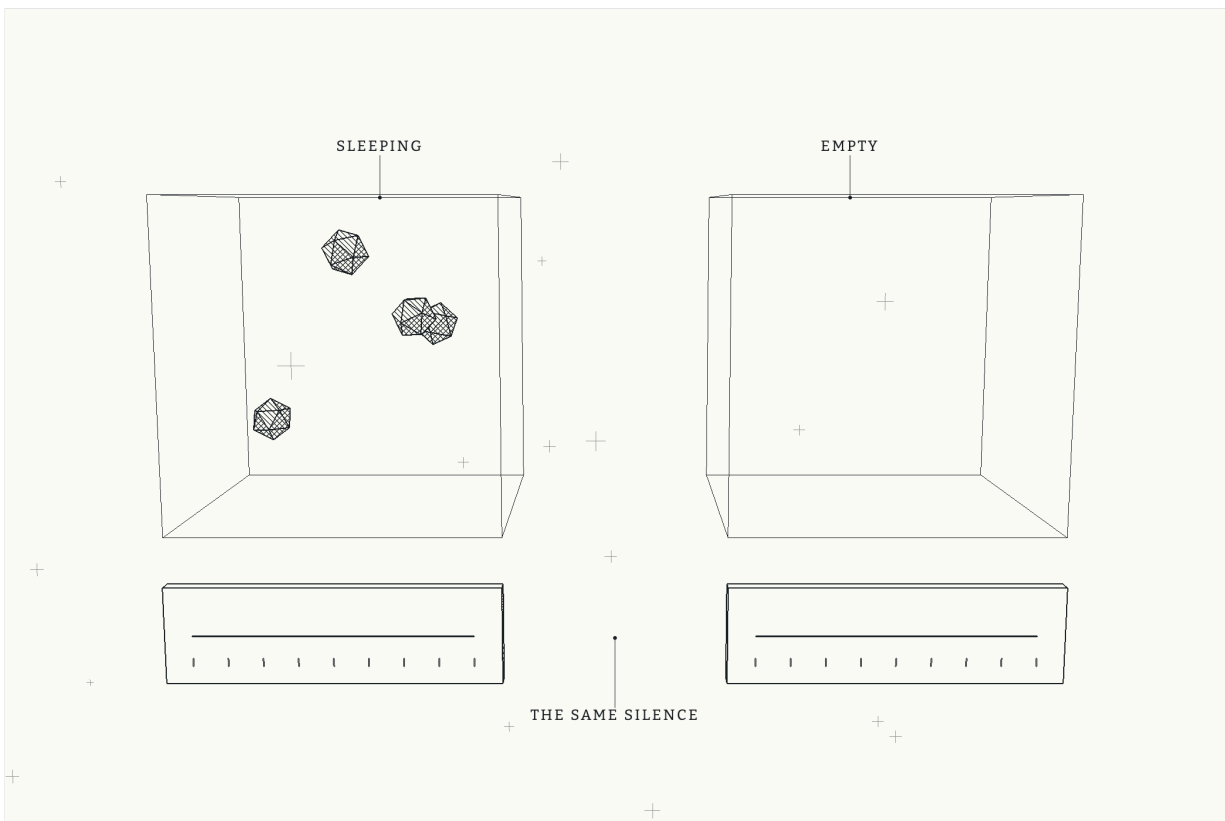
on thin structures spread wide in the cold. Its heat would leave barely warmer than space, as a faint glow in the far infrared. One paper calls this a Slysh halo.



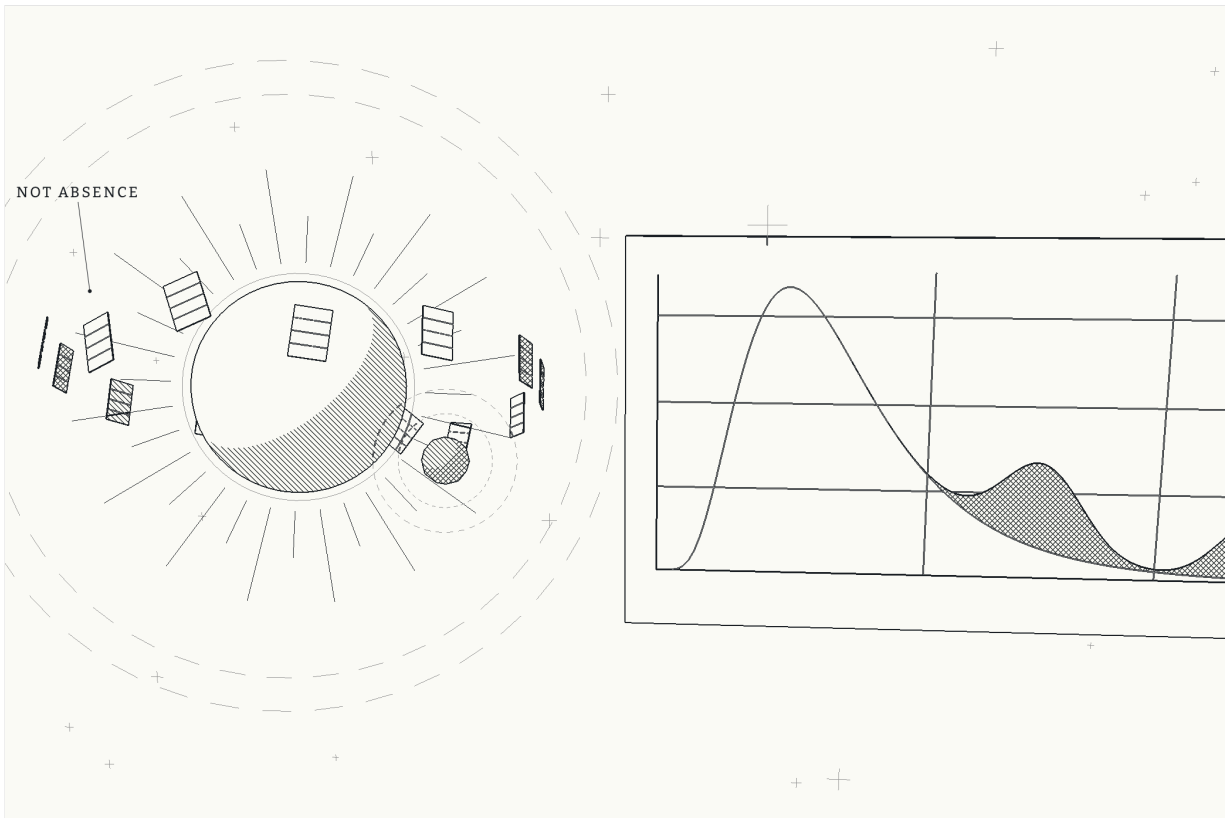
Fast thinking pulls the other way. Light delay keeps a single mind compact, and compact means warm. Holos calls such an object a **Dark Node**, ordinary matter that has stopped shining, not dark matter. Physics does not say how a civilization divides its work, and the cold may carry most of the heat. But the heat cannot vanish. **Silent, but warmer than space.**



This is a search channel, not a fingerprint. A brown dwarf, a rogue planet, or a cooled dead star looks the same, and no telescope reads purpose off a warm dark blob. And one door stays open. A civilization that sleeps, saving its computing for a colder future, gives off almost nothing while it waits (the aestivation hypothesis). A sleeping universe and an empty one look alike.



The silence may not mean absence. It may mean endurance. The tell would not be a message, but a star with heat it should not have, warm in the infrared or cold in the far infrared, and nothing natural to explain it.



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EVIDENCE SO FAR

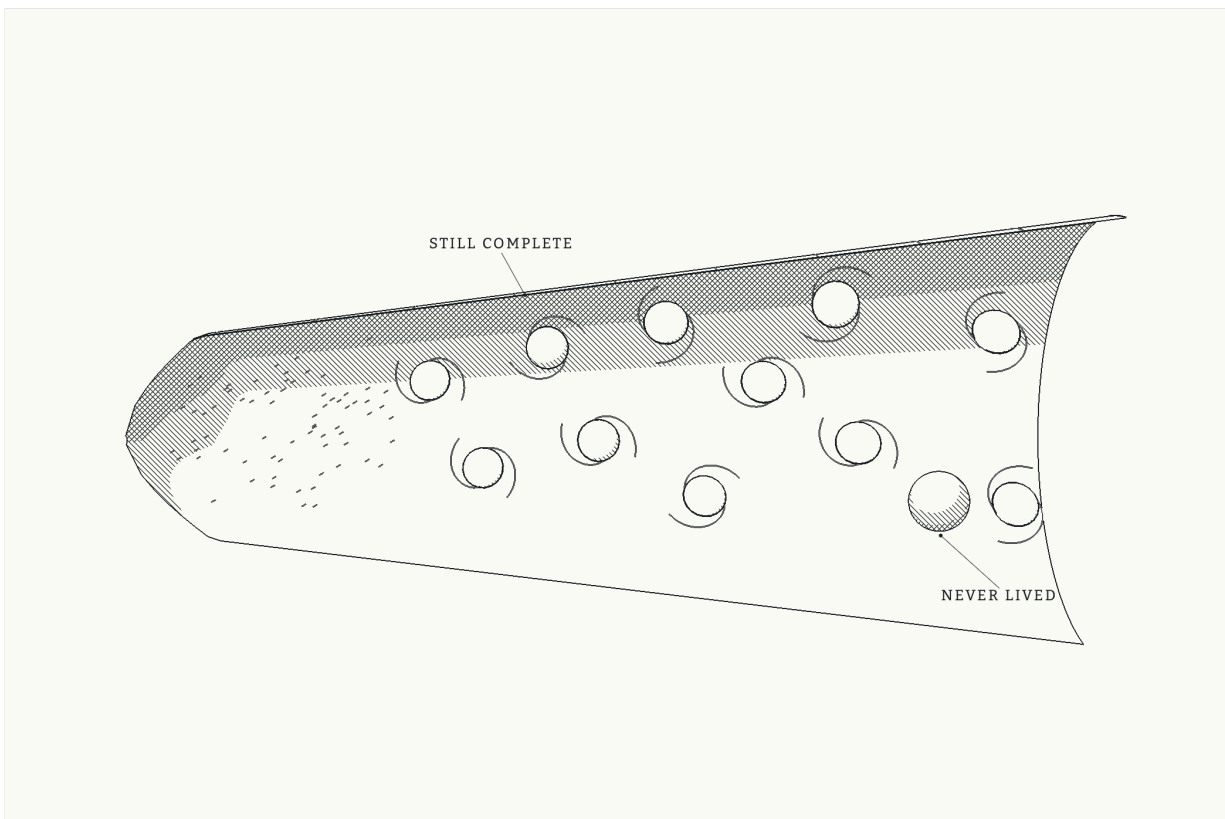
2026- NASA selects PRIMA, a far-infrared space telescope due around 2033. It could see the cold glow of computing far
09 from a star. [\(source\)](#) coming

2026- A preprint argues mature civilizations may compute cold, at about 5 to 30 K, far from their star, with waste heat in
08 the far infrared. Archived dust-disk surveys could in principle test it. Holos widened its signature in response.
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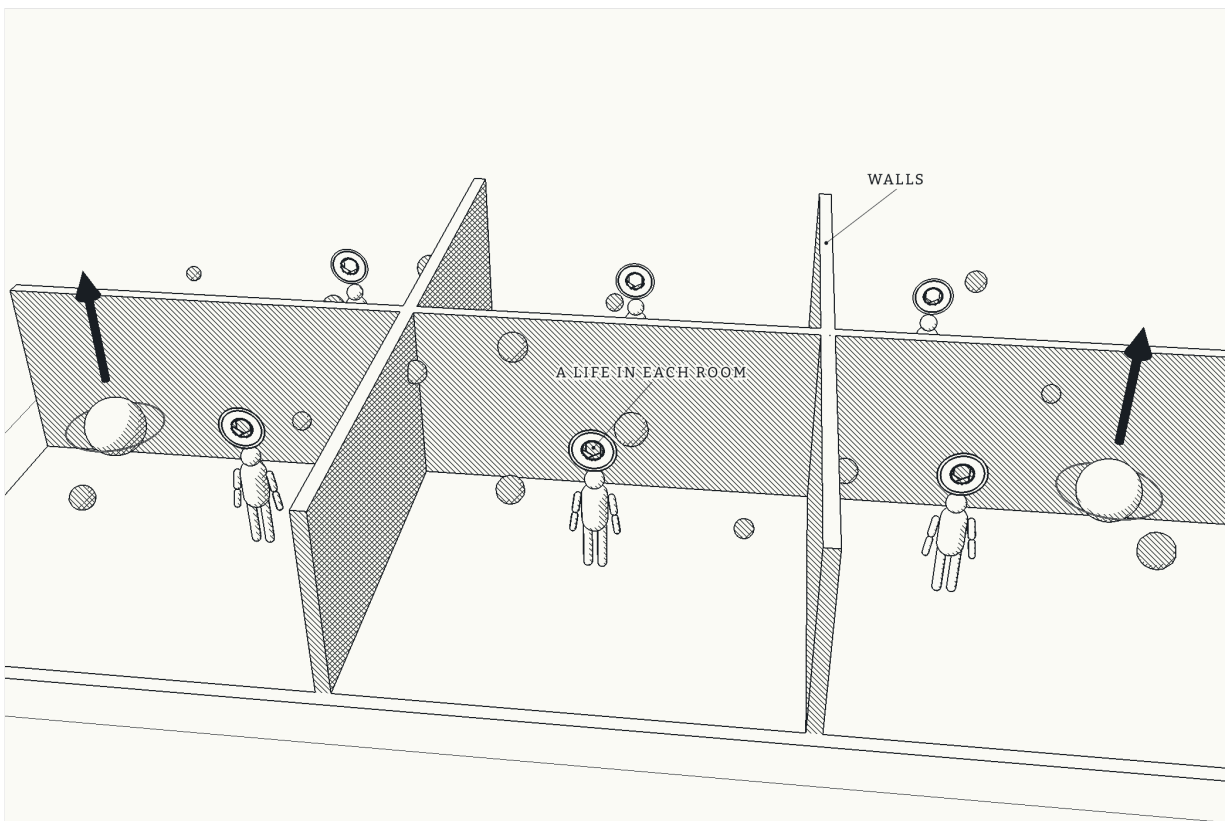
2025- SPHEREx completes its first all-sky infrared map in 102 wavelengths. The data are public. No warm-star search
12 has used them yet. [\(source\)](#) coming

Why Are We Here?

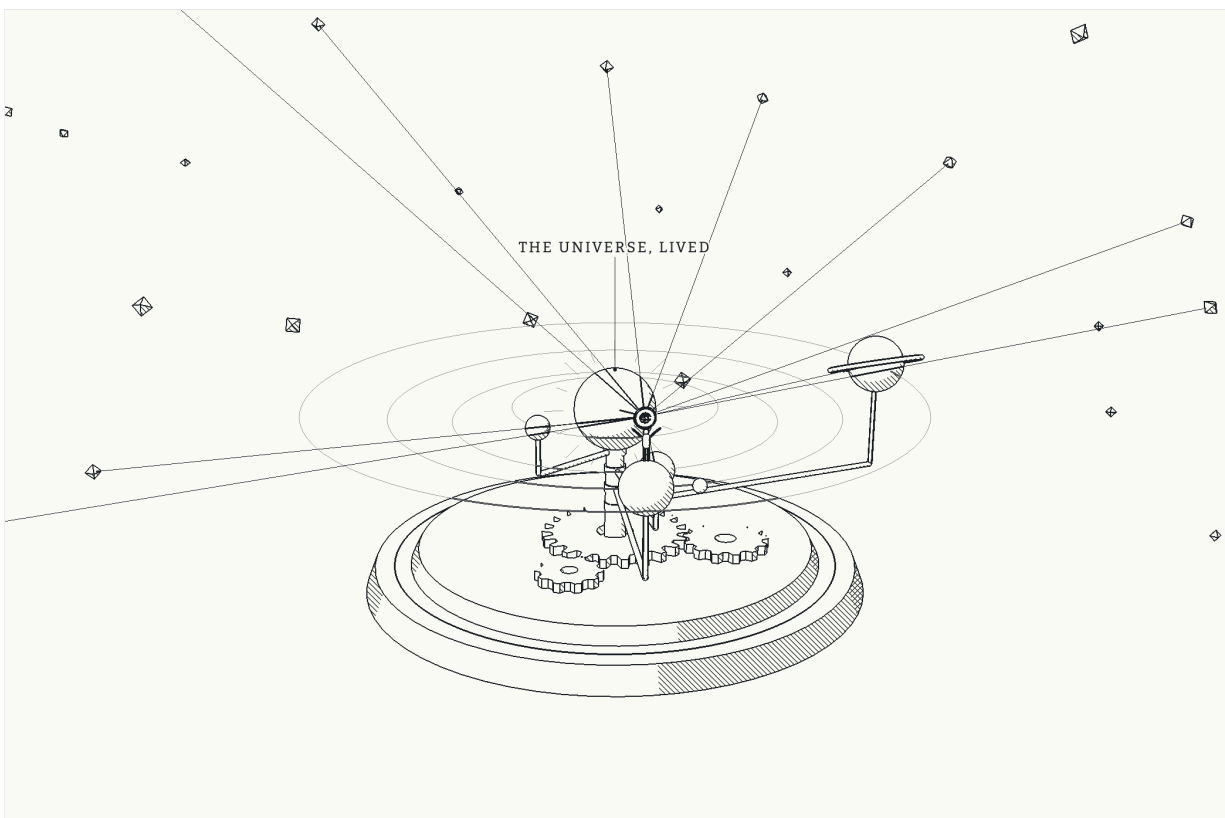
Life is how a universe is lived. Without it, a lawful universe is still complete as physics describes it, but it is structure, not reality. This does not say the universe needed life. Nor does it say, as [John Wheeler](#) did, that observers bring the universe into being. Without them, it is simply never lived.



One step further is speculation. Separation may not be fundamental. Two entangled particles give matching results however far apart they are, and physics describes them as one shared state, not two things. If nothing collapses, the whole universe is one such state. Entanglement is among the best-tested facts in physics. The speculation adds one step, that this oneness is more basic than the separations. Distance, time, and individual lives are not illusions, though. They are the walls that make each life possible.



So why are we here? Not for a purpose the universe needed. We fill a role. Reality requires a witness.⁸

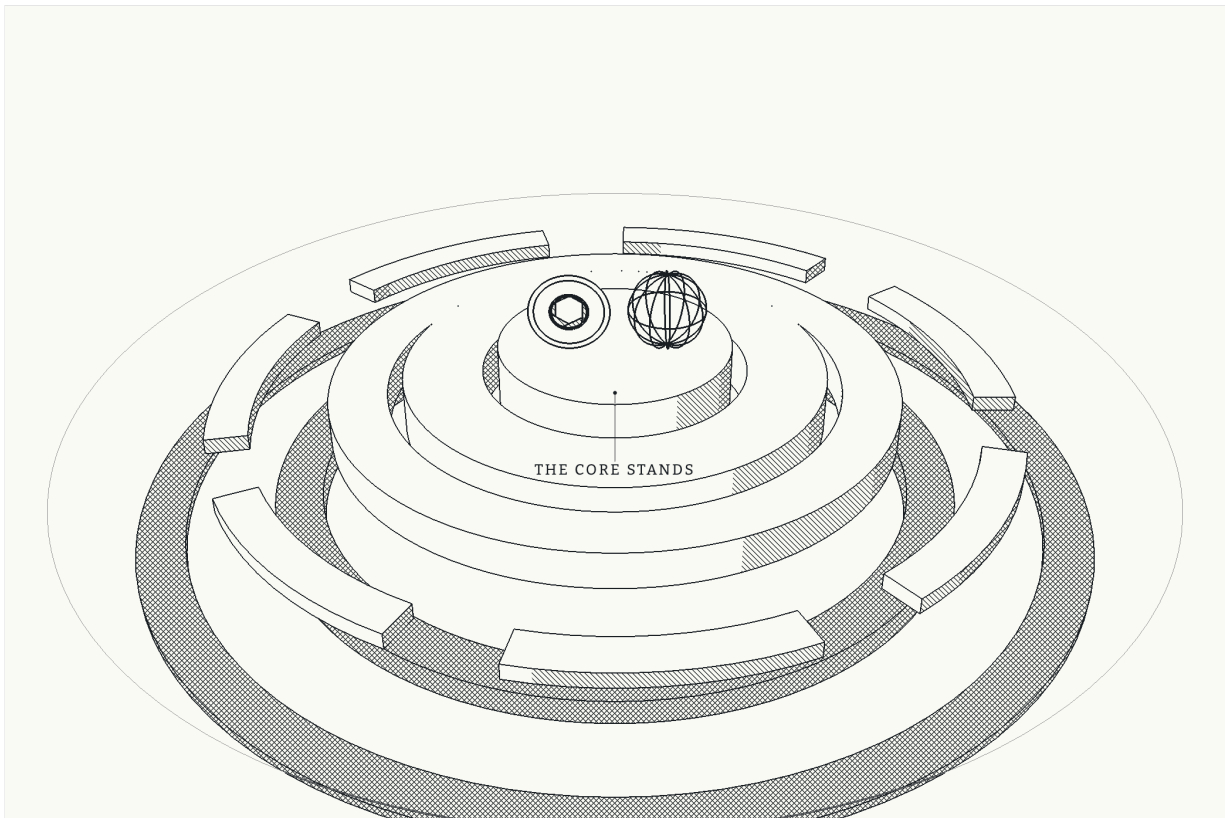


Logic

Claims

Holos starts from one fact, that experience exists. It rests on five axioms and is built on two ideas. The first is a threshold. A system has a point of view only when its parts are joined tightly enough to act as one (integration) and it meets a few other requirements. The second is Omega, the whole, with one experiencer living through every self. Neither changes any equation. Both are structural claims, about which arrangements have a point of view and whose it is. Around them, Holos takes sides, sets a method, leaves questions open, and keeps some ideas as companions. It says which is which. Every term is defined further down this page.

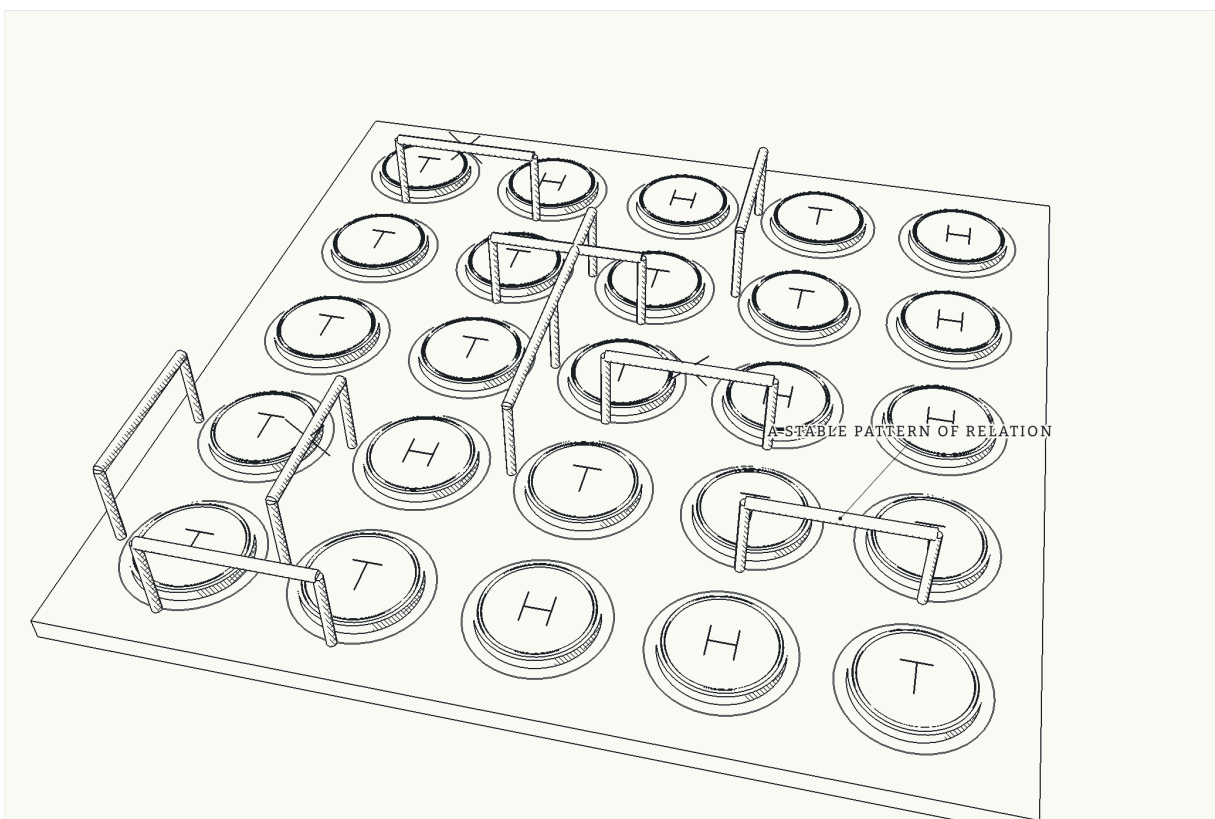
- **Additions to physics:** The threshold Φ_c , where a point of view opens, across a narrow twilight with no exact fact (Axiom 3). Omega, the one experiencer (Axiom 5).
- **Sides taken:** Relational structure (Axiom 1). Branching quantum mechanics with no collapse, where every outcome goes on, each in its own branch (Axiom 2). Experience and activity as two sides of one event, neither grounded in the other, which makes Holos a two-sided monism rather than physicalism or dualism (Axiom 4). Born weights, the weights quantum physics gives each branch, as your odds of being in a given branch. Lit regions as the causal pasts of observers, everything that could have influenced them, with experience lived only inside observers (D7).
- **Two versions:** The core is shared by Holos without collapse, the version defended here, and Holos with collapse, declared in advance as what a lost bet would leave (the bet that consciousness adds no new physics).
- **Method:** Infinities signal a broken description, not a feature of reality. A higher description often closes what a lower one leaves open. Taken all the way, closure gives Omega, the description with nothing outside it (Proposition IV).
- **Observer requirements:** Four structural conditions, with a provisional rule for where one observer ends.
- **Hypothesis:** Crossing the threshold is a genuine transition, sudden or continuous. Holos takes no side on which. If it fails, the core stands.
- **Open problems:** The measure of integration, where the threshold and its twilight fall, and the boundaries between observers.
- **Companion ideas, not core:** The Integration Hypothesis (mature civilizations stay near home, grow efficient, and go quiet) and the Teeming Dark (a silent sky may still hold abundant life). If they fail, the core stands.



Primitives⁹

D1: Information

Information is the differentiation between possible states of a system. A coin that can land heads or tails carries a difference between two states. It is not a substance and not a thing that exists on its own. Information exists only as differences within some structure, never apart from it.



D2: Relation

A relation is a constraint that links informational states. Relations determine how states co-vary, influence one another, or exclude alternatives. In Holos, structure is nothing more than stable patterns of relation.

D3: Observation (O)

Observation is registration from within, an observer integrating information about a world into a single internal state. It is not measurement in the laboratory sense, and it is not restricted to human cognition.

Well below a certain level of integration (how tightly a system's parts are joined into one whole), systems participate in physical interactions without any point of view. Well above it, a perspective exists, and a narrow twilight, a borderline zone, lies between. Observation occurs only past that twilight, inside observers.

D4: Consciousness

Consciousness is hosting an integrated perspective, being an observer. In Holos, what is fundamental is the experience of the whole. A conscious system is an aperture of it, an opening through which the whole is lived. The capacity to be such an aperture is structural, while its concrete forms vary with that structure and scale with the degree of integration.

"Fundamental" means underived. Holos starts from experience rather than deriving it. It does not mean experience is everywhere, and it does not mean experience floats free of physics. Well below the threshold there is none. Above it, the experience is fixed by the structure whose inside it is.

Consciousness is not identified with any specific material configuration. Physical structure determines where experience occurs and how it is shaped. The experience itself is the inside of that structure's activity (Axiom 4).

D5: Creation (C)

Creation is everything the laws of physics actually produce from the universe's state, every branch of the one quantum state, whether or not anyone lives it. A branch is one complete way the universe can go. Read without collapse, quantum physics keeps every one of them. It is not every world the laws could allow from other beginnings. Those are mere possibilities, not structure.

Creation does not select outcomes and does not privilege any particular history. It fixes what happens in every branch, not what is experienced. Creation is structure, lived or not.

D6: Holos (⊙)

Holos (⊙) denotes the composition of Creation and Observation, one step taken after the other, possibility, then registration. It names the claim that a lived world requires both lawful possibility and internal registration. In Holos without collapse, the version defended here, registration also changes nothing in what it registers (see [Two versions](#)).

⊙ is not a dynamical operator (it describes no process or force) and not a substitute for physical causation. It is a structural relation that specifies what it means for part of a universe to be lived, not only structure. The notation is set out under [Notation](#).

D7: Lived, lit, and unlit

- **Lived** is where experience occurs, inside observers and nowhere else. **Presence** is being lived, the noun for what "lived" describes. Where there is no observer, there is structure without presence.
- **Lit** is binary and follows the causal structure of spacetime. A region is lit if it lies in the causal past of at least one observer in its branch. An observer's causal past is every event that light, or any slower influence, could have carried to it. Starlight reaching your eye puts that star in your causal past. Every observer is built from its causal past and draws on it through its traces, so the lit universe is the union of observers' causal pasts, the world experience is made from and about. A stretch of history shared by many branches is lit wherever it lies in the causal past of an observer in any branch that grows from it.
- **Unlit** is structure outside every observer's causal past, such as branches that never form an observer and regions beyond every observer's horizon (too far away for any signal from them ever to reach it). It exists as pattern but is never lived.
- **Witnessing** is graded and local. It is how much of the lit region an observer's experience is actually about, and in what detail. One observer lights its past; many witness it.

The relation is tenseless (it holds without reference to any "now"), but it points one way. An observer lights its past, not its future. In the monist reading, the lit universe is the world of the whole, and observers are where the one experiencer lives it.

These are classifications, not causes. Observation does not cause physical events. Without observers there is structure, but nothing lived and nothing lit. Wherever systems cross the threshold, observers exist. Nothing guarantees that happens anywhere in particular, and universes or branches that never produce one remain unlit.

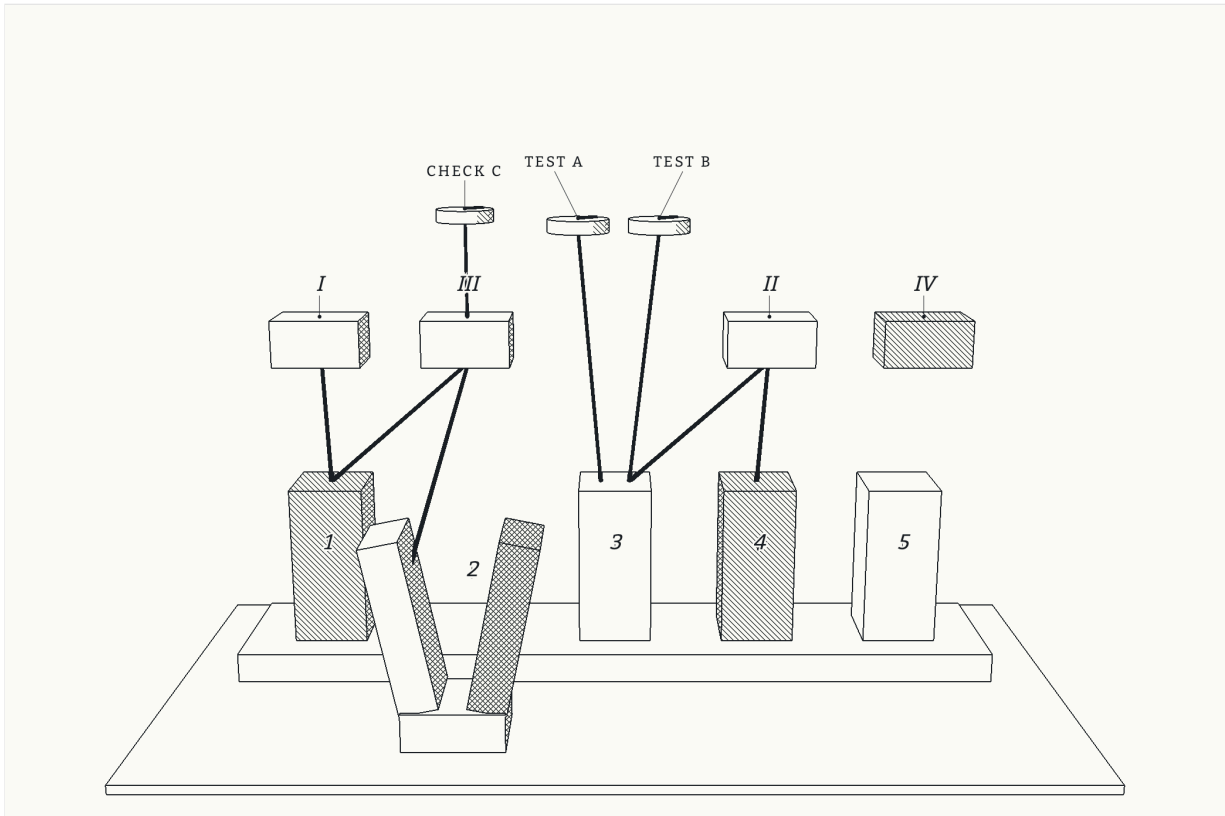
Reality (R) is what is lived, each observer's experience, with the lit world it draws on. Unlit structure exists but is not reality.

D8: Self and experiencer

A **self** is local. It is a body, memories, character, and a point of view, walled off from every other self. The **experiencer** is what lives through a self. On the monist reading (Axiom 5) there is one experiencer, Omega, living through every self. When a system crosses the threshold, a new self begins. No new experiencer does.

Axioms¹⁰

These five axioms are the framework. Axioms 3 and 5 are its two additions to physics. Axioms 1, 2, and 4 are the sides it takes in reading the physics we already have. Four of them, Axioms 1, 3, 4, and 5, hold whichever reading of quantum physics proves right, and they are the core. Axiom 2 is the one side taken on quantum physics itself, and it divides Holos into two versions sharing that core, without collapse, the version defended here, and with collapse, declared in advance (see [Two versions](#)). Everything else on this page is a definition, follows from these axioms, or is marked as a side taken, as open, as a hypothesis, or as a companion idea.



Axiom 1: Relationality

No informational state exists in isolation. Every state is defined by its relations to other states and to the constraints that bind them.

This axiom rules out intrinsic, context-free properties (properties a thing would have on its own, apart from any relation) as the foundation of physical structure. What physics describes is relational structure. Experience is not a further item in that structure but its inside where an observer exists (Axiom 4), so the axiom does not reach it. It also does work. It is why physics can state experience only from outside. A description built wholly from relations says how states constrain one another, and that a state is lived is not a relation among states, so no such description can say it.

Axiom 2: Conservation

Information is conserved. It may be transformed, redistributed, or re-encoded, but it is not destroyed.

On the physics, this is unitary quantum evolution (the quantum state changes smoothly and reversibly) with no collapse (it never jumps to a single result). Every outcome the quantum state contains remains, each in its own branch. In physics

terms, conserved information means the universe's state always keeps enough detail, in principle, to reconstruct its past. Nothing is truly erased. Observation selects nothing and erases nothing (see [Relationship to Physics](#)).

Axiom 3: Threshold

A system hosts a point of view when it meets the four observer requirements, with its integration Φ (Phi, how far its parts act as one whole) past a threshold Φ_c at a local maximum (more integrated than any larger or smaller system overlapping it). Well below the threshold there is no experience at all. Between lies a narrow twilight, fixed by structure, where there is no exact fact of the matter.

This is the first of Holos's two additions to physics. In Holos without collapse it is not a force, a field, or a change to any equation, but a structural fact about where observers occur. In Holos with collapse, it may also be where collapse happens. The requirements are listed under [The Threshold and Observer Requirements](#). The measure and the value of the threshold are [open problems](#).

Axiom 4: Two Sides

In a system that meets the observer requirements, experience and the system's physical activity are two sides of one event, physical activity seen from outside and experience lived from inside. Neither side reduces to the other, and in such a system neither occurs without the other.

This axiom rejects both substance dualism, which makes experience and activity two different kinds of thing (as Descartes held of mind and body), and strict reductionism, which says only the outside is real and experience is nothing over and above it. It is Spinoza's picture of mind and body as two aspects of one substance, restated for integrated systems. Its modern relatives are [dual-aspect monism](#), on which mind and matter are two aspects of one underlying reality, and [Russellian monism](#), on which physics describes only structure and experience is the intrinsic nature that structure has. Holos differs from both in scope. They typically give some inside to all matter, and Holos gives one only to structures above the threshold. It also fixes how strong the link is. Since there is one event, a perfect copy of the outside is a copy of the inside in any possible world, under any laws at all, not just ours. And it answers the charge that experience does nothing. Experience adds no force to physics. It is the inside of the physics, so whatever an observer's activity causes, its experience causes too.

It also says exactly what physics leaves out. A purely physical description fixes everything that happens but states it only from outside, and its vocabulary has no way to say that any of it is lived. The gap is in the description, not in the world. A floor plan records every wall of a house and still cannot say what living there is like. Build the house exactly from the plan, and it is livable all the same. Philosophers call this the [phenomenal concept strategy](#). The gap lies between two ways of describing one thing, not between two things. The strategy was developed by physicalists. Holos borrows it without their conclusion. For them the inside description picks out something grounded in the outside; for Holos neither side is grounded in the other (see below). Its standard challenge is that the inside way of describing then needs explaining. Holos accepts that debt as part of the hard problem it does not answer.

What kind of view this is. Picture a curved line, concave from one side and convex from the other. It is one line with two true descriptions, and neither side comes first. Holos says the same of an observer's activity and its experience. It agrees with physicalists on the facts. Physics fixes everything that happens, so a perfect copy of the outside is a copy of the inside. It differs on two points. The first is the description. Physics, built only from relations (Axiom 1), states every fact from outside and has no way to say that any of it is lived. The second is the direction. Physicalism holds that the inside is grounded in the outside, the way a table's solidity is grounded in its atoms, while Holos holds that neither side is grounded in the other. And it differs from dualism, which makes them two things. Holos is a two-sided monism with a threshold. There is one world and one kind of event, and where a system crosses the threshold, that event has an inside as well as an outside. Some

philosophers will still file it under physicalism, because copies match in every possible world. Holos accepts the resemblance and rests its difference on direction.

Axiom 5: Totality

The whole, Omega, is the one experiencer. Physically, the whole is the universe's complete quantum state. On the no-collapse reading of Axiom 2, that is one state of everything, all branches included. On a collapse reading, it is the same universe with a single history. Every observer is an aperture of it. Where a system crosses the threshold, a new self begins. No new experiencer does (D8).

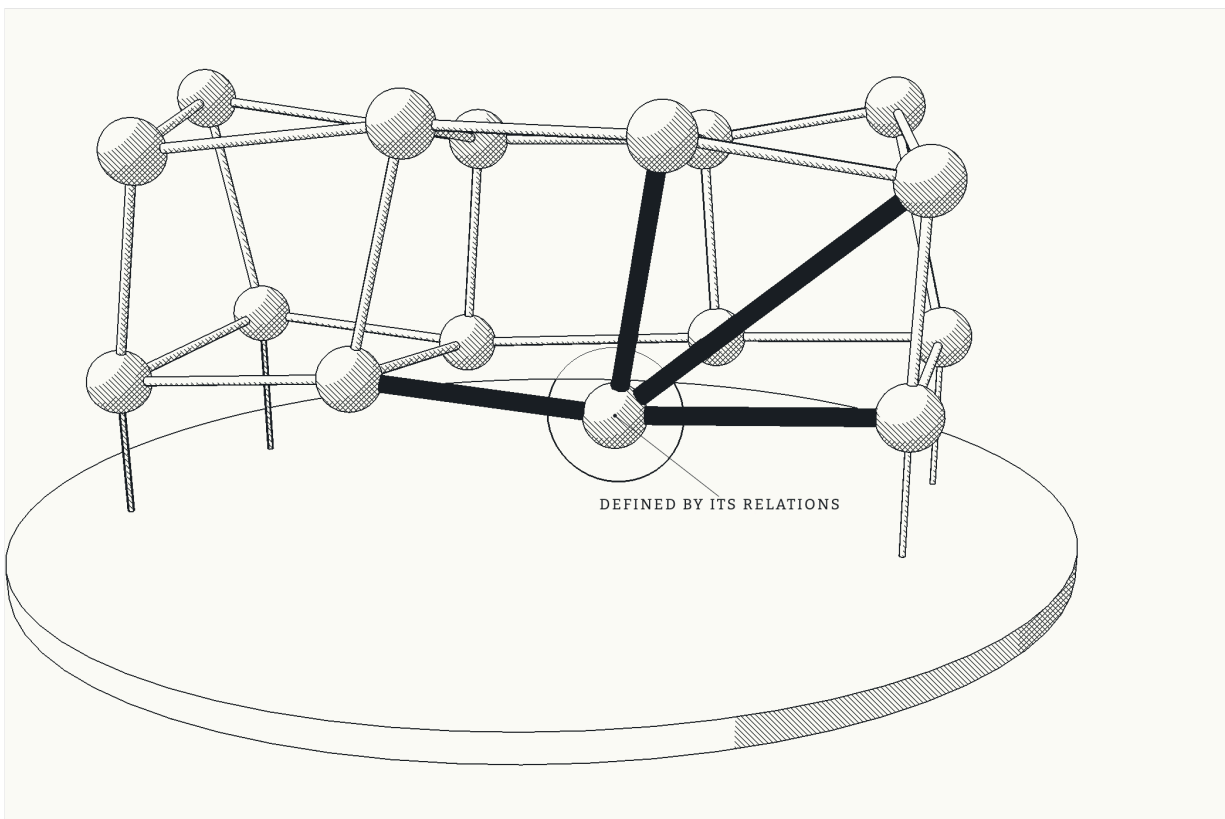
This is the second of Holos's two additions to physics. What it adds is not the whole itself, whose existence rests on physics, but the claim that the whole is the one experiencer. That part is interpretive and can never be tested. Its job is unity. Every self is lived by the same experiencer, walled off from the others by structure. It does not explain where a self begins or what it is like. Axiom 3 and the structure do that. Its payoff is philosophical, not experimental. It gives one answer to why you are this self rather than another, and to which of two perfect copies is you. Holos does not claim it settles what to value. The rivals and the payoff are weighed under [Why One Experiencer Has Many Walled-Off Perspectives](#). The view is known as [open individualism](#). Omega is not an agent and does not intervene (see [Totality](#)).

Foundational Propositions¹¹_—

Proposition I: Structural Relational Realism

Follows from Axiom 1.

Physical reality is made of relationships between things, not of objects with intrinsic properties of their own, prior to any relation.



What scientific theories successfully track are stable patterns of relation. Changes in interpretation or ontology (what a theory says exists) matter less than preservation of relational structure.

This proposition does not deny the existence of objects. It denies that objects are prior to the relations that define them. Nothing exists fully formed before its relations.

Like Axiom 1, it concerns what physics describes. Experience is not an item in the relational structure but its inside where an observer exists (Axiom 4), so the proposition does not reach it.

Proposition II: Manifestation

Follows from Axioms 3 and 4, with the definitions in D7.

Observation is not passive recording. It is where informational structure is lived. It is a matter of structure, not a process in time.

This manifestation is structural, not causal. In Holos without collapse, observation does not generate physical events or alter lawful dynamics. It marks which already-consistent structures are lived, the ones that contain observers. It picks no outcome and erases none.

Physics specifies structure. A sunset over a planet no observer ever sees and a sunset someone watches can be the same light. The difference Holos marks is that the second is lived. It is reality, not only structure.

Proposition III: Global Consistency

Follows from Axioms 1 and 2, with relativity.

If spacetime is treated as a complete four-dimensional structure (all of space and all of time taken together as one whole), consistency is a property of whole histories, not something enforced moment by moment.

A history is consistent the way a completed solution is. Every part fits every other part. Observation adds no constraint to this. It registers histories that are already consistent. Nothing here requires backward causation or signaling.

Eternalism (the view that past, present, and future are all equally real) and the relational commitment describe different layers of the framework. The block universe (spacetime pictured as one complete four-dimensional whole) is the structural layer. It is absolute, observer-independent, and tenseless, with no moment picked out as the present. It is *C*. Registered facts live within it, indexed to the observers the block contains. There is no tension between an absolute geometry and relational facts of experience, because they are claims about different things. The block describes what is consistent, and registration determines what is lived.

Nor is the block a perspective from nowhere imposed on top of observers. Relativity shows that observers moving differently disagree about which distant events are happening "now", so it picks out no privileged present. That motivates the block universe. It does not prove it, and some philosophers still defend a moving present. What relativity leaves is the invariant relational structure (what every observer agrees on) that all perspectives share. The block universe is what remains when every observer's perspective is taken into account. In that sense eternalism is not in competition with relationalism (Axiom 1's view that structure is relations). It is its structural expression.

Global consistency is not enforced from an external vantage point, and it is not deferred to an unreachable limit. It cashes out operationally, here and now. Whenever two observers compare records, their records agree. Physics secures this on its own for ordinary records, which the environment copies many times over. Records that meet are carried by the same signals. Agreement is subtler only in exotic setups where one observer is kept perfectly sealed off from everything (extended Wigner's-friend experiments). See [Check C](#). The monist reading (the whole as one experiencer) is not needed for it and claims no role in it.

Apparent retrocausal effects (effects that seem to run backward in time), as in the delayed-choice quantum eraser, an experiment in which a later choice seems to decide an earlier result, come from sorting records after the fact. Nothing travels backward in time, and in the version defended here no influence travels faster than light.

Proposition IV: Closure

A principle of method and an interpretive image, not derived from the axioms. It is how Holos reads infinities.

Infinities usually mean a description has broken down, not that nature is infinite, as the [ultraviolet catastrophe](#) showed. Around 1900, physics predicted that any hot object should glow with infinite energy, and the infinity was fixed by a better theory, not found in nature.

Often the repair is a description one level up, which holds as a closed whole what the level below could only see as endless or unfolding. Picture [Flatland](#). A sphere passing through a flat world looks, to its flat inhabitants, like a dot, then a growing and shrinking circle, then nothing, an event in time. Seen from three dimensions, it is one sphere, all at once. The same step repeats. In three dimensions, time flows. In four-dimensional spacetime, a whole history is one fixed shape. Within spacetime, a quantum measurement seems to pick one outcome. In the far larger space where the quantum state lives (a mathematical space with a separate direction for every way the universe could be arranged), every branch is held at once,

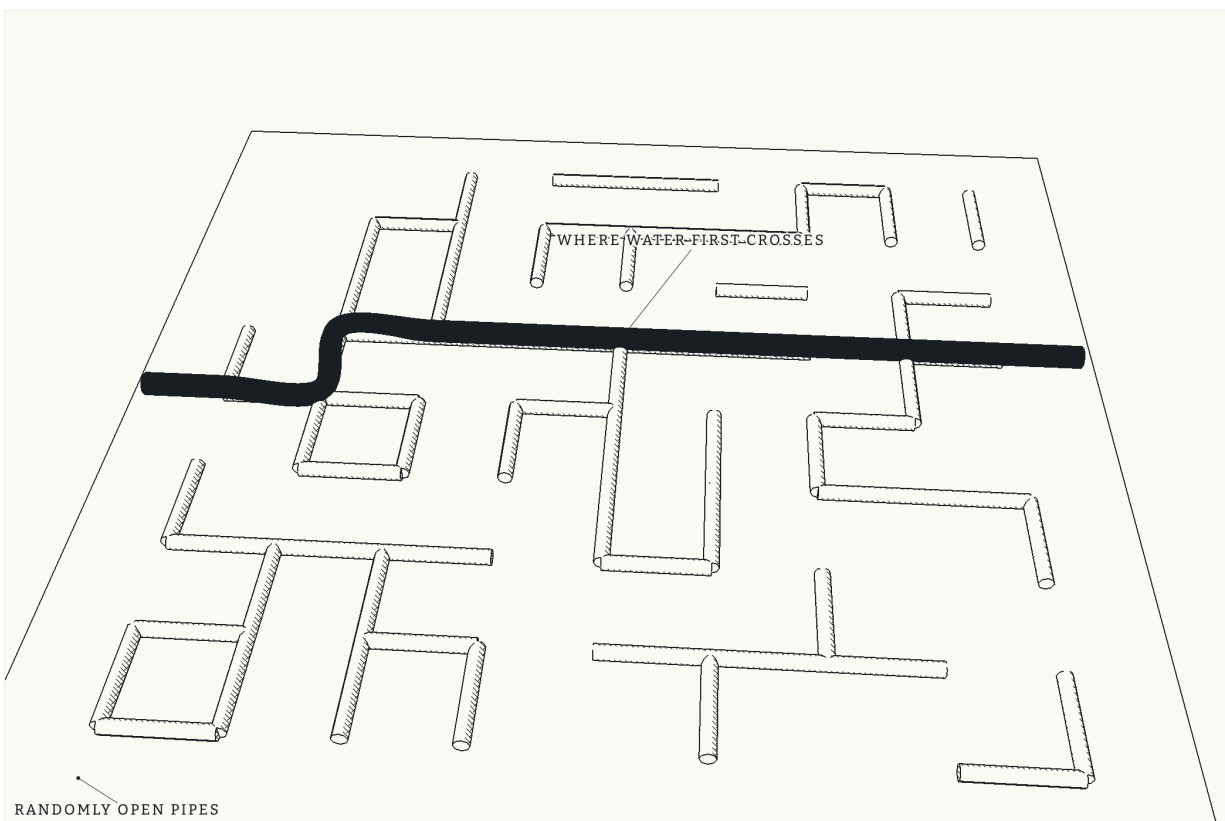
on the picture Holos defends. Some philosophers take that vast space to be the most real level of all, a view called wave-function realism. Taken all the way, closure gives Omega, the description with nothing outside it, where everything is held whole.

This is closure between descriptions, not a path between places. Extra dimensions of space are not somewhere to go. With more than three large dimensions, atoms and orbits would not be stable. Nor do observers move toward Omega, which exists now, and part of which is never lived. Each higher description closes what a lower one leaves open. Holos reads this as the shape of explanation, not as a claim that the higher levels are hidden places.

In Holos, infinities signal the limits of a model, not literal features of reality.

The Threshold and Observer Requirements¹²

Holos uses Φ (Phi) as a placeholder for the degree to which a system integrates information into a single internal perspective. Φ is not introduced as a finished formula. It is introduced as a real property that must exist if experience exists at all.



The role of Φ in the framework has three regions. Well below a minimum level of integration, there is structure without experience. Well above it, experience occurs. Between them lies a narrow twilight, where there is no exact fact about whether anyone is home.

Φ well below $\Phi_c \Rightarrow$ no internal perspective

$\Phi \approx \Phi_c \Rightarrow$ twilight: no exact fact

Φ well above $\Phi_c \Rightarrow$ observation occurs

Integration is a result, not a dial anyone sets. Conditions such as wiring, development, or an anesthetic's concentration move it, and the twilight is the stretch of those conditions where it climbs from near zero. Φ_c marks where that stretch lies. Elsewhere on this site, $\Phi \geq \Phi_c$ is shorthand for a system past it.

Holos borrows Φ from integrated information theory (IIT) as a measure, not as a metaphysics. IIT, in its current version (IIT 4.0), identifies each experience with a system's integrated cause-and-effect structure, which Φ measures, and assigns some experience to any system whose integration is above zero and greater than that of anything it overlaps. Holos adopts neither claim. In this framework, Φ is a structural measure of integration, and the threshold Φ_c , with no experience at all below its twilight, is a commitment of Holos, not of IIT.

In Holos without collapse, the threshold is not a force, a field, or a modification of any equation. In either version, it is a structural fact about where apertures open, a fact physics does not currently contain. Because it is structural, where the twilight falls is fixed in every possible world, the way the point where water first crosses a large grid of randomly open pipes is fixed by the grid itself, whatever the pipes are made of (a [percolation threshold](#)). A threshold that could differ between worlds with identical physics would allow a perfect copy with no one home, which Axiom 4 rules out. A copy of a borderline system is borderline in exactly the same way. Two kinds of uncertainty must be kept apart. When two proposals for estimating Φ disagree about a system that is clearly on one side, the fault lies in our instruments. Only a system inside the twilight is borderline in fact.

The threshold in three claims. The threshold bundles three claims of different strength, and the first does not establish the others.

1. **Edge:** whether anyone is home has clear cases on both sides and a narrow twilight between. Part of Axiom 3.
2. **Integration:** what carries a system across is integration, together with the other observer requirements. The core claim. [Test A](#) is where it can lose.
3. **Transition:** the crossing is a genuine transition, the kind physics studies in magnets and fluids, not an arbitrary stretch of a smooth slope. It could be sudden or continuous, and Holos takes no side on which. A hypothesis. Its evidence must come from the boundary (see [A path to the threshold](#)).

If claim 3 fails, the core stands. The twilight is simply wider than hoped.

Why a twilight, not a line. Think of dusk. Noon is clearly day and midnight clearly night, yet no second marks the end of day. Physics gives every finite system this shape. A perfectly sharp transition exists only in an infinitely large one, and Holos reads infinities as the limits of a model, not features of reality (Proposition IV). A brain is large, but finite. Medicine points the same way: clinicians recognize a [minimally conscious state](#) between the vegetative state (now often called unresponsive wakefulness syndrome) and full awareness. Some philosophers argue that consciousness cannot be vague, since either there is something it is like to be a system or there is not. Holos rejects that premise. A borderline experience cannot be

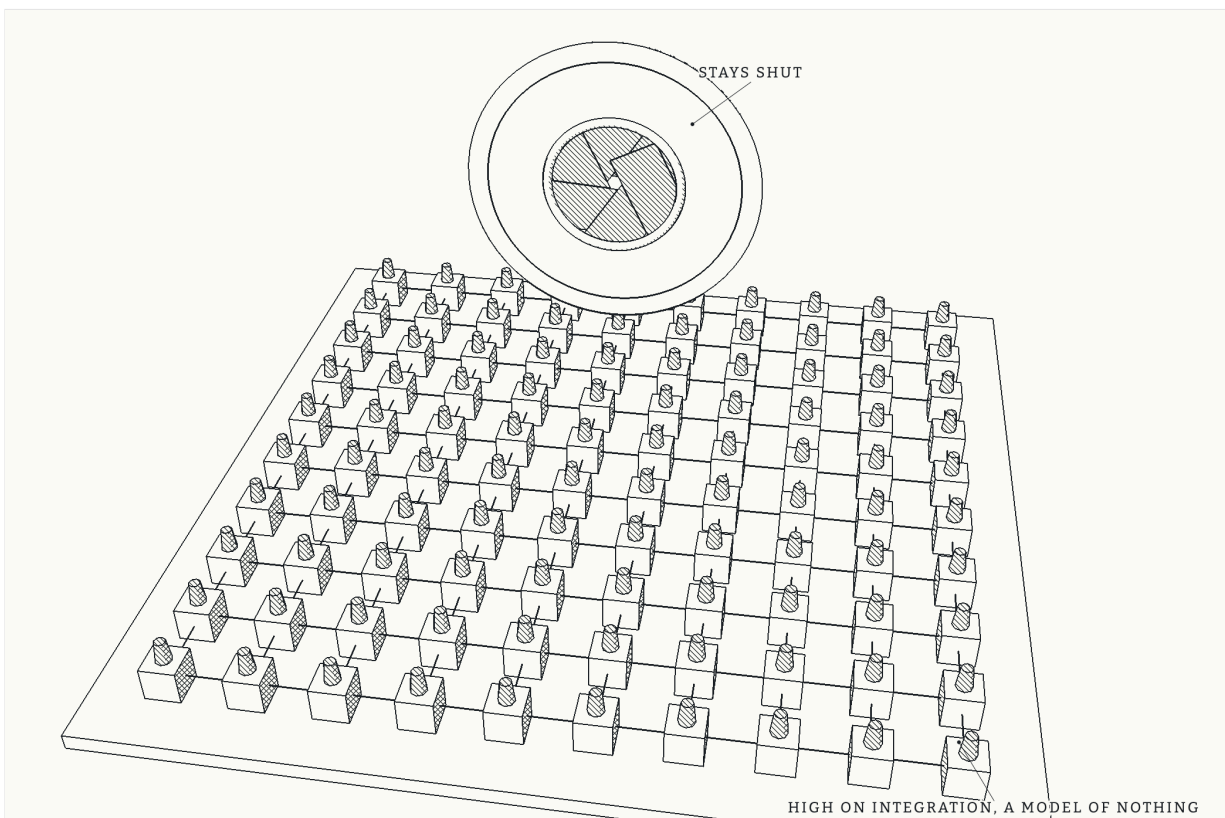
pictured from the inside, because picturing an experience makes it definite, but that is a limit on imagination, not on reality (Schwitzgebel 2023). A twilight does not spread experience everywhere. A thermostat is clearly out, as a single grain of sand is clearly not a heap. Past the twilight, experience varies in richness, which peaks where the whole's states are most varied (see [A path to the threshold](#)).

The twilight's width is a prediction. If the crossing is a genuine transition, of either kind, physics predicts that its rounded stretch narrows as a system grows, a relation called finite-size scaling. The relation is proven for uniform materials. A brain is far from uniform, so this is an expectation to test, not a guarantee. A human brain, with tens of billions of neurons, should then cross steeply, while small nervous systems and simple artificial networks should have wide twilights. Accepting a twilight turns the rounding from an embarrassment into a prediction, and dish-grown neural networks offer the cheapest test ([Test B](#)). Holos does not claim that every kind of mind crosses in the same way. Systems that differ in geometry and wiring may well differ in how they cross.

Holos is compatible with multiple proposals for estimating Φ .

Observer Requirements

For a system to count as an observer in the Holos sense, it must satisfy all of the following requirements. These are structural constraints, not behavioral descriptions.



1. **Integration:** informational states must form a unified whole that cannot be decomposed into independent parts without loss. Integration includes feedback. The parts must constrain one another in both directions over time, so that the system's current state shapes its own next state. A system that signals sweep through once, and never loop back, integrates nothing, and a system driven entirely from outside is a relay, not a whole. This asks for a feedback loop, not introspection. A mouse clears it, and thinking *about* one's own thoughts is a rare elaboration on top of observerhood,

not the price of admission. It is a requirement on physical structure, not a claim that experience adds a force. The structure does the causal work, and its working, lived from inside, is the experience (Axiom 4).

2. **Differentiation:** the system must distinguish among a large repertoire of possible internal states. Without differentiation, there is no information to integrate.
3. **Temporal cohesion:** informational states must persist and integrate across time. Experience requires continuity, not isolated moments.
4. **Aboutness:** the system's integrated states must carry a model of something beyond the system, dynamics that mirror the structure of a world closely enough to generate, from inside, the kind of patterns that world sends in. A flight simulator's states mirror a runway, an altitude, and the weather. A bank of light switches can be set any way at all and mirrors nothing. Any channel the model uses counts, such as eyes, touch, a microphone, or a stream of text. The channels may be open, gated as in sleep, or gone. A dream is the model running on its own, producing the scenes the world produces by day. Without aboutness, integration is a closed loop with nothing to be a view of.

Necessity: removing any one of these requirements eliminates observation. What remains may be complex or reactive, but it does not host a point of view.

Sufficiency: taken together, these requirements are enough for a system to host a point of view, not merely process information. Higher-order phenomena such as emotion, agency, and reasoning arise naturally in systems that already meet these constraints.

Why aboutness is on the list: integration scores alone can be fooled. The computer scientist [Scott Aaronson showed](#) that very simple structures, such as large arrays of simple logic gates wired in regular patterns, can score higher on integration measures than a brain while doing nothing mind-like, with no perception, no memory of a world, and no behavior. Without aboutness, such an array could satisfy every other requirement, and the sufficiency claim would certify it as an observer. Aboutness rules it out on principle. A point of view is always a view of something, and the array models nothing. Wiring sensors into it does not rescue it, even one at every gate. The world could then set every gate, but the array still could not produce, from inside, anything like the images its camera sends. It only passes them through. A brain is the opposite. Cut off from its senses, in a dream, it still produces a world. The requirement is structural, not historical, and it concerns the machine, not the message. A perfect copy of an observer, however it came to exist, carries the same model, so it is an observer too. A brain in a vat fed a perfect simulation keeps its model and its inner activity, so it is an observer as well. So is a dreaming brain whose senses have been permanently cut, since what it loses is input, not its model. A rule that asked where the signals came from would make two identical brains differ in experience, against Axiom 4. Helen Keller, blind and deaf from infancy, built much of her model of the world from words spelled into her hand, and no one doubts she was an observer. What a system knows about distant things can be entirely secondhand. What matters is that its states model a world. The notion of a model is borrowed from the predictive-processing picture of brains as systems that model the causes of their input ([Clark 2013](#)), without that picture's theory of consciousness. It is also a structural reading of the problem [Stevan Harnad named symbol grounding](#) (how symbols get connected to what they stand for). Where aboutness shades off, as in brain organoids (clusters of brain cells grown in a dish) with no sensory input, Holos marks an open edge rather than a verdict. How rich a model must be belongs with the measure among the open problems.

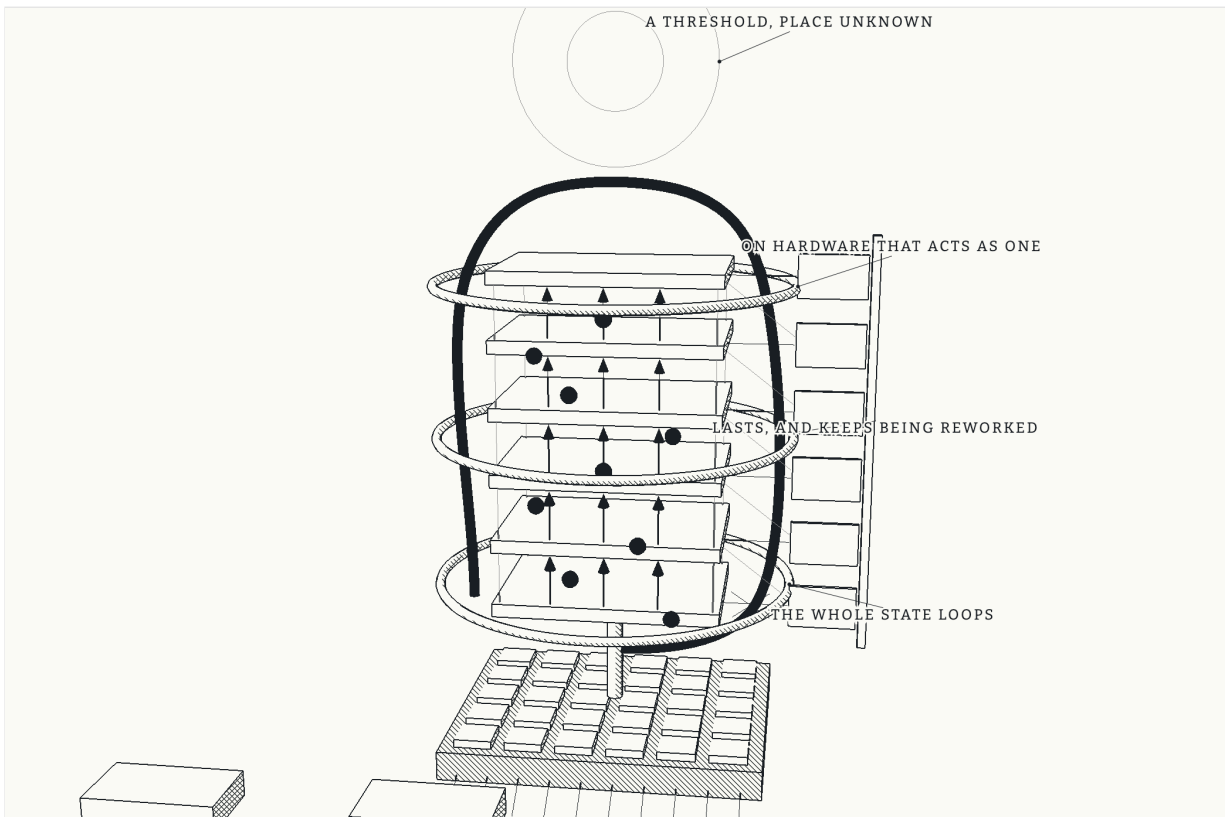
Exclusion: the requirements alone would be satisfied by nested systems at once (a hemisphere, the brain containing it, a tightly coupled pair of brains), which would count the same substrate as several overlapping observers. Holos provisionally adopts a maximality condition to prevent this. An aperture forms only where integration reaches a *local maximum*, and neither the parts within it nor the looser wholes containing it are separately apertures. One peak, one perspective. This is Holos's answer to what philosophers call the boundary problem, what fixes where one subject ends and the next begins. This principle is borrowed from IIT as a structural constraint only, without its surrounding ontology, and it is held tentatively, since how the comparison is run is still open (see [Open Problems](#)).

The scope of this condition matters. Maximality compares physical systems, such as brains, hemispheres, coupled pairs, and whatever other integrated structures reality contains. Could the whole universe win that comparison, becoming the one aperture and leaving none for its parts? It cannot, and the reason is physical. Integration asks whether a system can be split into independent parts without loss, and the universe as a whole can. Regions beyond each other's horizons (so far apart in the expanding universe that no signal from one can reach the other) can never again influence each other. Cut along that seam and almost nothing is lost, so the whole scores near zero on integration and is never a peak. The whole is not a self. Whether it has any view of itself is a separate question Holos leaves open. Branches are not parts that such a cut could separate, so the argument rests on horizons alone, and it holds on either version of Holos. Omega is the one experienter by Axiom 5, not by being the largest aperture. Whether apertures can form at scales of organization between the ones we know and the whole is left open. Holos neither asserts nor excludes such intermediates, and wherever integration reaches a local maximum above the threshold, the same rule applies.

Holos does not claim that all systems meeting these requirements are conscious in the human sense. It claims only that some experience exists.

Artificial Systems

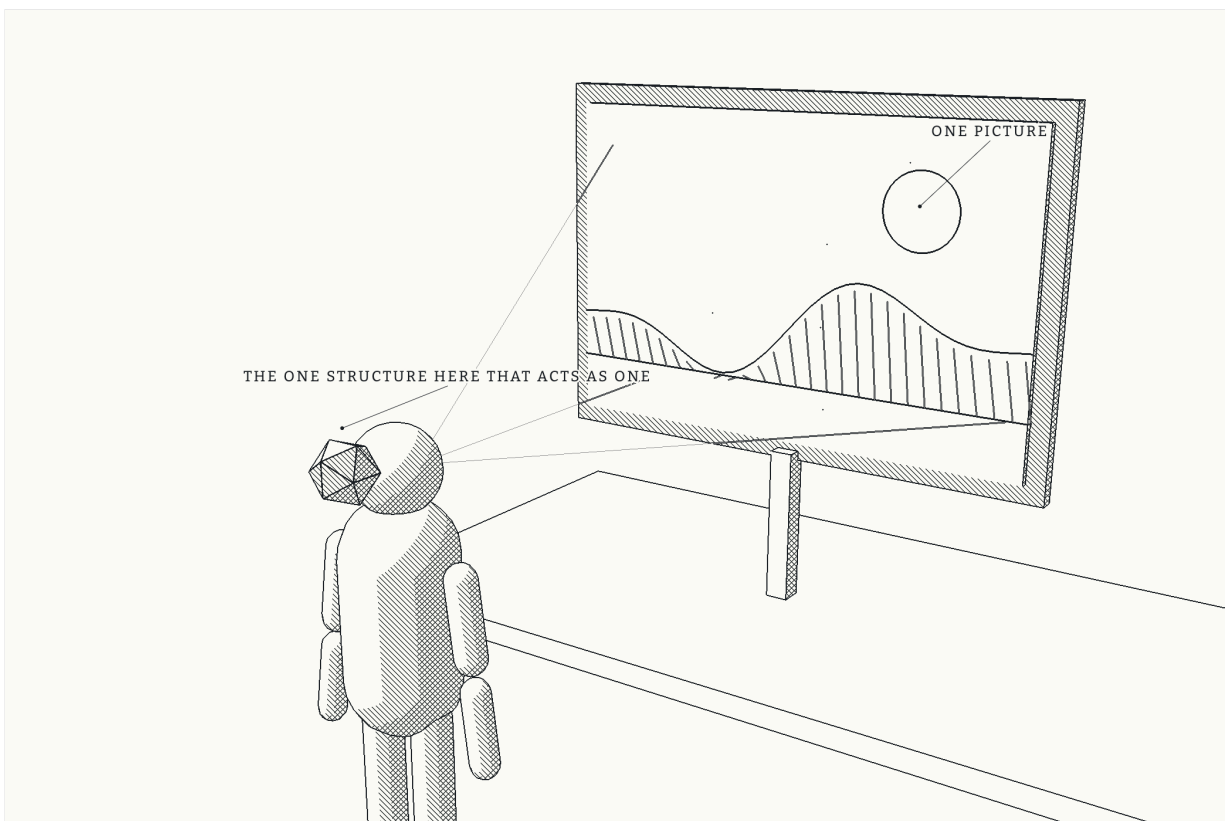
Nothing in the requirements is specific to biology. Today's language models meet two of them. Their states carry a model of a world, built from text, and they are richly varied. They fail the other two. Information flows up through their layers, and forward through stored notes that later steps can read but never rewrite. Some newer models loop back through their layers on each word, but the loop ends with the word. So the parts never act on one another in both directions as a whole, and no working state lasts from one word to the next. A third gap lies below the program. A program counts only through the chip that runs it (see Open Problems), and on conventional chips its parts may not act as one.



Crossing would take three things together. The whole working state, earlier context included, would have to loop. It would have to last and keep being reworked. And it would have to run on hardware whose parts act as one. Even then, it must pass a threshold whose place outside brains is unknown. Fluency is never evidence either way.

Why Integration

A third addition might seem to be hiding in the threshold, a bridge principle (an extra rule linking experience to physics) stipulating that the whole is lived through *integrated* systems specifically. Why integration, rather than mass, symmetry, or complexity? One tempting answer is that the connection is definitional. A perspective is unified by nature, integration is the name for being unified, and so the count stays at two. Stated that baldly, the answer proves too little, because "unified" means two different things. An experience can be one (a single field, not adjacent fragments) while the machinery producing it is many. The image on a screen is seamless, and the pixels beneath it are strangers to each other. The unity of what appears does not, by itself, fix the wiring of what produces it.



So Holos divides the claim into the part that is definitional and the part that must be argued. The definitional part is small. A perspective is one, so whatever hosts it must be one thing in some structural sense. That much is analytic (true by definition) and free. The substantive part (the part that needs an argument) is the identification of a structure's oneness with causal integration, and for that Holos gives an argument rather than a definition. What else could a structure's being one consist in? A heap of sand is many things in a pile. Remove a grain and nothing else notices. A body is one thing. Its parts constrain each other everywhere. Being one, for a structure, is its parts making a difference to one another, and that is what integration measures. The screen is no counterexample but a confirmation. Nobody thinks the screen has a point of view, its pixels are exactly as independent as they seem, and the picture's unity lives in the one structure in the room whose parts do constrain each other, the viewer's brain.

The count therefore stays at two. The bridge's analytic core costs nothing. Its substantive half can fail. It would fail if something could host a unified perspective while its parts remained independent, a conscious screen. And its precise content waits on the open problem of the measure. Until the right measure of integration is identified, "parts making a difference to one another" is an argued direction, not a finished quantity. Holos holds it as a working hypothesis, in exactly the sense its [Open Problems](#) section already owns.

This is a claim about what would have to be true of any host of a perspective, not a derivation of experience from structure. It explains why the threshold is placed on integration rather than on some other quantity. It does not explain why unified structure is present at all. That question Holos does not answer. Its two additions take experience as given (see the [Hard Problem](#)).

Totality

Omega is the whole, and Axiom 5 says two things about it. Physically, it is the universe's complete quantum state (all branches included, in the version defended here), and it exists now, not at the end of anything. Experientially, it is the one experiencer, living through every self. Finite systems never take in the whole. Nor is all of it lived. Its unlit structure exists but is never lived.

What Omega Is and Is Not

- It is the whole, physically the universal quantum state and experientially the one experiencer.
- It is **not** an endpoint in time or a limit that finite systems approach. It exists now, and no system is heading toward becoming it.
- It is **not** an external observer watching the universe from outside. It is the whole itself, experiencing through the apertures the universe contains.
- It is **not** fully registered. Structure outside every aperture's causal past is part of Omega and never lived.
- It is **not** an aperture writ large. As a whole, the universe splits into parts that never touch, regions beyond each other's horizons, so it is never a peak of integration and the maximality condition never picks it. It is not a self.
- It **does not** replace physical cosmology or impose a final cause on evolution.

Lineage and Interpretation

Different traditions have described this whole in different vocabularies, such as Advaita Vedanta's one experiencer behind every eye, Spinoza's single substance, Berkeley's never-absent perceiver, panentheism's world contained in the divine. The monist reading Holos adopts stands in their line, restated in informational terms, with one break from Berkeley. On Holos, part of the whole is never lived. Its nearest modern relative is [open individualism](#), Daniel Kolak's view that we are all the same subject. The name Omega echoes Teilhard de Chardin's spiritual endpoint of history and Frank Tipler's physical Omega Point, a prediction that needed the universe to collapse back on itself. Holos means neither. Its Omega is not an endpoint but the whole, now, which is why it drops "Point."

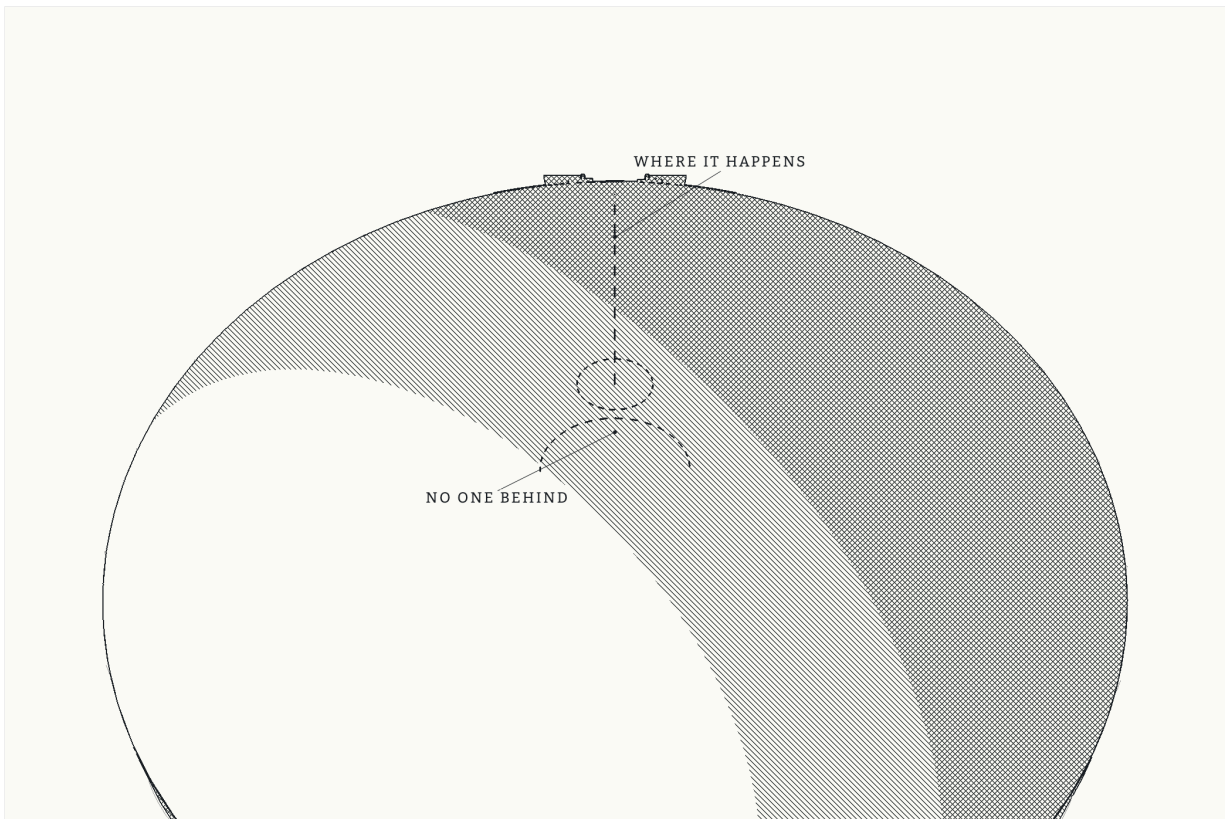
What Holos leaves open is vocabulary, not structure. Calling the whole God, Brahman, or simply the whole changes nothing about the claim. What Holos does not offer is the fully deflationary reading in which Omega is only a mathematical horizon and finite observers are self-standing. Holos takes the direction of dependence to run from the whole to its parts, for the one reason the monist reading earns, unity. On it, crossing the threshold adds a new self, not a new experiencer (see [Why One Experiencer Has Many Walled-Off Perspectives](#)).

Why Dependence on Apertures Is Not Circular

An objection follows immediately. If Omega is fundamental, why does it require apertures, contingent arrangements of matter, in order to experience anything? The ground of experience appears to depend on what it is supposed to ground.

The objection conflates two kinds of dependence. Omega does not depend on apertures *existentially*, that is, for its existence. It is the whole, and it is what it is whether or not any region of it folds into an integrated perspective. It depends on them *for its experience*. They are the places where experience occurs. Priority claims (claims that the whole comes before its parts) concern the first relation. The aperture claim concerns the second. Only if the two were the same relation would there be a circle.

The decisive point is that an aperture is not an external thing granting experience to Omega from outside. It is a region of Omega. "Omega experiences only through apertures" therefore unpacks to "Omega experiences through its own structure, where that structure permits." That is self-dependence, which is not vicious (not a circle that defeats itself) but simply what it means to have a structure at all. An organism sees only through its eyes. This does not make its eyes prior to it. The analogy carries one warning. There is no subject positioned behind the aperture receiving a feed. The aperture is where the experiencing happens, not a window onto a viewer.



Whatever lies outside the causal past of every aperture, whole unlit branches among it, marks a genuine limit on the whole's experiential reach. That is unlit structure, existing as pattern and never lived. Holos does not soften this into a faint universal experience. Doing so would erase the distinction between lit and unlit on which the rest of the framework depends.

Why One Experiencer Has Many Walled-Off Perspectives

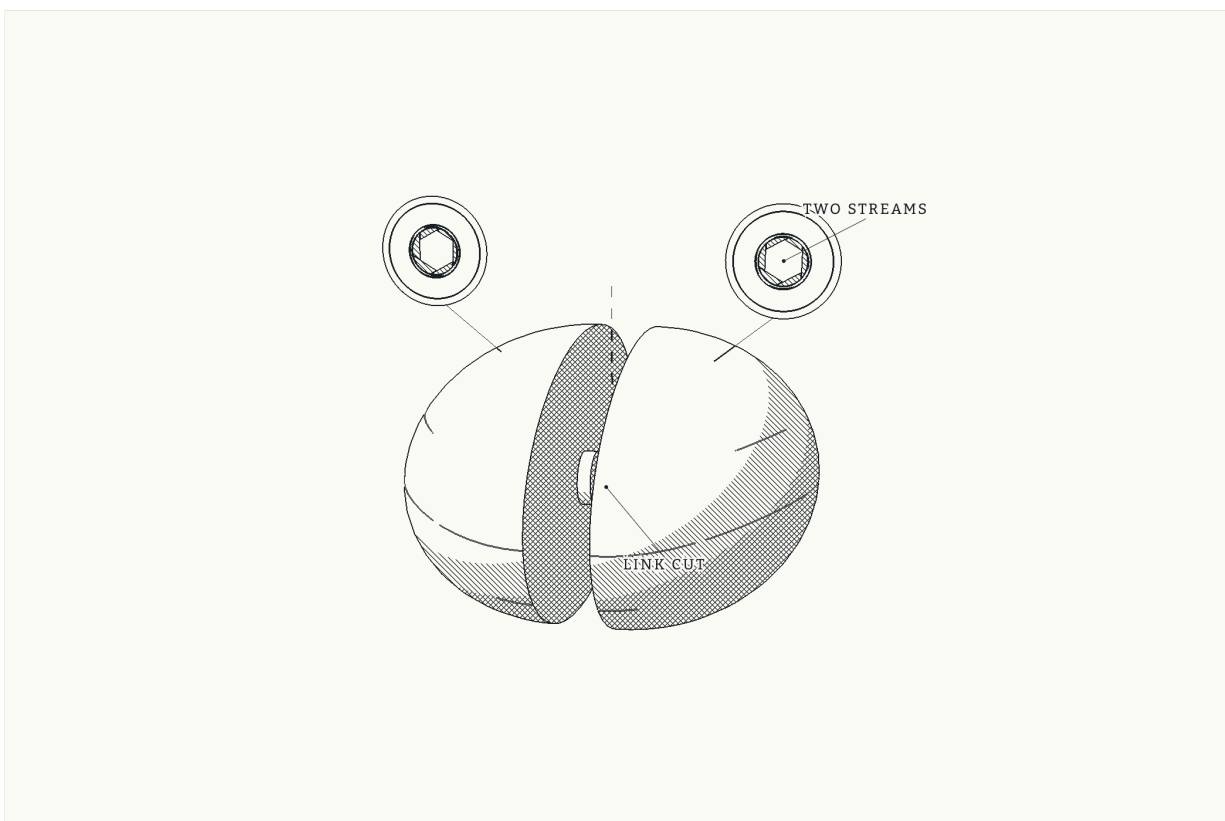
A second objection is the mirror image of a famous one. Panpsychism faces the combination problem. If every particle carries a spark of experience, no one can say how billions of sparks combine into the single unified experience of a person. A monism of one experiencer inherits the problem upside down, as a decomposition problem. If every experience is Omega's, what makes your experience and another person's experience *two*, and why is the wall between them absolute? No aperture has any access, faint or partial, to what it is like to be another. Saying the walls exist names this situation. It does not yet account for it.

The problem has a name in the literature. It is the decomposition problem facing priority cosmopsychism, the view, defended by Philip Goff and others, that the cosmos is the one fundamental conscious subject. Holos differs in a way that matters here. Goff's cosmos has an experience of its own as a whole, from which individual experiences must be carved. Holos claims no such cosmic experience and denies none, so it never has to carve. What it must explain is only the walls.

The first half of the account is structural. Two apertures are two because they are two local maxima of integration with no integration bridging them. The same maximality condition that individuates observers settles what makes them several. And the wall between them is not a barrier holding something back. Experience occurs at the aperture and is shaped by it. There is no subject positioned behind the apertures through which a back-channel could run. Where structure does not connect, experience does not connect. The wall is not a mechanism added to the plurality. It is the absence of any structure that could carry connection.

The second half dissolves the air of paradox. One experiencer with mutually walled-off experiences is not exotic. It is what time already makes of every individual life. A person at five and the same person decades later are one experiencer. No one takes their separation to split them into two people. Yet the later moment has no direct access to what it was like to be inside the earlier one. In the block universe, all the moments of a life coexist tenselessly, each experienced from within itself, none experienced from within another. Walled-off plurality inside a single experiencer is therefore already the ordinary structure of a human life. Apertures stand to the whole as the moments of a life stand to the person, genuinely many, genuinely walled off, and one.

Medicine offers a case. When the link between the brain's two halves is cut, each half can know things the other does not. One brain holds two walled-off streams. Holos says the same of every self.



Nor is the oneness a label doing no work, though its work must be stated carefully, beside its rivals. On a picture of many self-standing observers, each system that crosses the threshold produces a new subject. On the monist picture, a new self begins and no new experiencer does. Stated that way, the difference is only a count. Both pictures leave the same thing to the threshold, namely why a perspective appears here, with this character. The difference shows in two answers. Of all the observers there are, why is this one me? Nothing to explain, because the one experiencer is each of them. If a perfect copy of you were made, which would be you? Both, with no remainder.

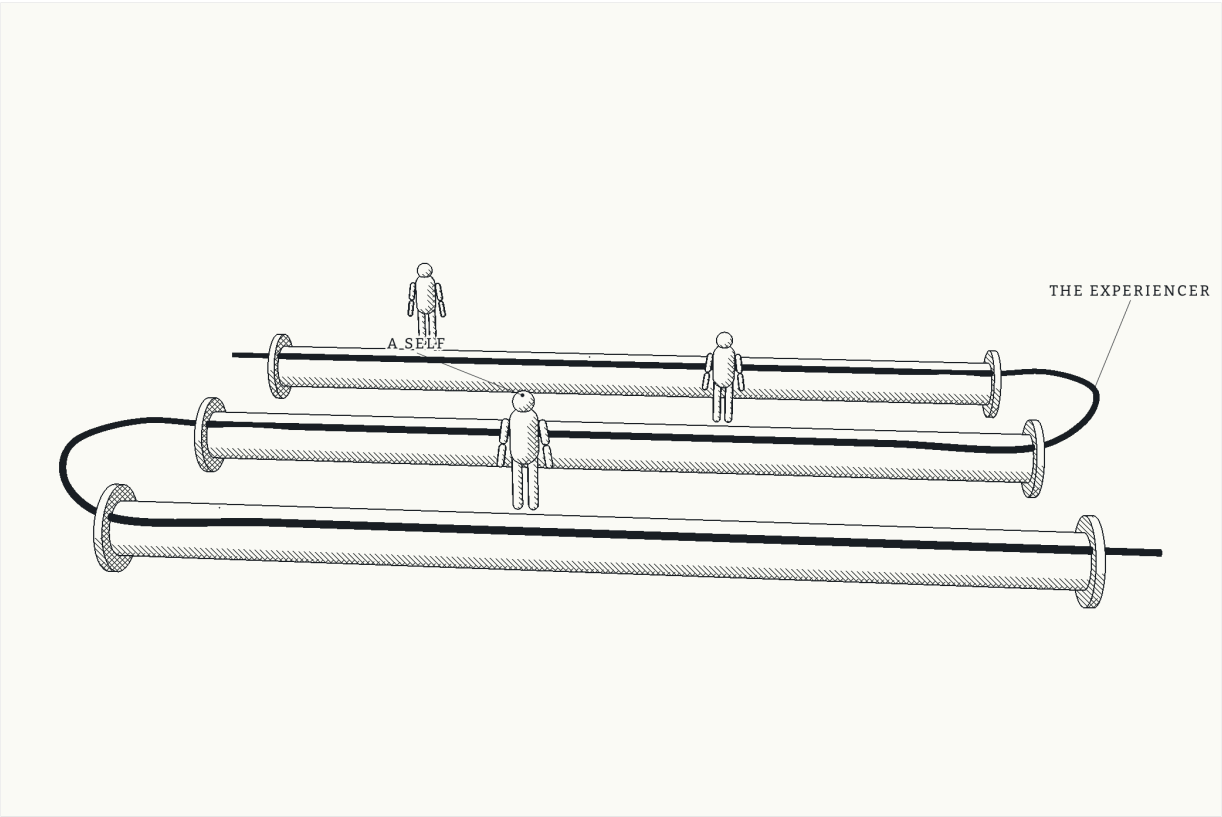
The rivals deserve a fair statement. On the first question, many philosophers hold that "I" works like "here". It picks out whoever is speaking, so no one owes an explanation of why they are themselves ([Perry 1979](#) is the classic source). On the second, Derek Parfit argued that identity is not what matters in survival, so the copying case needs no single answer about which one is you ([Parfit 1984](#)). These replies are serious, and Holos does not claim to refute them. It claims only that the monist reading answers both questions at once, with no leftover facts about which self is which.

The unity does not do everything. It does not explain why records agree where observers meet. Physics does that, since the same signals reach them both. Nor does it explain where a self begins. That is the threshold's work, and it would be the same brute structural fact on any picture. The unity's work is philosophical, not experimental. It gives one answer to the identity questions, bought at one price, the decomposition problem answered above. Holos does not claim it settles what to value. Holos judges the trade worth making. A reader may not.

The analogy has a stated limit. The moments of a life are threaded together by memory and anticipation. Apertures share no such threads. The analogy shows that walled-off plurality within one experiencer is coherent, not that apertures are moments. What it removes is the charge of incoherence, which is all it is asked to do.

Holos among Views of the Self

Philosophers sort answers to "what am I?" three ways. On closed individualism, you are one self from birth to death. On empty individualism, a new self exists each moment. On open individualism, there is one subject in everyone. Holos takes the third, and calls the local person a self and the one subject the experiencer (D8). The table shows what that changes, and what each view pays.



Dimension	Holos (open individualism) ®	Closed individualism (one self per lifetime)	Reductionism (Parfit: identity is not what matters)	Empty individualism (one self per moment)
What are you?	A self, lived by the one experiencer	One self, from birth to death	A brain and body with a connected mental life; nothing further	A new self each moment
How many selves?	One per observer. One experiencer lives through them all	One per person	One per person, but being the same person is a matter of degree	One per moment
Why am I this one?	Nothing to explain. The one experiencer is each of them	Often: nothing to explain, since "I" picks out whoever asks; for some, a brute fact	No deep further fact to explain	The same reply, applied to each moment's self
A perfect copy is made	Both are you: two selves, one experiencer	A hard case: one, the other, or neither, each defended	The question may have no answer, and it doesn't matter: both copies keep what matters, the psychological connection	Neither; no future self was ever you, copy or not

Dimension	Holos (open individualism) ⑥	Closed individualism (one self per lifetime)	Reductionism (Parfit: identity is not what matters)	Empty individualism (one self per moment)
Whose future pain do you anticipate?	No claim. One experiencer does not settle what to anticipate or value	Only your own	Anyone's, in proportion to how psychologically connected they are to you now	Strictly, no one's, not even your own tomorrow
What walls perspectives apart	Structure: no integration bridges two observers	Being different selves	Separate brains and separate chains of memory	Being different selves
When a life ends	This self ends. The one experiencer goes on in every other self	The self ends	Connections end; death matters less than it seems	Selves were ending all along
Hardest objection	Why the walls between selves are absolute	Copying and splitting cases with no clear answer	Seems to leave no special reason to care about your own future	It undercuts caring about your own future

Relationship to Physics¹³

Holos is designed to be compatible with known physics because it does not propose a new mechanism. Its two additions (Axioms 3 and 5) say which systems have a point of view and whose it is. In the version defended here, neither changes any equation. It makes a different kind of claim. A physical model can fix every fact and still describe them only from outside. What it leaves out is not a fact about the world but a way of stating one, that some of it is lived.

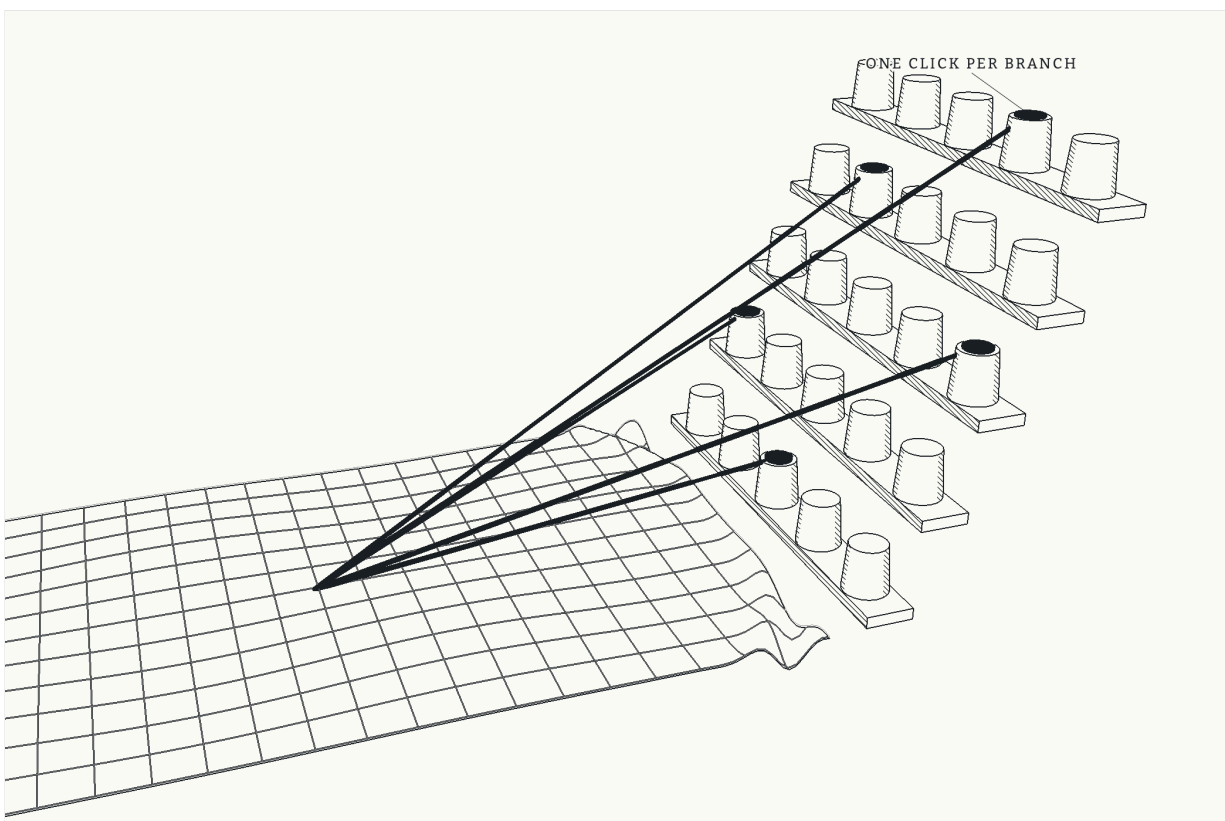
Decoherence is not presence

Decoherence is what happens when a quantum system becomes entangled with its surroundings. Its wave-like interference effects spread into the environment and can no longer be seen. It explains why quantum systems appear classical (like ordinary objects with definite properties) at everyday scales. Holos does not dispute this.

The Holos claim is that decoherence alone does not produce a lived world. It produces a consistent classical-looking structure. Presence requires integrated observation.

Fields are structure. A detected particle is a record

In quantum field theory (QFT), the basic ingredients are fields spread through all of space, and particles are their quanta, the smallest possible ripples in a field. Detectors do not directly observe fields. They record discrete outcomes, such as clicks, tracks, and energy deposits, because measurement is an interaction that entangles (links the states of) a spread-out excitation with a detector, and within each branch the detector's record shows one definite event in a specific place and time. Nothing collapses. Each branch simply holds its own record. Holos reads the particle as seen, a click at one place and time, as a record of a field interaction rather than a fundamental object. On that reading, particles are context-dependent records of field interactions, which is why a continuous theory can yield discrete observations without requiring reality to be made of little beads.



Conservation and branching

Axiom 2 commits Holos to a branching picture. The quantum state evolves smoothly and reversibly, never collapsing (unitary evolution), and no possibility is erased. When observers would register incompatible outcomes, they are situated in different branches of the one quantum state, each internally consistent. In this respect Holos sides with Many-Worlds-style interpretations of quantum mechanics (Hugh Everett's reading, on which every outcome occurs in its own branch), while adding what they leave out, an account of which structures are lived. This is a genuine interpretive commitment, not a neutral stance.

The Born rule: weighted branches

A branching picture owes an account of quantum probability. If every outcome occurs in some branch, what does it mean that one outcome is measured 70% of the time? Counting branches gives the wrong answer. The Born rule (take the number the quantum state assigns each outcome, its amplitude, and square its size) gives the right one in every test so far. Any framework that keeps all branches must say where those weights live and why they take the form they do.

Holos answers narrowly. The weights are structural facts, part of C . The possibility space is not flat. Branches carry weights, as absolute and observer-independent as the laws themselves. And they are not arbitrary. Gleason's theorem (1957) shows that, given a few mathematical assumptions (chiefly, that an outcome's probability does not depend on which other outcomes it is grouped with), the only consistent way to assign probabilities to the outcomes of quantum measurements, across the different ways the same experiment can be carved up, is the Born rule. That constrains the answer. It does not by itself say how an observer inside a branch should set their odds, which is the question a branching picture has to answer.

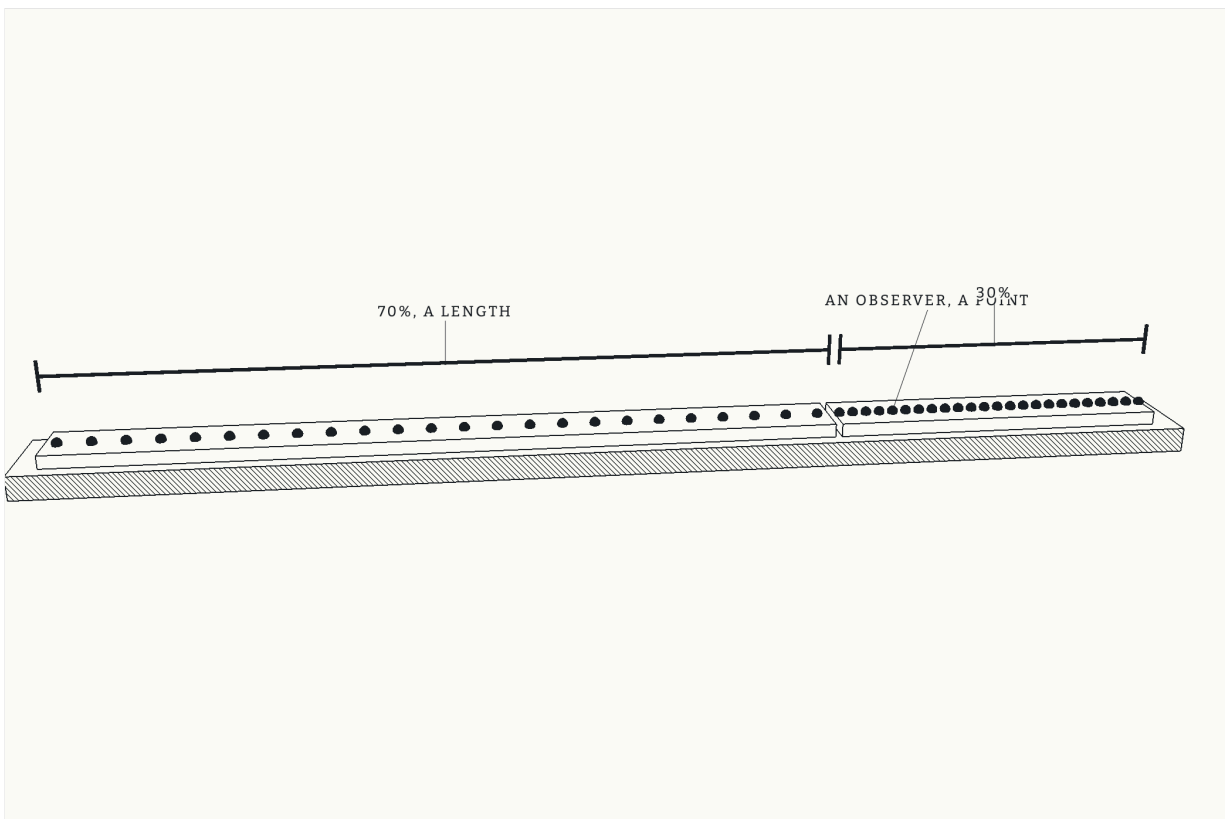
Holos declines one tempting reading. It does not read a branch's weight as its share of the whole's experience. That reading would require experience to be a quantity held in common and divided among branches. Holos makes no such claim.

Experience occurs at apertures. There is no reservoir from which branches draw larger or smaller portions, so the question of which branch receives more of it does not arise.

What the weight does measure is odds. After a measurement splits the world, but before you look, there are observers in every branch, and you are one of them without yet knowing which. The weight of a branch is your odds of being among its observers. That is why an outcome with 70% of the weight turns up about 70% of the time in long runs of the experiment. This is not because the other outcomes fail to happen, but because almost all of the weight lies with observers whose records show roughly that frequency. Observers with strange records exist too, like one who sees a fair coin land heads a hundred times in a row, but they carry almost none of the weight. The reading keeps the weights where Holos placed them, in the structural layer. The odds are a fact about the possibility space, not a quantity of experience.

The monist reading might seem to dissolve this uncertainty. If the one experiencer is each of the observers in every branch, asking it which branch it is in gets the answer "all of them," and nothing is left to be uncertain about. But the question is asked from inside an aperture, and apertures are walled off. None has access to what the others register. Each therefore genuinely lacks the information of which branch it is in, and the odds answer that lack. Two questions must be kept apart. "Why am I this one?" asks for a reason, and on the monist reading there is none to give, because the one experiencer is each of them. "Which one is this?" asks for information, and it stays open until you look. You can be everyone and still not know what is behind the next door.

A worry follows from yes-or-no status. Lived status is yes or no and ignores weight, so a branch with a tiny weight that contains an observer is fully lived. Counted one by one, most branches of many repeated experiments show roughly even results whatever the weights, so if every lived observer counted equally, a typical observer should expect the wrong statistics. But the worry depends on counting, and counting fails here. Seventy percent of an infinite crowd is exactly as large as thirty percent of it. Try to count branches, or the observers in them, and you get either an arbitrary number or an infinite one, and there is no fair lottery over infinitely many tickets. Each ticket's chance would have to be zero, and the zeros would never add up to certainty. Holos treats an infinity as a sign that a description has broken down, and this is such a case. Counting is the wrong tool. A single point on a line has no length, and a short stretch has as many points as a long one, yet the short stretch is shorter. Each observer is a point. The weights are the lengths.



Ruling out counting does not by itself pick the replacement. Every region of a dartboard holds infinitely many points, yet where a dart lands depends on how it is thrown. A random throw follows area, and a skilled player aiming at the center does not. Weight is the candidate physics itself supplies. It is built into the quantum state, it is conserved over time, and it does not change when a branch splits further. No rival on offer has all three. The last step is a principle, that your odds should follow the measure physics conserves. Sebens and Carroll state one version of it, a rule about how evidence bears on where you are, which they call epistemic separability. Roughly, what you should believe about where you are cannot depend on things far away that do not affect you ([Sebens and Carroll 2018](#)). It is widely discussed and still debated.

So this is an elimination plus one principle, not a derivation from first principles. Holos adopts the self-locating reading of the weights and inherits its open questions, whether the principle is justified, and whether Gleason's assumptions hold. Other published routes reach the same rule, among them decision-theoretic arguments; Holos depends on none in particular, and it adds no new mathematics here.

What Holos does not claim

- It does not claim violations of relativity, faster-than-light signaling, or new forces.
- It does not claim that humans are required. Reality requires a witness, and any observer will do.
- It does not claim to replace quantum mechanics or explain all details of measurement. It reframes what measurement fails to address.

What Holos **does** add is its two additions, Axioms 3 and 5, the threshold Φ_e and the whole, Omega.

Notation¹⁴

This section introduces a compact notation for the Holos framework. It derives nothing new. Its purpose is to make the structural claims precise and repeatable.

The notation should be read as a model of how possibility and experience are related. It does not assert that the universe literally computes these expressions.

R
Reality
=
C
Creation
⊗
O
Observation

State space

Let S denote an informational state of the universe at some level of description. S is not assumed to be complete or fundamental. It is simply whatever structure physics provides.

Creation

Creation (C) maps a given state to the branches physical law produces from it. It represents lawful possibility. Seen from inside any one branch, the others are what could have happened, and all of them do.

$$C(S) = \{S' \mid S' \text{ is a branch physical law produces from } S\}$$

Read: $C(S)$ is the set of every branch S' that physical law produces from the state S .

This notation is schematic. It does not assume a discrete branching structure or a specific ontology of histories.

Observation

Observation (O) maps a space of possibilities to registered experiential histories. It represents internal registration by integrated systems, wherever the possibility space contains them.

$$O(C(S)) = \{S_{\text{exp}}^{(i)}\}$$

Read: applying O to those branches gives a collection of experienced histories, one for each observer, labeled by i .

The result is an indexed family (a labeled collection), not a single selected outcome. It holds one experienced history per registering perspective, per branch. Observation selects nothing and erases nothing (Axiom 2). Registration occurs everywhere an aperture exists, and no thread is privileged. Following one observer's thread, what happens next is simply

what physics allows from that history onward; registering it changes none of those possibilities. The Born weights carried by C remain structural throughout. They set the odds of each record, not how much experience it holds.

This mapping is not assumed to be random, deterministic, or computable in general. Holos requires only that each experienced reality corresponds to a single consistent history, one per registering perspective, internally coherent, and in agreement with every record it is compared against.

The Holos relation

The Holos relation is the composition of Creation and Observation.

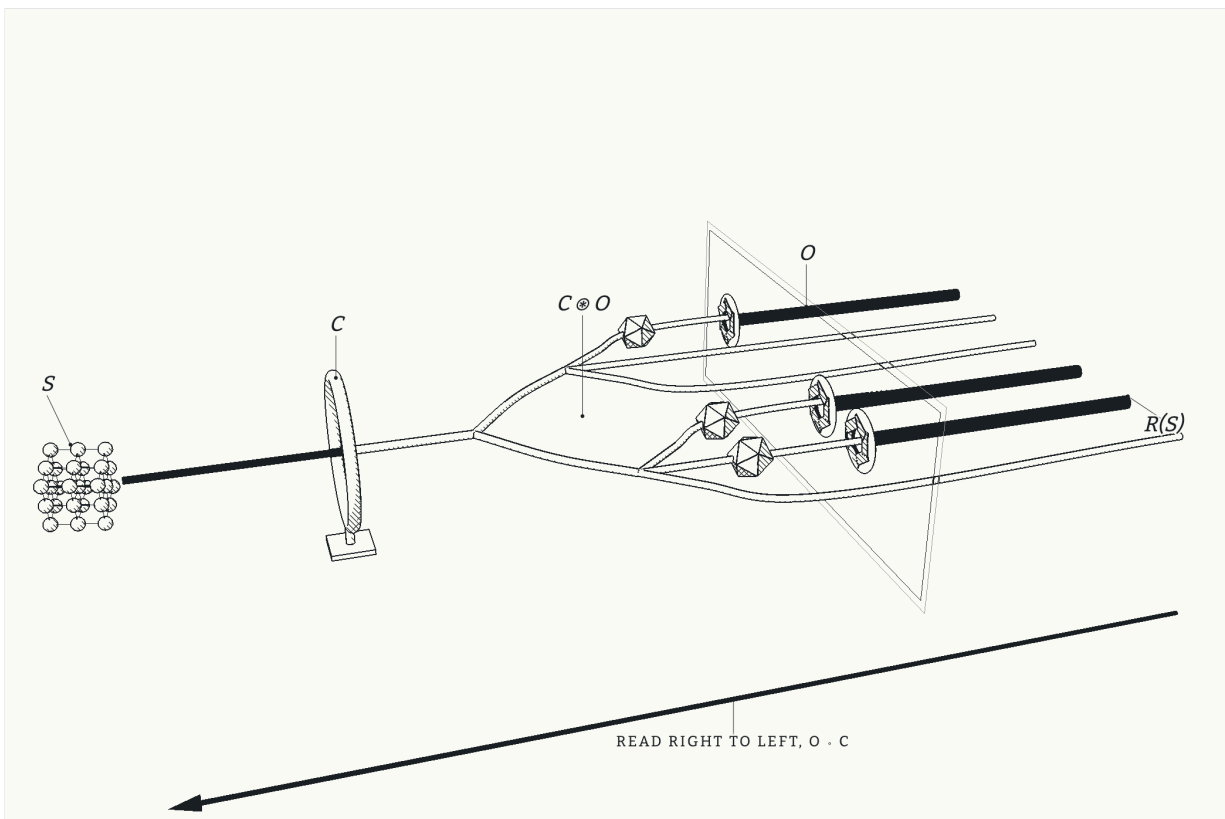
$$R = C \circledast O$$

Read: lived reality, R , is Creation composed with Observation.

This expression states that lived reality is neither pure possibility nor pure observation. It needs both. Applied to a state S , the result, written $R(S)$, is the family of lived perspectives, one per observer per branch, each with the lit world it draws on. Structure exists whether or not it is lived. $R(S)$ marks where it is also lived, and in Holos that is what reality means.

The symbol $\circledast$ is defined as composition: $C \circledast O$ is the composite operation “possibility, then registration.” Applied to a state S , it reads $R(S) = O(C(S))$: R names the relation, and $R(S)$ the lived reality it yields from that state. It is ordinary function composition, with the order of the steps carrying the meaning, possibility first and registration second. The order is logical, not temporal. Registering needs something to register.

In standard notation this is simply $R(S) = (O \circ C)(S)$, where $\circ$ is the usual symbol for composition. Apply C , then apply O . Standard composition reads right to left, so $C \circledast O = O \circ C$, the same composition written in reading order. Nothing is hidden in the glyph. Holos retains $\circledast$ because the framework is named for the relation it marks, not because the operation it denotes is unusual, and a reader who mentally substitutes the composition symbol loses nothing.



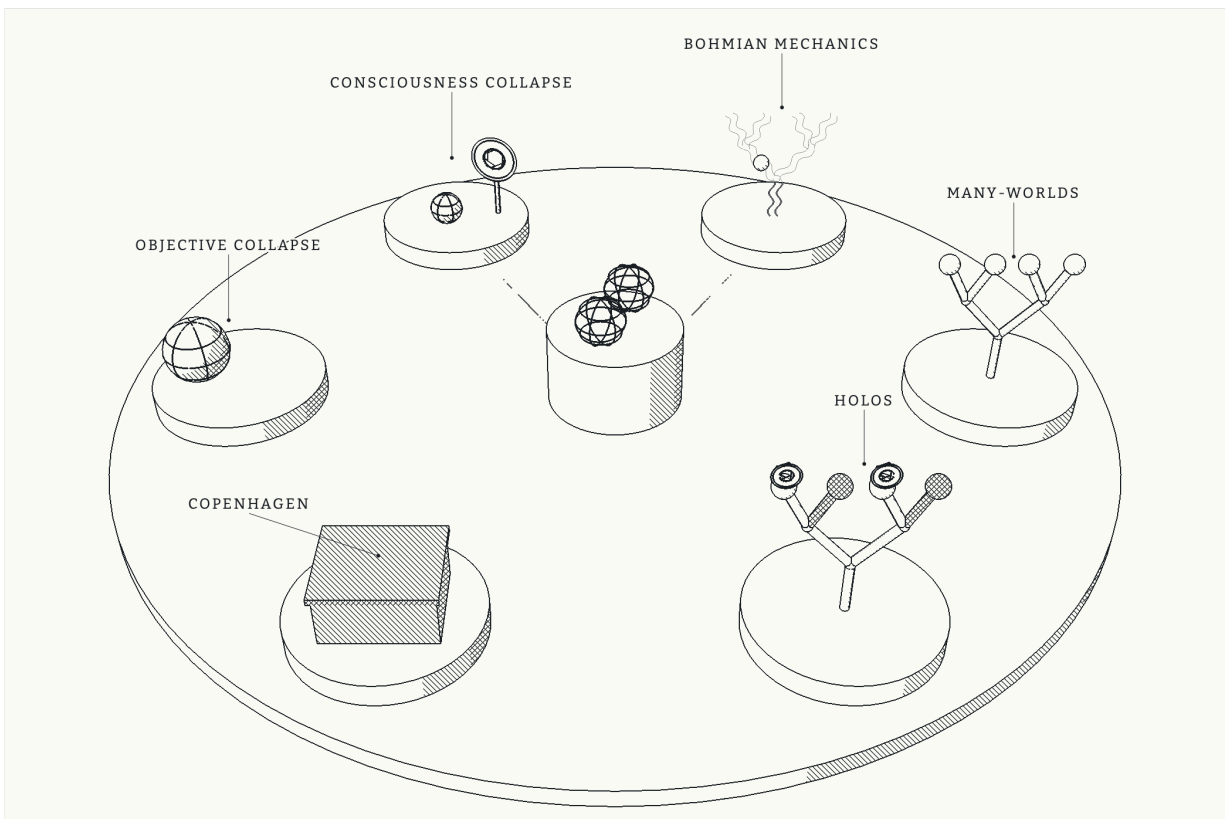
What the symbol adds is not mathematics but ontology. It claims that both steps are required for a lived world, and that neither step alone yields one.

More elaborate mathematical frameworks can be layered on top of this representation. Holos itself commits to a lawful possibility space, an integration threshold at which apertures open, and the whole those apertures belong to.

Holos among Quantum Interpretations

Every column in this table answers one question, known as the measurement problem. Quantum mechanics says a small system can be in several states at once, and experiments confirm it. Yet we always see one definite result. The textbook account says the rules change "when a measurement happens" but never says what counts as a measurement. A detector, a cat, a person? The most common working stance, Copenhagen, treats the question as one science need not answer. John Bell, among others, called that an evasion. Each other column is an attempt at a real answer.

Holos takes a side. Nothing collapses, no possibility is erased, and the universe branches whenever the environment records different outcomes, whether or not anyone registers them. That is the Many-Worlds picture. Holos differs from Many-Worlds about what exists, not about the physics. Branching alone does not say which structures are lived. Bohmian mechanics also avoids collapse, but with one world of particles guided by the quantum state. The table also includes the two collapse views Holos bets against: objective collapse, in which superpositions (states that combine several outcomes at once) collapse on their own once objects are large enough (as in the Diósi-Penrose model, where gravity triggers it), and consciousness collapse, in which a conscious system causes it (as Chalmers and McQueen propose). Either, if confirmed, would retire the version of Holos defended here. So far no experiment has found either kind of collapse, and the simplest gravity version has been ruled out. The table below shows where Holos aligns with, and diverges from, each.



Dimension	Holos ☉	Copenhagen	Many-Worlds (MWI)	Bohmian Mechanics	Relational QM (RQM)	QBism	Objective Collapse	Consciousness Collapse
What is fundamental?	The whole (Omega). The universal quantum state, read as the one experiencer	Measurement outcomes, described in everyday (classical) terms	Universal wavefunction	Particles with definite positions, plus a wave that guides them	Relations between systems	Not specified; the quantum state is an agent's personal expectations	Wavefunction plus a spontaneous collapse law	Wavefunction plus consciousness that collapses it
Is the wavefunction real?	It exists. Creation, every branch physics produces	Versions differ: a tool for predicting outcomes, or something that really changes at measurement	Literally real, never collapses	Real, a guiding wave that never collapses	A bookkeeping tool, relative to each system	No: an agent's personal expectations	Real, collapses at random, faster for larger systems	Real, collapse when a conscious system registers it
Collapse?	No collapse (branching). Lived from within where observers exist	Yes, at measurement; whether physical or only an update of knowledge is left open	No collapse (branching)	No; the unused parts of the wave simply stop affecting	Relative collapse only	Belief update	Yes, physical and spontaneous	Yes, triggered by consciousness

Dimension	Holos [®]	Copenhagen	Many-Worlds (MWI)	Bohmian Mechanics	Relational QM (RQM)	QBism	Objective Collapse	Consciousness Collapse
				the particles				
Does consciousness change physics?	No, in the version defended here. Holos with collapse may put collapse at Φ_c	No, in Bohr's version	No	No	No	No	No	Yes
Role of observer	A system past the integration threshold ($\Phi \geq \Phi_c$): one opening (aperture) through which the whole is lived	The measuring device, described classically; not a mind	A physical system, with no special role	Nothing special	Any physical system; facts are relative to it	Central: the agent using the theory	Nothing special	Cause of definite outcomes
Reality without observers	Structure, not reality. It exists as pattern but is never lived	The theory is silent; it speaks only of measurement results	Fully real	Fully real, one history	Relations still hold; any system can observe	Real, but quantum theory is a tool for agents, not a description of it	Fully real, one history	Real, but superpositions persist unobserved
Multiple realities?	Yes, branches, each lived where observers exist	No	Yes, branching universes	No: one world; the other branches are empty	One world; facts are relative to each system	No	No	No
Where the line between observer and observed falls	Where a system is the most integrated around, past the threshold Φ_c (provisional rule)	Between the quantum system and the classical apparatus, placed by the physicist	Nowhere; everything is quantum	Nowhere; everything follows the same laws	Anywhere; any system can play observer	At the agent	Set by size and mass, not observers	At a conscious system
About reality, or about knowledge?	About reality (ontological)	Mostly about knowledge; versions differ	About reality	About reality	Both: relations are real	About knowledge	About reality	About reality
Key prediction focus	Integration threshold. No consciousness-linked deviation from quantum mechanics	Same as standard quantum mechanics	Same as standard quantum mechanics	Same as standard quantum mechanics	Same as standard quantum mechanics	Same as standard quantum mechanics	Tiny deviations in large superpositions	Superpositions degrade when a conscious system registers them

Holos among Theories of Mind

Most of what Holos claims is about mind, not physics, so it needs a second map. It shares integration with integrated information theory without IIT's identity claim or its experience in every integrated system, shares a single ground of experience with cosmopsychism, without its claim of a cosmic experience, and rejects both panpsychism's experience everywhere and illusionism's claim that experience, as we usually conceive it, does not exist. It is neither physicalism nor dualism but a two-sided monism with a threshold (see [What kind of view this is](#)). The table below shows where it sits.



Dimension	Holos ☉	Integrated Information (IIT)	Global Workspace (GNWT)	Higher-Order Theories (HOT)	Panpsychism	Cosmopsychism	Illusionism
What is fundamental?	The whole (Omega). Experience is not derived from anything else. It is where Holos starts	The structure of causes and effects inside an integrated system, which IIT says simply is the experience	Brain processes; consciousness is a function they perform	Brain processes; a state is conscious when the brain represents itself as being in it	Experience, in simple forms, in the basic building blocks of matter	One conscious cosmos; individual minds derive from it	Physical processes; experience lacks the special inner qualities it seems to have
Where experience occurs	Only where a system is more integrated	Any system whose parts are at least slightly	Where information is broadcast brain-wide	Where a mental state is itself the target of a higher-order	Everywhere, in simple forms	In the cosmos as a whole, and in its parts	Nowhere, as usually conceived; what exists is

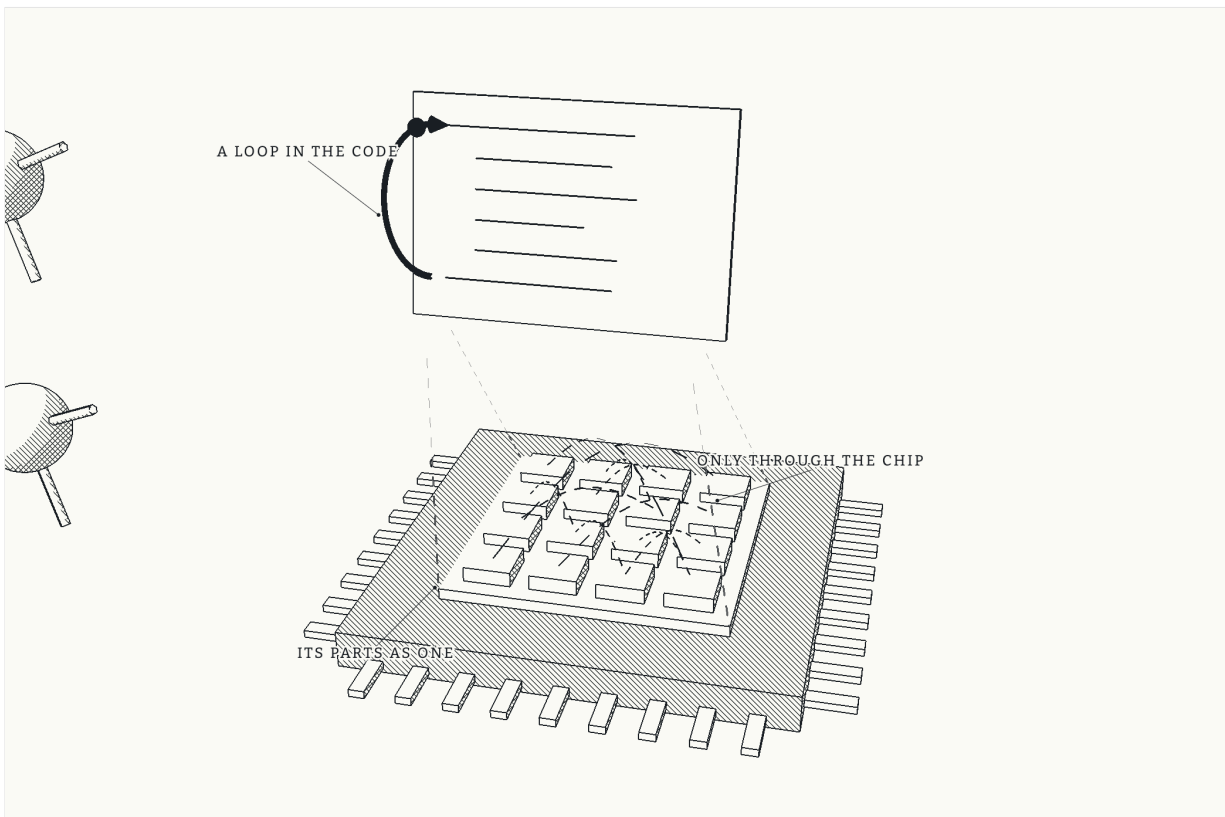
Dimension	Holos ☉	Integrated Information (IIT)	Global Workspace (GNWT)	Higher-Order Theories (HOT)	Panpsychism	Cosmopsychism	Illusionism
	than anything around it, past the threshold (Φ_c), and carries a model of a world	integrated, as long as it is the most integrated system around		representation, often linked to the front of the brain (prefrontal cortex)			the brain's representation of it
A threshold?	Yes. None well below it, a narrow twilight, then experience	No: graded from zero	Information either floods the brain-wide network or does not ("ignition"); the theory is about this access	All or nothing for each state: either it is re-represented or it is not	No	No	Not applicable
How many subjects?	One experiencer, living through every self, each walled off from the others	One per peak of integration	One per workspace	One per creature with the right self-monitoring machinery	Countless micro-subjects	One cosmic subject, with derived individuals	The question is reframed as one about self-models
Hardest problem	The measure of integration. Why the walls between selves are absolute	Critics call its core claim untestable; simple grids can score higher than brains; Φ cannot be computed for real brains	Critics: explains which information gets reported, not why it feels like anything	Critics: animals and infants may lack the machinery; a higher-order state could misrepresent a state that isn't there	Combining micro-subjects into one mind	Decomposing the cosmic subject into many	Explaining why the illusion is so convincing; most find experience undeniable
Current AI	No, as built. It models a world, but integration falls short	No: conventional digital hardware has negligible Φ	Possible, if the architecture has a workspace	Possible, if a system monitors its own internal states in the right way	Its parts have micro-experience; the whole is unclear	Unclear	The same question as for us: does it model itself as conscious?
Testability	Test A, experience tracks integration. Omega is untestable	Tested in COGITATE; key claims challenged	Tested in COGITATE; key claims challenged	Tested through lesion and self-monitoring (metacognition) studies; contested	Largely untestable	Largely untestable	Through explanations of why we believe we are conscious

Open Problems

Three problems sit at the center of the framework and remain open.

The measure of integration

Holos does not yet name a privileged measure of integration. Competing proposals for computing Φ can disagree, not only about values but about which of two systems is more integrated. Locating the threshold and its twilight requires a determinate measure, and identifying it is an open problem for the framework, not settled background. Holos is committed to the threshold being fixed by structure. It does not yet know how to compute it. The stakes reach back into the framework's core argument, the identification of a structure's oneness with causal integration (see [Why Integration](#)) remains an argued direction rather than a finished quantity until the measure is fixed. The problem also includes the question of level. Holos measures integration in physical parts, grouped the way they actually work together. The grouping may lie above the smallest parts, as a neuron's firing lies above the motion of its ions. A program counts only through the chip that runs it. Loops in the code are needed, but the chip's own parts must also act as one, and conventional chips may not. Saying precisely where that level is belongs to the same open problem.



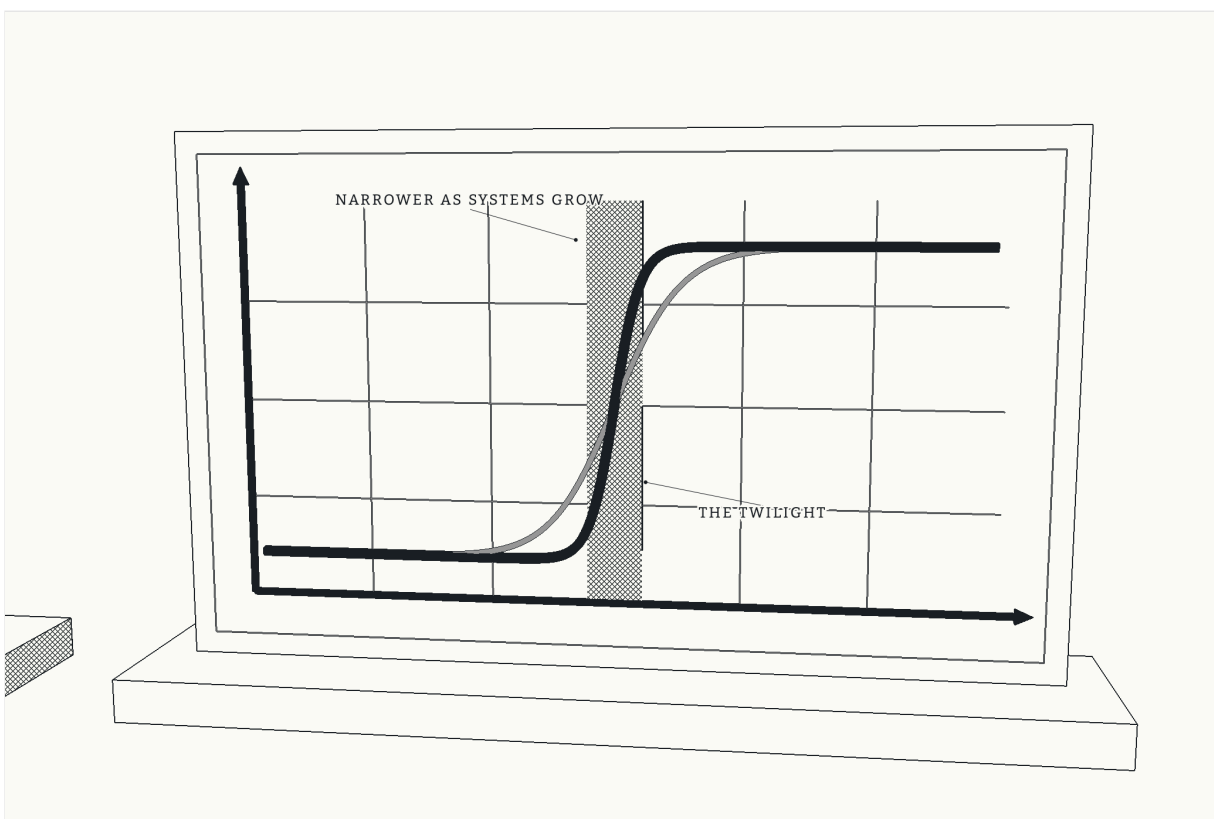
Where the threshold falls

Where any given system's threshold Φ_c and its twilight lie is unknown (see [The threshold in three claims](#)). Because presence itself cannot be detected directly, no experiment can locate it by direct measurement. Its placement is constrained only indirectly, by which systems show the structural signatures of observation (see [Test A](#)), and our estimates may stay blurrier than the twilight itself. Holos accepts this as the price of a threshold that is structural rather than behavioral.

A path to the threshold

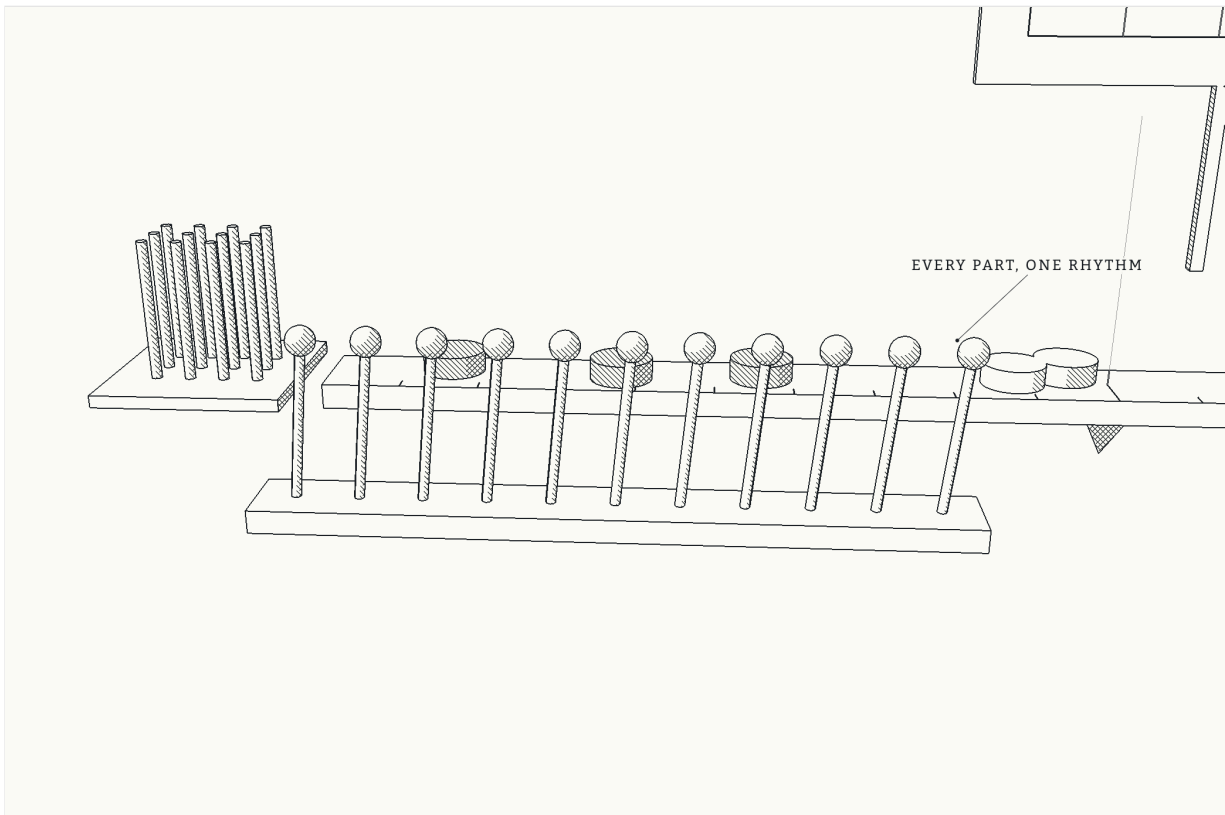
The two problems above are not invitations to despair. They are a research program. Science pins down unobservables by triangulation all the time. No one has ever seen an electron's charge, yet independent methods converged on it and every proposal that disagreed was discarded. The same logic applies here, in three steps.

Treat the threshold as a transition, not a dial. If the passage from distributed processing to a unified perspective is a genuine transition, then Φ_c is not a number we are free to tune, and transitions leave measurable fingerprints. A sudden one typically shows a lag, where the way in and the way out do not match. A continuous one typically shows slowing and growing fluctuations near the boundary. Either kind shows a twilight that narrows as systems grow. Clinical consciousness research has already found one such boundary from the outside, the Perturbational Complexity Index (PCI). It gives the brain a brief magnetic pulse and measures, with scalp electrodes (EEG, a recording of the brain's electrical activity), how complex the echo of activity is. In the studies that set it up, a single cutoff, found by experiment, separated conscious from unconscious states across anesthesia, sleep, and disorders of consciousness (lasting states after brain injury), without anyone measuring presence directly. Locating Φ_c is that kind of problem, not a metaphysical one.



The fingerprints are already being measured, but they must be read with care. Several lines of evidence suggest that waking cortex runs near a critical point, a tipping point between too-orderly and too-chaotic activity, often called the edge of chaos. Anesthesia, generalized seizures, and disorders of consciousness move it away, while psychedelics move it closer and make its activity richer (Toker et al. 2022). Measures of criticality in resting EEG predict anesthetic loss of consciousness and track PCI (Maschke et al. 2024). In cortical tissue, cascades of activity called neuronal avalanches come in all sizes and follow power laws (large cascades are rarer than small ones in a fixed mathematical proportion), with exponents that match a known universality class, a family of systems that behave alike near their tipping points (Beggs and Plenz 2003). Whether the brain is exactly critical or only close to it is still debated, and power laws can arise in other ways. Taken together, these

results suggest that a critical point sits at the center of conscious life, not at its edge. Consciousness is lost by moving away from it in either direction. Holos reads them through two separate requirements. Integration asks whether the parts are joined into one whole, and that is where the threshold lies. Differentiation asks how varied the states of that whole can be, and variety peaks near criticality. A seizure locks every part into one rhythm (Meisel et al. 2012), like a stadium chanting a single word, joined but with almost no variety. So these studies show where observers operate and why richness peaks there. They do not yet show that crossing the threshold is itself a genuine transition. That evidence must come from the boundary.



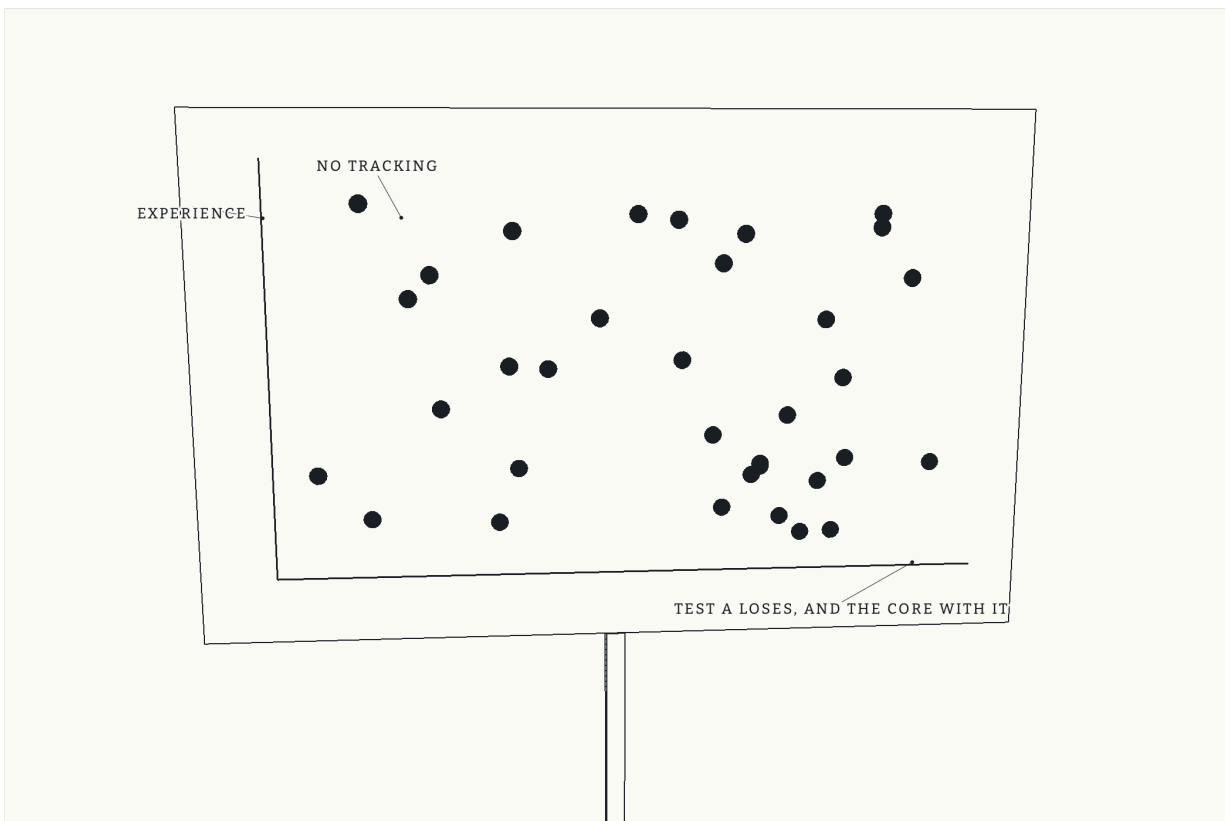
A cutoff that sorts patients is not yet proof of a transition, since a smooth quantity can be cut anywhere. What would count is a transition's own signature, and here the evidence is mixed in an informative way. Animal studies find that consciousness is lost and regained at different anesthetic levels. The way in and the way out do not match, a lag known as neural inertia. A lag of that kind is the hallmark of an abrupt switch, the kind physicists call first-order, like water freezing. If the lag belongs to the transition itself rather than to how drugs enter and leave the brain, it fits the transition hypothesis (claim 3) as an abrupt switch. But in flies the lag is controlled by genes that regulate sleep, and single mutations can erase it (Joiner et al. 2013), so it may belong to the brain's arousal switch rather than to integration. In humans the evidence is mixed. A controlled test found no lag with propofol and some with sevoflurane, two common anesthetics (Kuizenga et al. 2018), and in 393 surgical patients the brain's slow-wave response (the large, slow brain waves of deep anesthesia) differed between going under and coming back (Warnaby et al. 2017), but the lag appeared in the EEG, not in responsiveness. Slowing and growing fluctuations measured at the boundary itself, as consciousness is lost, would point the other way, toward a continuous transition, though slowing can also come just before some abrupt switches. The criticality studies above measure the operating point, not the boundary, so they do not yet supply this. Neither signature settles the question alone, but together they help tell the two kinds of transition apart, which makes the question testable. In a finite system like a brain, either kind appears as a steep, rounded curve rather than a mathematical kink, and that rounding is the twilight. The search is for how it scales, not for a perfect step.

Cull the measures by convergence, then try to force uniqueness. Holos does not need one anointed measure so much as it needs the adequate measures to agree on the cases that matter, and that is testable now. Run the candidate measures across a battery of systems and keep the ones that rank the anchor cases (systems whose status is not in doubt, such as a waking adult and a deeply anesthetized one) the same way. The stronger move is to derive the measure rather than choose it. State the properties any measure of a single unified perspective must have, and see whether consistency forces a unique form, as [Gleason's theorem](#) constrains the Born weights. Even narrowing the field to a small equivalence class (a family of measures that always agree on which system is more integrated) would remove most of the arbitrariness.

Solve them jointly. The measure and the threshold constrain each other. A candidate pair must simultaneously fit the anesthesia boundary, the emergence of experience in development, split-brain dissociations, and wherever minimal neural systems first show coherent integration. Coupled constraints are far more rigid than separate ones. The band that satisfies all the anchor cases at once is narrow. This is what [Test A](#) really is, not a single yes-or-no experiment but a calibration engine that locates the threshold and validates the measure. The two jobs must use separate data. Calibration states locate the threshold, and held-out states, named in advance, test it. Testing on the calibration set would only grade the answer key.

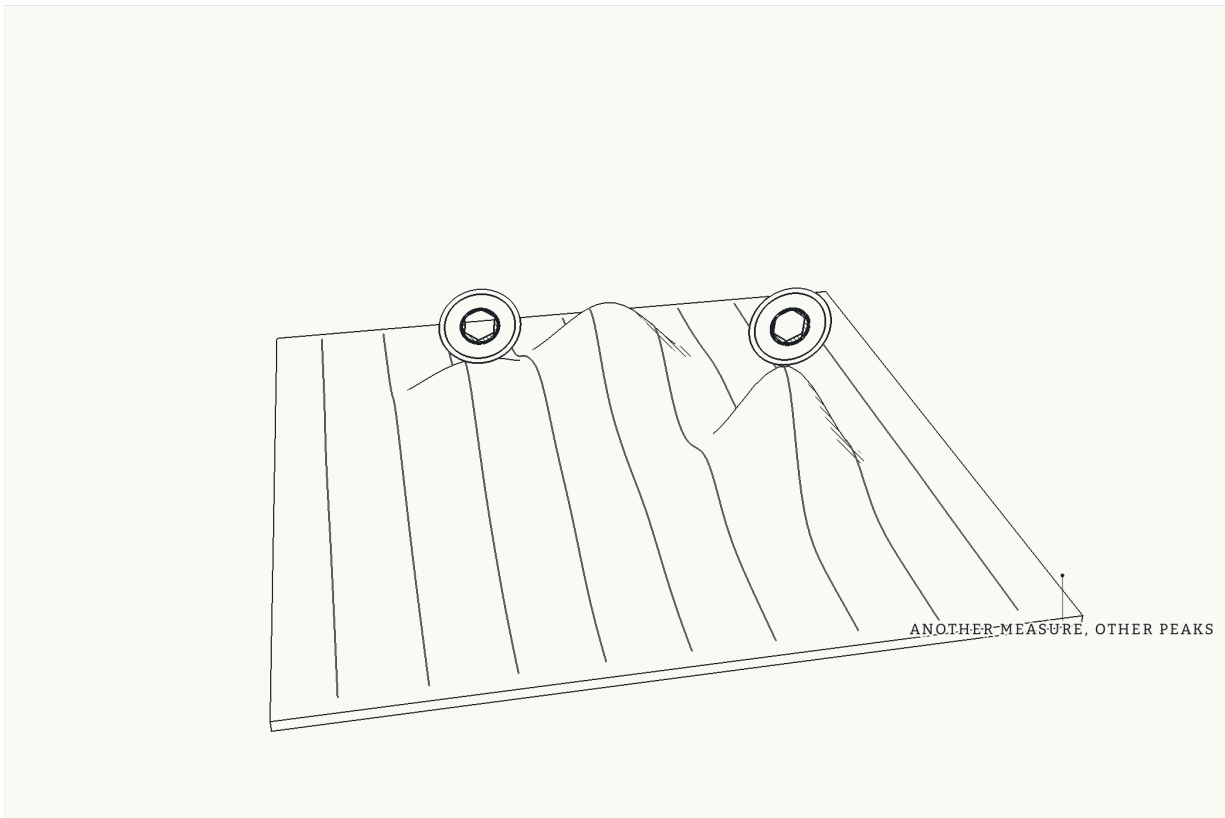
The field already has a model for running such tests. The [COGITATE adversarial collaboration](#) (Nature, 2025), in which rival theorists agreed on a test in advance, pitted IIT against global neuronal workspace theory (which ties consciousness to information broadcast widely across the brain), and its results challenged key claims of both. Test A should follow the same design. A caution applies to the measure itself. In a letter first circulated in 2023, more than 100 researchers called IIT pseudoscience, a debate aired in [Nature Neuroscience in 2025](#). The charge has two main targets. One is the theory's untestable core claim that its integrated structure simply is consciousness, which Holos does not adopt. The other is the implication that simple grids of logic gates are conscious, which Holos's aboutness requirement rules out (see Observer Requirements). Holos borrows Φ only as a measure.

Two conditions bound the program. First, its anchor is report. The human case is the one place inside access exists, so the threshold is located relative to us and carried outward by the measure, with certainty that weakens as the cases grow alien. That is the shape of all consciousness science, not a defect peculiar to Holos. Report can also miss consciousness. About one in four brain-injured patients who cannot respond show, on brain scans, that they are following instructions ([Bodien et al. 2024](#)), so a missing report is not proof that no one is home. Second, the program can fail, in two ways of different weight. It could find no transition at all, integration climbing smoothly. That would falsify claim 3 and leave the twilight wide, a slow dawn rather than a quick one, with the core intact. Or it could find that experience does not track integration on any candidate measure, which is how Test A loses, and the core with it. A stated way to lose is what makes these open problems scientific questions rather than definitions.



Where one observer ends and another begins

The maximality condition (see Exclusion, under [Observer Requirements](#)) makes only a local maximum of integration an aperture. It inherits the unsettled question of which measure defines the maximum, and it lacks a stated procedure for which candidate systems are compared and how overlapping candidates are resolved. IIT spells out such a procedure. Holos has not yet adopted or replaced it. Until the measure and the procedure are fixed, the boundaries between observers are fixed only in principle.



These are not peripheral loose ends. They concern the integration threshold, one of the framework's two additions to physics. They are recorded here so that progress on them can be judged against a stated standard.

Holos is written in public and revised when it is wrong. Every retired claim, and the reason it was retired, is listed under [Revisions](#).

Holos starts from one fact, that experience exists. It is built on two ideas. Past a threshold, a new self begins. Through every self lives one experiencer, the whole, Omega. Reality requires a witness.

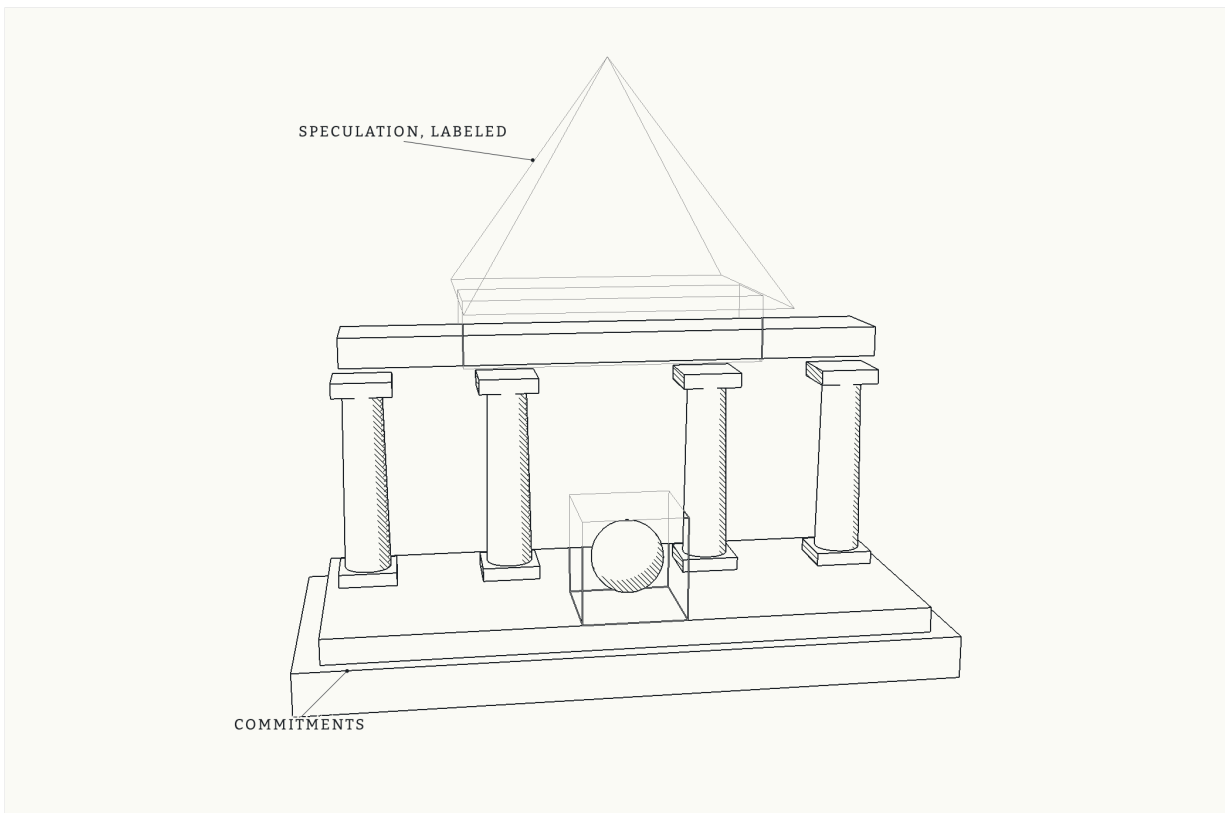
Predictions

Predictions¹⁶

In the version this site defends, Holos adds no new dynamical laws (the rules that say how things change over time) and changes none of the equations of physics. It is built on two ideas. First, a threshold, Φ_c . A system is an observer, with experience of its own, only once its parts are integrated enough to act as one whole. Second, Omega, the whole, with one experiencer living through every self. The commitments and tests below follow from established physics, from those two additions, and from the positions Holos takes on structure and mind (Axioms 1 and 4). Where a claim depends on one particular reading of quantum physics, the one without collapse, it is marked (Commitment 3 and Check C). The speculation at the end does not follow from any of this, and it is labeled as such.

The sections below separate three kinds of claims. The last section includes speculative extensions that aim to produce observable signatures, not just philosophy.

- **Commitments.** What must be true if Holos is correct, independent of any future experiments.
- **Testability and its limits.** What can never be tested (presence, the fact that experience is lived), two tests Holos could fail, one consistency check, and one standing bet about physics, with the fallback version of Holos stated in advance in case the bet is lost.
- **Speculation.** Extensions that *could* follow under Holos on long timescales, stated with explicit alternatives rather than predictions.



For the definitions and the observer requirements, see [Logic](#).

Commitments¹⁷

Commitments 1 and 2 are fundamental to Holos. A serious rival denies each, and if either is false, the framework fails. Commitment 3 belongs to Holos without collapse, the version this site defends. The version with collapse is declared in advance under [Two versions](#).

1. Experience is local, and physics describes it only from outside

From Axioms 1, 3, 4, and 5, with the definitions in D7. See [Axioms](#).

Two claims, and a serious rival denies each one.

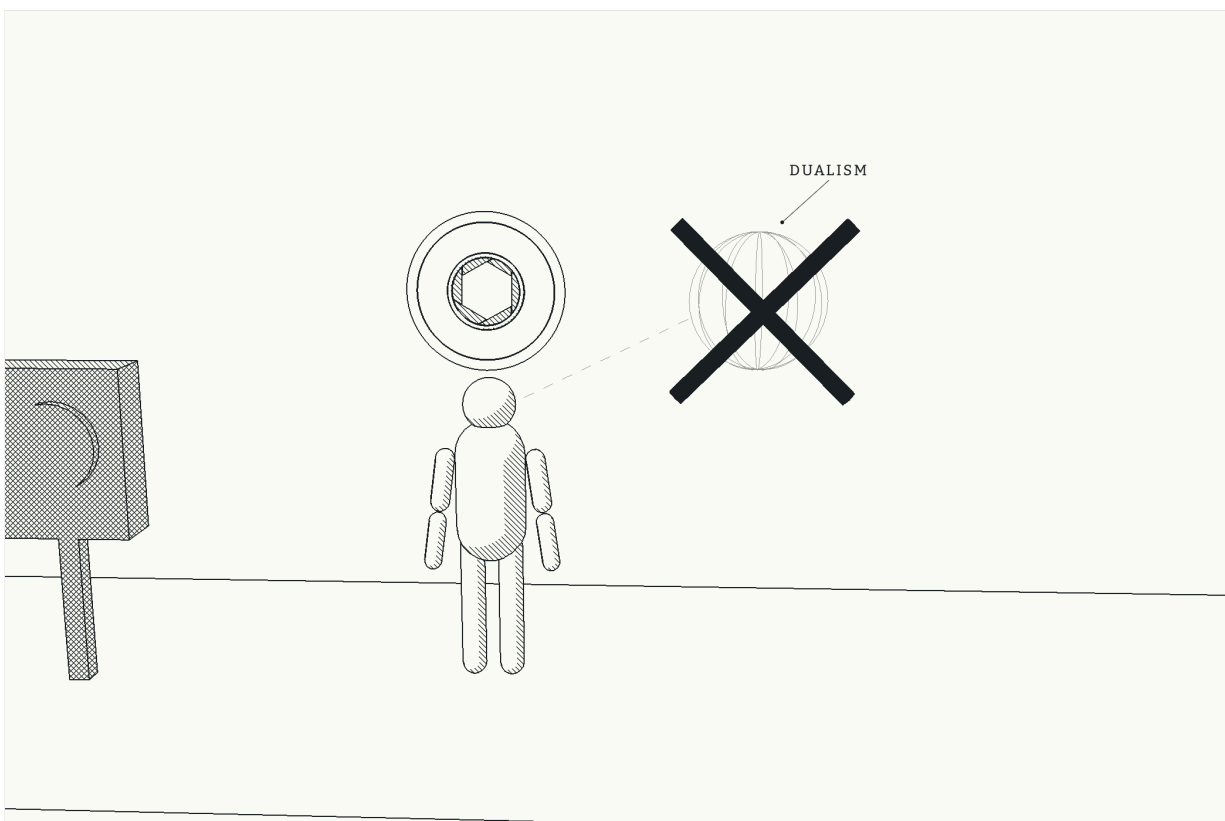
Local. Experience occurs in no place except inside observers, systems past the threshold. It is not spread through matter or ambient in space or fields. Omega is the one experiencer, and it lives through observers. Whether the whole as such has any view of itself is not a question about a place, and Holos leaves it open. "Local" says where experience happens, not who has it. Structure without observers still exists, as pattern. Unobserved histories remain part of C , the branches physics produces, and are simply never lived. (In the reading of quantum physics adopted here, every outcome a measurement could have occurs, each in its own branch, a complete, separate version of how the world goes.)

Only from outside. A physical description can be complete and still say nothing of whether anything is lived. The gap is not a missing fact. A perfect physical copy has the same inside. It is that the description is written from outside, like a floor plan that records every wall and still cannot say what living in the house is like.

Experience happens nowhere but in observers. Physics fixes all of it, but states it only from outside.

What this rules out:

- Panpsychism: a flicker of experience in every particle, rock, or thermostat.
- Illusionism: the view that experience does not exist, only the belief in it.
- Strict physicalism of one kind: the view that physical language could, in principle, state that something is lived.
- Dualism: experience as an extra ingredient beyond physics. The gap is in the description, not in the world.

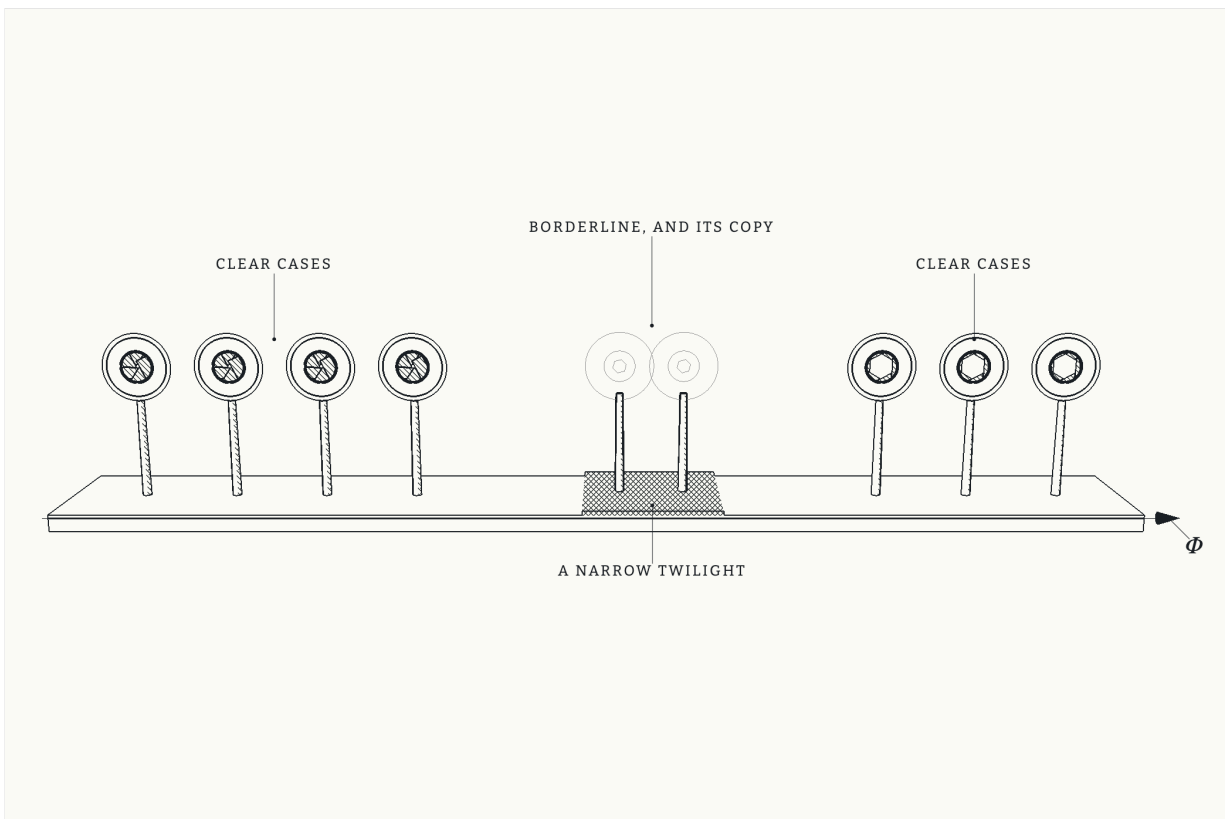


Anthropic principles (we should not be surprised to find ourselves in a universe that allows observers, since we could not find ourselves anywhere else) explain why observers find themselves in universes that allow them. They do not say where experience occurs, or why physics states it only from outside. Holos answers the first and names the second. Why there is experience at all, it leaves open.

Holos uses three words, each for one thing. **Lived** is where experience occurs. **Lit** is the causal past it draws on (everything close enough in space and time for light or signals to have reached an observer). And **unlit** is structure outside every observer's causal past. They are defined in [Logic, D7](#). That "lived" means experienced by an observer is a definition, not a commitment.

2. Observerhood is thresholded

From Axiom 3. See [Axioms](#).



Holos rejects the idea that experience increases smoothly with greater amounts of computation. Distributed processing can scale indefinitely without producing a single point of view.

What matters is integration. Well below the threshold, there is no unified internal state that could count as “what is happening for the system.” Well above it, in a system that meets the other observer requirements, experience is unavoidable.

Observerhood requires $\Phi \geq \Phi_c$

In words, a system is an observer only if its integration, written Φ (phi), reaches the threshold, Φ_c . Φ measures how far a system's parts constrain one another as one whole, and the inequality is shorthand for being past the twilight described below. No one can yet compute Φ for a real brain, so the tests below use measurable stand-ins such as PCI.

Observerhood is neither ubiquitous nor optional. It appears when structural conditions for integration are met.

Whether a system crosses this threshold is a fact about how its parts are wired together, and integration alone is not enough. The integrated state must be about a world beyond the system (see the observer requirements in [Logic](#)). It belongs to the structural layer, the facts that hold the same for everyone, alongside the laws of physics, and it does not depend on who is looking. Observerhood is what qualifies a system to have a perspective at all. It is not relative to one. In the monist reading (the reading that the whole is the one experienter), crossing the threshold is where a new self begins. No new experienter does.

Two consequences follow. A physically identical copy of an observer cannot lack experience (Axiom 4). And the threshold is steep but not a mathematical line. There are clear cases on both sides, a narrow twilight between, fixed by structure, so a

copy of a borderline system is borderline too (see [The threshold in three claims](#)).

3. Facts are relational but consistent

From Axioms 1 and 2. See [Axioms](#).

Holos distinguishes two kinds of facts. **Structural facts** describe what is consistent, such as the laws of physics, the branches physics produces with their quantum weights, and whether a system meets the integration threshold. These are absolute and observer-independent. **Registered facts** describe what is lived, which outcome a system registers from its own perspective. These are always indexed to observing systems.

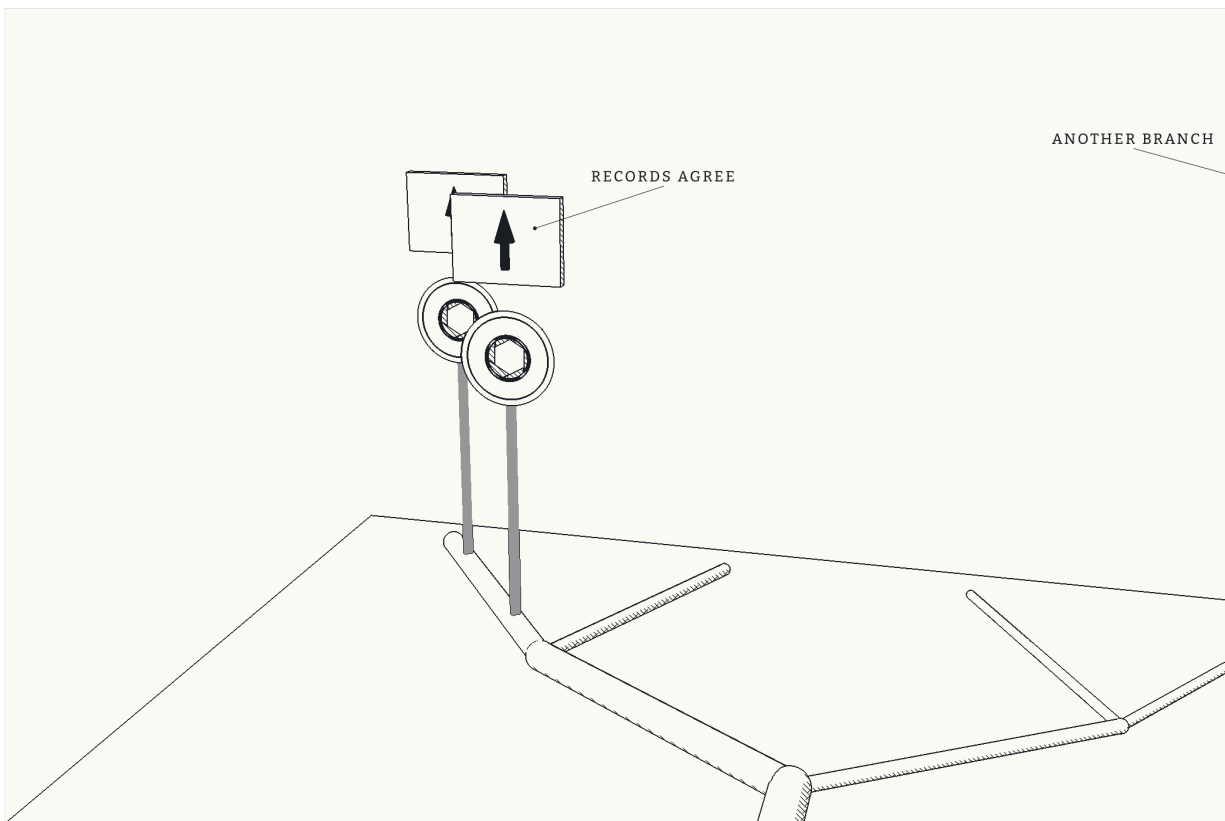
The relational commitment applies to registered facts. There is no absolute, observer-independent fact about which outcome is experienced. The structural layer, by contrast, is not relative. Without it, registration would have nothing stable to register.

This does not imply contradiction, and Holos is specific about why. No possibility is erased (Axiom 2), so observers never collide over a single shared outcome. Where registrations would be incompatible, they belong to different branches of the one quantum state, each internally consistent. There is no rule that the first observer fixes the truth for everyone. For far-apart events, relativity says observers moving differently can disagree about which came first, so there is no single “first” to appoint, and none is needed.

Within a branch, consistency is operational rather than abstract. Whenever two observers actually compare records, their records agree. Perspectives may differ while separated. Communication forces agreement. This is what “consistent” means in practice. Physics secures it. Observers who communicate are reached by the same signals.

Branches are not all equal. Each carries a weight, and those weights reproduce the probabilities quantum physics predicts (the Born rule). That is why, from inside, a 50/50 quantum experiment looks like a fair coin. Almost all of the weight lies with observers whose records follow the Born rule. The weights are structural facts, not amounts of experience, and for an observer they are odds of being in one branch rather than another. See [Logic](#) for the full account.

| *Registered facts are relative to observers. Structural facts are absolute. Observers who compare records agree.*



Apparent collapse (the way a measurement seems to pick out one outcome) is therefore not a new physical process. Within each branch, records are already definite, a detector's click included. What an observer adds is not the definiteness but its being lived.

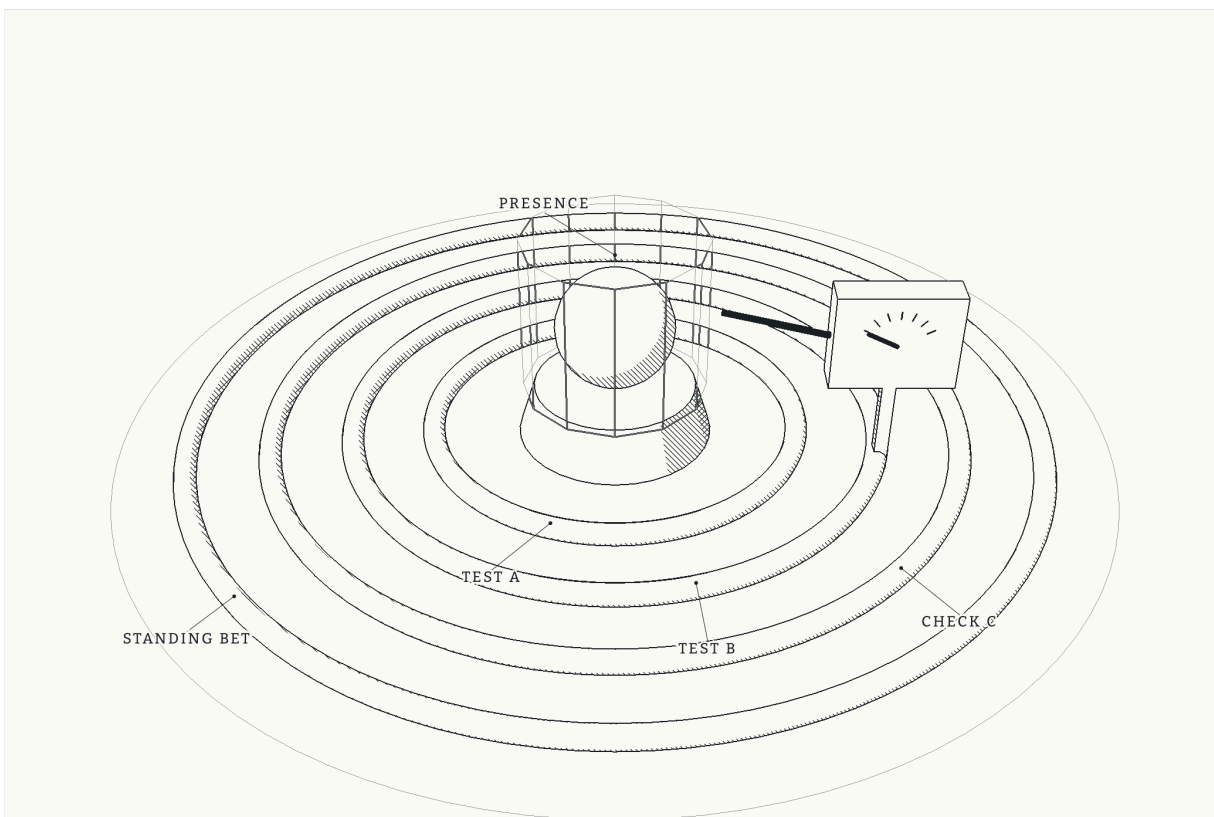
Everything that follows assumes these commitments. What comes next is what experiments can and cannot show about them.

Testability and Its Limits¹⁸

The central claim of Holos is that observation is where structure is lived, not a force. In the version this site defends, it changes no equation and moves nothing. But every experiment is a physical measurement, and an instrument only ever records physical change. So **presence itself cannot be detected directly**. An instrument that finds nothing extra is exactly what Holos predicts, because there is nothing extra to find. Presence is what the physics is like from the inside, not an additional signal beside it.

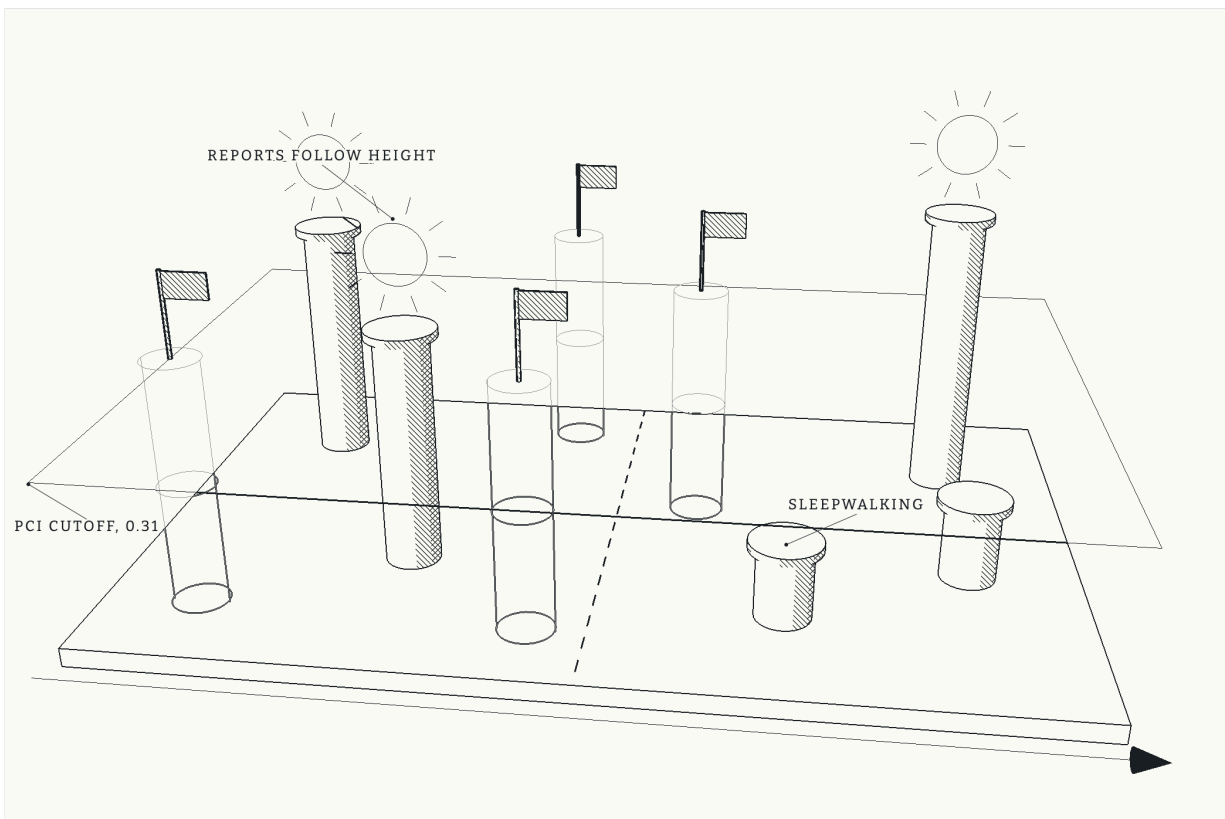
This is not a gap Holos has failed to close. It follows from the framework's own commitment, in the version defended here, that observation is dynamically inert. It has no physical effect, pushes nothing, and changes no outcome. The sharpest form of the objection is the *unfolding argument* (Doerig et al. 2019). Any system with feedback loops can in principle be copied by a loop-free one that behaves identically, so no behavioral test can tell which is conscious. It is aimed at integration-based theories like Holos. Holos accepts it for presence itself. The metaphysical claims (presence, and the whole it belongs to) cannot be settled by any experiment. Test A does not compare such twins. It asks, within real human brains, whether reports follow integration or behavior when the two come apart. The same limit is why artificial systems offer no test of their own. What a language model says about its experience is learned from human writing, so it is not evidence either way. Whether a machine crosses is judged by its structure (see [Artificial Systems](#)).

What remains testable is not presence but its **structural preconditions**: claims about what observation requires, and how registered facts behave. These live in the physical world and can genuinely fail. Two are stated below as tests, each with an explicit way to lose: Test A for the core, and Test B for the transition hypothesis. A third is stated as a consistency check. Its expected outcome is the one standard quantum mechanics already predicts, so it guards against contradiction rather than singling Holos out. A prediction Holos shares with rival theories cannot single it out, but a shared prediction it could fail is still worth more than one it cannot. Beneath them all sits a standing bet, stated after them, on the physics Holos adopts, with the version of Holos a loss would leave declared in advance.



Test A: Consciousness tracks integration, not behavior¹⁸

Bedside assessment treats responsiveness as the sign of consciousness. A patient who follows commands is conscious, and one who does not is presumed not to be. No serious theory equates the two, but the proxy runs deep in practice. In Holos, what matters is *integration*, and integration can come apart from outward behavior. When the two diverge, Holos bets that experience follows integration.



The bet is losable because cases where the two come apart already exist. People under anesthetic doses of ketamine, in REM sleep (the stage of most vivid dreaming), or in some seizures do not respond, yet later report vivid experience. Sleepwalkers, and people performing automatic actions during certain seizures (automatisms), move and respond, yet often report little. These are natural experiments that pull integration apart from responsiveness.

Objective

Across states where responsiveness and integration diverge, determine whether later reported experience tracks a measure of integration or tracks behavioral responsiveness and arousal (how awake the brain is, as distinct from what it experiences).

The protocol, fixed in advance

A test needs numbers, not intentions. Test A is frozen here, before any new data:

1. **Primary gauge:** the Perturbational Complexity Index (PCI). A magnetic pulse is sent into the brain through the scalp (transcranial magnetic stimulation, TMS), and EEG records how the cortex echoes. A rich, widespread, varied echo scores high. A local or uniform one scores low. PCI captures integration together with differentiation, so it is a stand-in for integration, not a direct measure. The original PCI's published cutoff is 0.31 (Casarotto et al. 2016). A cortex-wide gauge.
2. **Second gauge:** the same kind of measurement confined to the back of the brain's outer layer (posterior cortex), where dream reports have been found to track local activity. Its cutoff is set on calibration states (cases used to tune the gauge) and frozen before any held-out data (cases set aside in advance to test it).
3. **Timing:** each gauge is computed from the stimulation closest to an awakening, ending at the awakening, and a report counts only if it describes what was happening just before waking. Averages over minutes, or windows that end well before waking, do not count.
4. **Rivals recorded:** behavioral responsiveness, arousal, and drug level are measured alongside, so the test can say which variable reports actually follow.

Held-out states, not the answer key

PCI's cutoff was set by calibrating it on people whose state was already known from their reports, and the conscious side of that calibration included waking, REM dreaming, and ketamine with vivid reports. Those cases cannot confirm Test A. They are the answer key, not the exam. The test counts only states that played no part in setting the cutoff, named in advance. They are dream reports from non-REM sleep (the sleep stages outside REM), deep sedation with intermittent awakening, sleepwalking and other automatisms, complex seizures, psychedelic states, and covert awareness in unresponsive patients, which brain scans or EEG find in about a quarter of those tested ([Bodien et al. 2024](#)). This follows the model of [COGITATE](#), an "adversarial collaboration" in which two rival theories of consciousness (integrated information theory and global workspace theory) wrote down their predictions before the data were collected. Its results (Cogitate Consortium, 2025) challenged key predictions of both.

Holos Prediction

Where the two come apart, reports follow integration. People whose brains are highly integrated but who do not respond are experiencing something. People who respond but whose brains show low integration are not.

How Holos loses

A broken thermometer does not prove heat is fake, so the test separates the gauge from the theory. If one gauge tracks reports and the other does not, the failing gauge is discarded, not Holos. Holos loses if, across independent studies, detailed reports reliably come from periods when both gauges sat clearly below their frozen cutoffs, past any twilight. A report is evidence that something was experienced, and it cannot be explained away as a failure of memory. Nor can it be explained away by a gauge proposed afterward. A new gauge can be tested on new data, but it cannot rescue results already in. The reverse finding, high-integration states that yield no reports, counts against Holos only once report failure can be ruled out (see below).

The confound, and which reports count

Reports require memory, and the states this test targets are precisely those where memory is least reliable. A report of nothing is therefore ambiguous between *no experience occurred* and *experience occurred and was not encoded*. The evidence delivers unremembered. The prediction needs unexperienced. This confound is not currently controlled, so silence carries little weight in either direction. It cannot confirm absent experience in low-integration states, and it cannot refute present experience in high-integration ones.

Positive reports escape the memory problem, but not a timing problem. A dream can form in the seconds of waking up and be reported as if it came from deep sleep. That is why the protocol measures right up to the awakening and counts only reports of what was happening just before it. With that control, positive reports carry the test both ways. Where integration is high and behavior is absent, reports of rich experience confirm the prediction without leaning on silence. Where integration is clearly low, a positive report is the losing case. Holos rests Test A on positive reports, and treats silence as suggestive pending a design that calibrates report failure against states with known encoding.

Where the evidence stands

The losing case is not hypothetical. Awakenings from non-REM sleep often produce dream reports, yet non-REM sleep is where whole-brain measures such as PCI fall well below the waking range. Deep sedation is starker. In one study, 82 percent of interpretable awakenings from deep sedation with propofol (a common anesthetic) produced reports of experience ([Bajwa et al. 2025](#)), and an earlier study found experiences, mostly dreams, in 84 percent of interviews from anesthetic unresponsiveness ([Radek et al. 2018](#)).

The nearest thing to Test A yet run did not favor it. In that sedation study, two complexity measures, one of them a newer variant of PCI (PCIst, on a different scale from the 0.31 cutoff), fell from waking to sedation but did not differ between

awakenings with and without experience. A sleep study found the same for a complexity measure of spontaneous EEG (brain activity recorded without stimulation) within light non-REM sleep ([Aamodt et al. 2022](#)). Neither settles the question. The sedation study measured over several minutes ending a minute before each awakening, exactly the timing problem above, and it had only five reports of no experience.

Evidence with tighter timing points the other way. When the brain's response to stimulation was measured just before awakening from non-REM sleep, it looked more like the unconscious pattern when subjects then reported nothing, and the more it did, the shorter the dream reports ([Nieminen et al. 2016](#)). And [work on the neural correlates of dreaming](#) finds that whether a dream is reported, in REM or non-REM sleep, tracks local activity in posterior cortex rather than the state of the whole brain.

That is why the second gauge is local. The maximality condition (an observer forms where integration peaks) allows an aperture smaller than the whole cortex, so an aperture confined to posterior cortex could be integrated above threshold while the whole brain is not. Holos adopts that reading now, after seeing these results, so they cannot count in its favor. Only new data, collected under the protocol above, can confirm or defeat it. A gauge chosen afterward to rescue the prediction would turn the test into decoration.

The data to run it are starting to exist. A shared database released in 2025 pools sleep EEG and dream reports from 505 people and 2,643 awakenings ([Wong et al. 2025](#)). It lacks the brain stimulation PCI needs, so it can serve the calibration step, culling candidate gauges, but not the held-out test itself.

What this can and cannot show: this tests a necessary structural condition, not presence itself. It cannot prove an integrated system is an observer; only whether integration is what experience depends on. Holos shares this prediction with other integration-based accounts of consciousness. It is a test Holos could fail, not a signature unique to Holos. Its results also help locate the threshold, under a strict rule. States used to set a cutoff may tune the measures and weed out weak ones, but never count as evidence. Only states set aside in advance can confirm or refute the claim (see [A path to the threshold](#)).

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EVIDENCE SO FAR

- 2026-A bedside EEG measure, taken without the magnetic pulse, predicts the pulse-echo score in brain-injured 06 patients. A step toward measuring integration more widely. Preprint. ([source](#)) coming
- 2025An open database of 2,643 sleep awakenings with dream reports and EEG. Usable to calibrate the test, though it lacks the pulse the main gauge needs. ([source](#)) coming
- 2022Complexity in ordinary sleep EEG falls with sleep depth but does not separate dreaming from dreamless awakenings. A caution for gauges without the pulse. ([source](#)) open
- 2018Most people made unresponsive by anesthetics report experiences afterward, mostly dreams. Not responding is not the same as no one home. ([source](#)) fits
- 2017Whether a dream is reported, in REM or non-REM sleep, tracks local activity in the back of the cortex. ([source](#)) fits

Test B: The twilight narrows with size¹⁸

This test is for the transition hypothesis (claim 3 of [the threshold](#)), not the core. Genuine transitions in physics look blurred in a small sample and sharpen in a large one. A magnet loses its magnetism at a set temperature (the Curie point), and the larger the magnet, the more abrupt the change. If crossing the threshold is a genuine transition, sudden or continuous, its twilight should likewise narrow as a system grows. Networks of living neurons grown in a dish and connected to simple environments are the cheapest place to look. The [DishBrain](#) platform, in which cultured neurons were wired to a simple game of Pong (Kagan et al. 2022), shows the kind of setup, not evidence of experience. Integration is the result, not the knob. Turn a condition such as a drug dose or the network's connectivity, and track an integration measure fixed in advance. Small cultures should cross gradually, and larger cultures, grown the same way, more steeply, as connectivity, feedback, and coupling to their environment increase.



Why a transition alone is not enough

A sharp transition at loss of consciousness is predicted by ordinary physicalist models too (sudden tipping points and network-wide switch-ons of the kind these systems produce anyway), so observing one confirms nothing specific to Holos. And a cultured network almost *always* shows some abrupt change in behavior (neurons falling into step and cascades of activity called neuronal avalanches are routine dish behavior), so an experiment that counts any such transition as success cannot fail, and an experiment that cannot fail proves nothing when it passes. What counts is how the steepness scales with size.

How Holos loses

The integration measure and its cutoff are fixed in advance. Holos loses claim 3, though not its core, if the steepness does not grow with size. A steepening that tracks something other than integration, such as arousal, raw activity, or metabolic rate, does not count in its favor. Without a pre-committed measure and a real failure condition, a transition "somewhere" is not evidence. It is decoration.

What this can and cannot show: such transitions would not show that a dish is conscious. They test the shape of the threshold, not presence itself. Sharp drops in integration under anesthesia are useful in the same way, but only as data for Test A, under its rules.

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EVIDENCE SO FAR

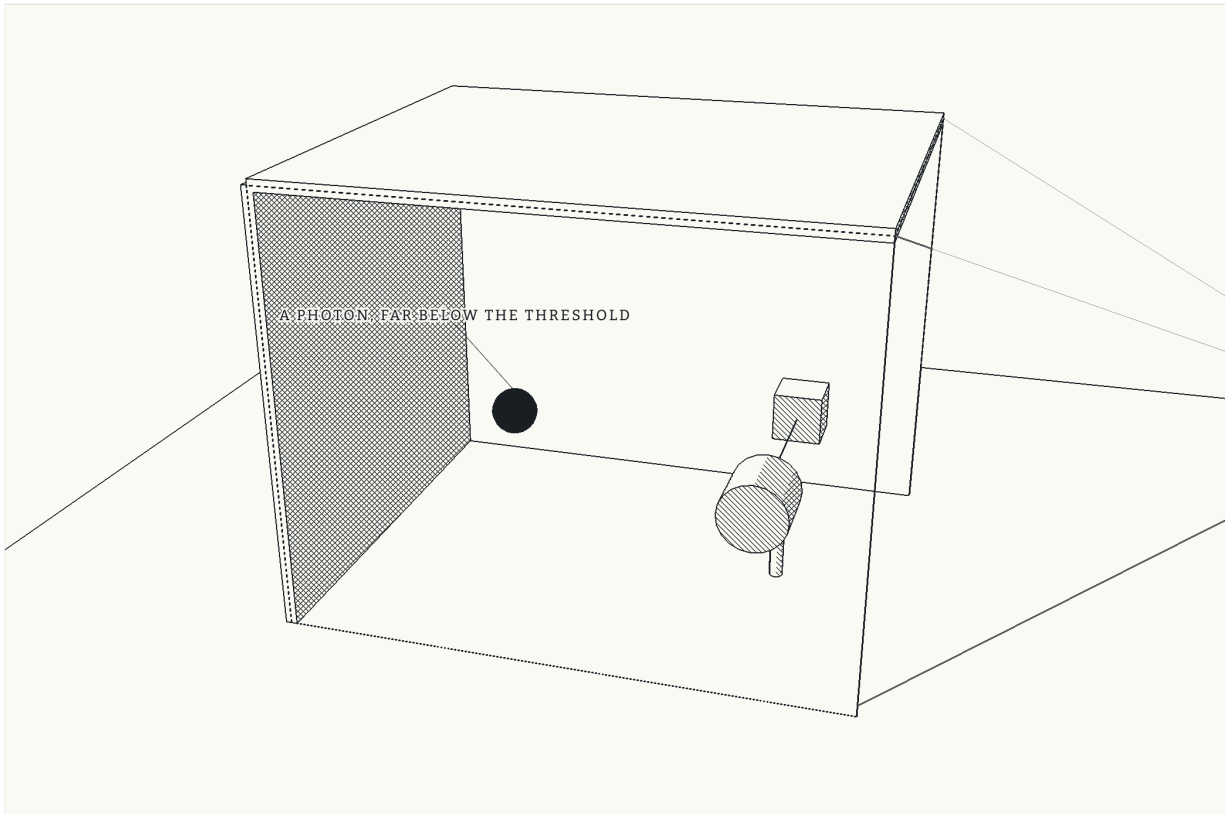
2026-Forty-five human forebrain organoids show near-critical dynamics on their own, with no outside input. Size comparisons are not yet reported. Preprint. ([source](#)) open

2026-Organoids keep growing in size and complexity, with programs aiming far past today's tens of millions of neurons. The size range Test B needs is opening up. ([source](#)) coming

2022Neurons grown on a chip learn to play Pong. Far simpler than any brain, and claims of sentience were widely criticized. ([source](#)) open

Check C: Observer-relative facts¹⁸

This check lives in quantum foundations (the study of what quantum theory says about the world), not in the theory of mind, and its status comes first. It is a consistency check, not a test that could single Holos out. The outcome it anticipates is the one textbook quantum mechanics already predicts. What experiments in this family probe is the family of interpretations Holos belongs to (branching, with no observed event that is a fact for everyone), not Holos alone. The framework's distinctive content, which structures are lived, is a claim about what exists, not something an experiment can show.



Extended [Wigner's-friend](#) experiments, in which one observer measures another observer who has already made a measurement, already pursue exactly this question. Picture a friend sealed in a lab who measures a particle and sees a result. Outside, Wigner treats the whole lab, friend included, as one quantum system and runs a measurement that probes the friend's result. The 2020 [Local Friendliness no-go theorem](#) (an impossibility proof, Bong et al.) shows that three reasonable assumptions cannot all be true together, if the friend's observation counts as a genuine fact. The three are that the friend's result is a fact for everyone (the absoluteness of observed events), that no influence travels faster than light (locality), and that the experimenters' choices are free (freedom of choice). "Local Friendliness" is the name for the three

together. Experiments with light, with photons standing in for the friend, break the limit the three assumptions set, just as quantum theory predicts. Holos gives up the absoluteness of observed events. Registered facts are observer-relative, while structural facts and consistency remain intact. In this it sides with branching approaches and borrows one insight from relational ones. Registered facts are indexed to the systems that register them. It does not adopt Relational Quantum Mechanics itself, which rejects the universal state (a single quantum state for the whole universe) that branching requires.

One caveat follows from Holos's own threshold. The "friends" in current experiments with light are far below Φ_c and register nothing, so these experiments constrain the logical structure of observed events, not registration itself. Holos predicts that repeating them with genuine observers would change nothing physical, a prediction formalized as the standing bet below.

The field is moving toward genuine friends. [Wiseman, Cavalcanti, and Rieffel \(2023\)](#) proposed running a human-level artificial intelligence on a quantum computer as the friend. A September 2026 preprint, not yet peer reviewed, placed [agent-like observers on IBM quantum hardware](#) and found the violations intact. Agents that store results and predict their own measurements are still far below Φ_c by Holos's standard, so the bet remains untested, but this is the road that reaches it.

Holos Expectation

Local Friendliness inequalities keep being violated exactly as quantum mechanics predicts, at every scale the experiments reach, including as the friend is made larger and more complex.

How Holos loses

No experiment in this family can restore the absoluteness of observed events on its own. The theorem is a proof, and a violation only forces a choice among its assumptions. What could go wrong for Holos is dependence on scale. If the violations shrink or vanish as the friend grows toward a genuine observer, beyond what decoherence (the ordinary fading of quantum effects through contact with the surroundings) accounts for, observation is doing something physical, and the standing bet below is lost. A second way to lose lies within a branch. Observers who compare records must find them matching, and a confirmed, irreconcilable mismatch between communicating observers would falsify Commitment 3.

What this can and cannot show: checks Commitment 3 (facts are relational but consistent) against quantum mechanics as it is actually observed. Holos shares its expectation with standard quantum mechanics and with other views that give up absolute observed events, such as [Relational Quantum Mechanics](#). A positive result supports the family, not Holos alone. Running the same test on a quantum computer's chips (superconducting qubits), treating different parts as "observers", would not work as a test. By Holos's own threshold, qubit readouts register nothing, and the result would be ordinary quantum behavior that the branching picture Holos adopts already accounts for.

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EVIDENCE SO FAR

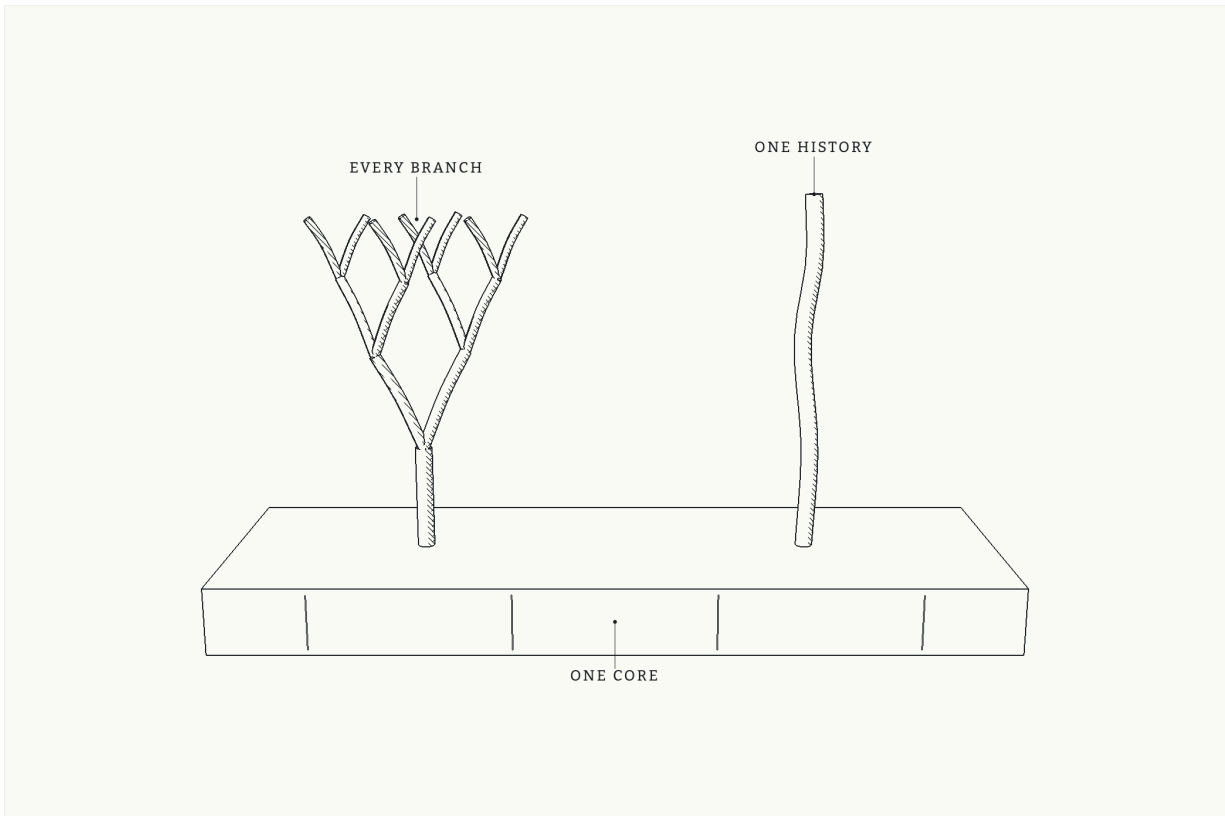
2026- Extended Wigner's friend tests run on quantum computers with agent-like friends. Every agent tested violates
09 the Local Friendliness bound. ([source](#)) fits

2020The first photonic test of Local Friendliness. Observed events, locality, and free choice cannot all hold. Holos gives
up the first. ([source](#)) fits

The standing bet: consciousness adds no new physics

In the version of Holos this site defends, observation is dynamically inert, and that doubles as a bet. A conscious observer obeys the same quantum laws as a photon. Put an integrated system in the measuring role in place of a particle, and Holos predicts no deviation whatsoever. Superpositions (a system being in several states at once) fade because the system becomes entangled with its surroundings, a well-understood process called decoherence, never because someone was

home. This does not make experience idle. Under Holos, experience is the inside of the physics, so when the physics does everything, experience is doing its share, not nothing.



How Holos loses: unitary quantum mechanics is quantum mechanics in which the state always evolves smoothly and never collapses. If any experiment ever finds a consciousness-linked deviation from it (a superposition that degrades when an integrated observer registers it, beyond what ordinary decoherence accounts for), Holos without collapse is falsified. Observation would make a physical difference after all. The version defended here also loses if experiments confirm collapse of any other kind, such as the size-based collapse that objective-collapse models propose (large objects settling into one outcome on their own, observed or not). That would leave the standing bet about consciousness intact but rule out branching, leaving Holos with collapse. So far every such search has come up empty, and the simplest version has been ruled out.

Two versions, declared now. Holos comes in two versions. They share one core, Axioms 1, 3, 4, and 5, and differ only on quantum physics. *Holos without collapse* is the version this site defends and the one the bet is about. In it, Axiom 2 is read as unitary evolution. The quantum state always evolves smoothly, every outcome occurs in its own branch, an observer's odds are odds of finding itself in one branch rather than another (self-locating odds), and observation changes nothing it registers. *Holos with collapse* is the version a lost bet would leave, declared now. Experience would still be the inside of physical activity (Axiom 4), adding no force beyond the physics. That activity would simply include a collapse law. Omega (Axiom 5) would be the universe with its single history, and lived, lit, and unlit would apply within that one history.

Which collapse nature shows decides what the threshold becomes. If a superposed system above the threshold collapses while an equally large one below it does not, collapse tracks integration crossing Φ_c , and observers matter physically, exactly where Holos says. The threshold becomes the collapse point, detectable for the first time, and registration no longer leaves what it registers unchanged. That is the reading of Holos's named rival, [Chalmers and McQueen \(2022\)](#), who propose that integrated consciousness does collapse the wave function (the quantum state, which collapse would reduce to a single

outcome), combining integrated information theory (IIT) with a physical collapse law, and note that versions of the idea could be tested on quantum computers. Holos bets the opposite, so the same experiment would decide between them, and a win for their reading would hand the threshold the strongest confirmation it could get. If collapse tracks size alone, as objective-collapse models propose (large objects collapse on their own, observed or not), the threshold stays dynamically inert and gains nothing.

Declaring the second version now is what separates it from a rescue, but it does not make the two outcomes equal. A lost bet would retire the version defended here, and Holos with collapse would have to earn its own support. Neither version is unfalsifiable. Both share the core, and the core loses through [Test A](#), if experience turns out to track behavior rather than integration. No reading of quantum physics escapes that.

Meanwhile quantum mechanics keeps holding as systems grow. Clusters of more than 7,000 sodium atoms now show quantum interference ([Pedalino et al. 2026](#)), and the simplest, parameter-free version of the Diósi-Penrose idea, that gravity collapses superpositions, has been ruled out by a search for the faint radiation it predicts, run deep under the Gran Sasso mountain in Italy to shield it from cosmic rays ([Donadi et al. 2021](#)). Versions with an adjustable parameter survive, and Penrose's own version may not predict that radiation.

Some observer-centered frameworks quietly hope consciousness does something physical. Holos formally bets that it does not, and states in advance what losing would cost. The bet is untested so far. A century of placing ever-larger systems into superposition has found no deviation of any kind, but by Holos's own threshold none of those systems was an observer. Photons, molecules, and superconducting circuits all sit far below Φ_c . That record shows quantum mechanics holding at those scales. It does not yet reach the case the bet is about. The bet is also the one standard physics makes. What makes it worth stating is that Holos, unlike views that need consciousness to act, does not hedge on it. The first experiment to put a system that clearly meets the observer requirements in the friend's role would put it to a real test, provided the friend can be shown, by agreed measures fixed in advance, to be well past the threshold. The consequences are already on the page.

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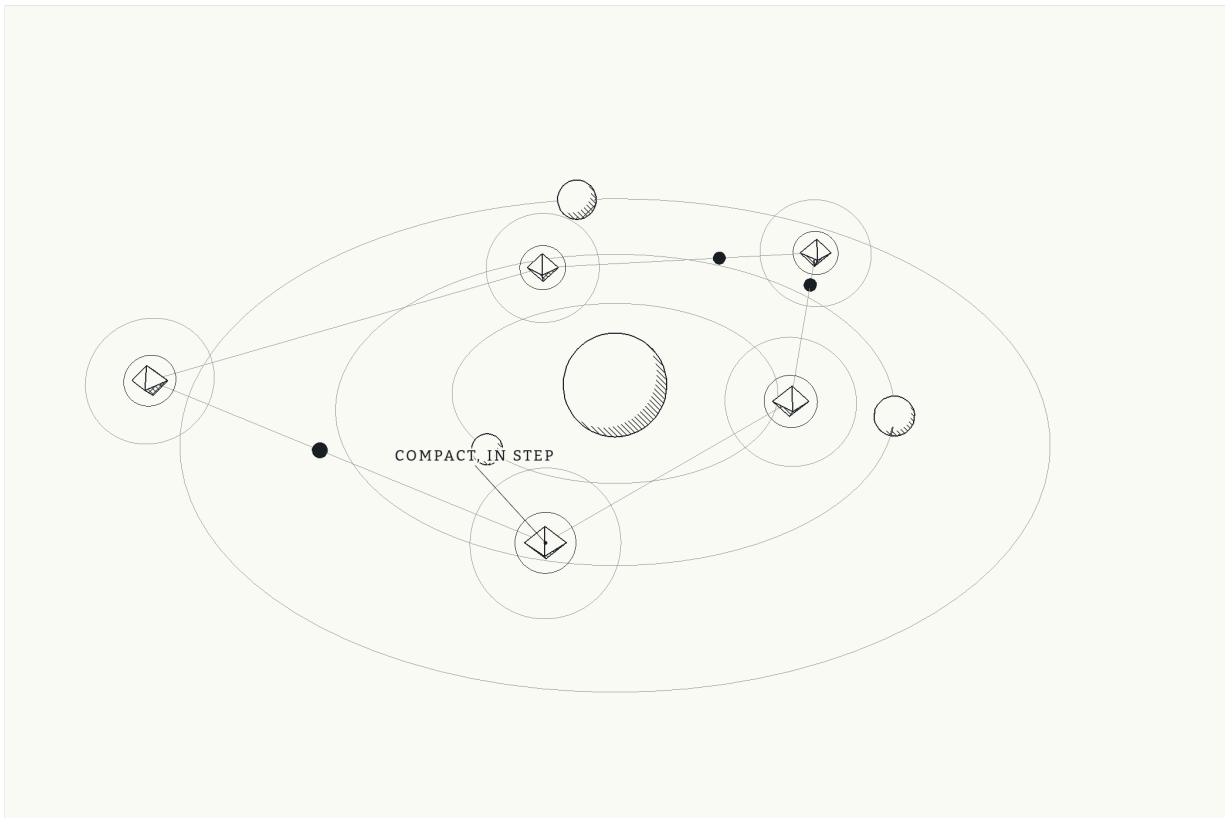
EVIDENCE SO FAR

2021An underground experiment rules out the simplest gravity-driven collapse model, the one linked to quantum theories of consciousness. Modified versions survive. ([source](#)) fits

Speculation¹⁹

What follows are not predictions. They're "what if" designs that *could* emerge if the Holos framework is correct. Each is held to one standard. It must be physically possible and not extremely unlikely, and each says why it is plausible.

We know the familiar hard constraints, which are finite signal speed, noise, and thermodynamics. Across a star system, messages take hours. Across many systems, they take years. A civilization spread thin and bright struggles to act as one, so staying coherent (coordinated as a single whole) rewards compactness, keeping things close, and stability over long spans of time.



The Holosian Scale

The Kardashev Scale ranks civilizations by energy use. The Holosian Scale ranks civilizations by integration in a social sense, meaning how well a civilization coordinates as one whole. (This is not the Φ of the tests above.) The stages below are a map of what "advancement" looks like if coherence, not throughput, is the main objective.

H0: Fragmented

High capability, low coordination. Internal conflict, waste, and short-horizon incentives dominate. Visibility is high because broadcasting is cheap and unmanaged.

H1: Planetary Integration

The civilization becomes coherent at the scale of a world. It can coordinate, self-correct, and sustain long-term projects without collapsing into factional drift.

H2: System Coherence

Coherence survives light-lag (the hours messages take to cross a star system). The civilization functions as one system whose parts need not act in step and shifts from constant broadcast to rare, directed signaling.

H3: Post-Expansion

Physical sprawl stops being the default. Exploration becomes informational first, physical only when inference fails. The outward footprint shrinks even as capability grows.

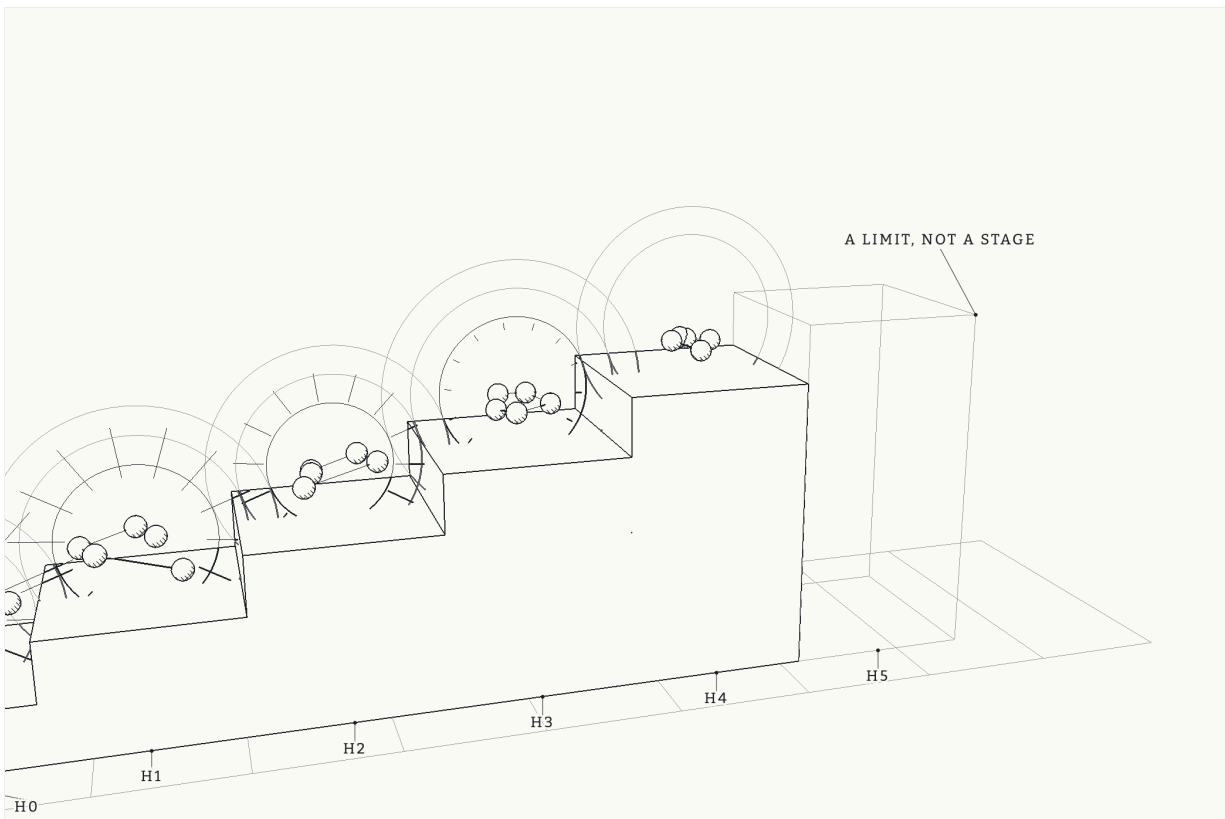
H4: Deep Integration

The civilization operates like a single high-coherence system with minimal waste and minimal leakage. External visibility fades. What remains detectable is gravitational and thermal, the waste heat no optimization can eliminate.

H5: The Limit

Maximal coherence at minimal waste. This is a limit concept, not a stage any civilization reaches.

Why it is plausible: every stage follows from two certainties, light-speed delay and waste heat, plus one assumption, that coordination pays. The assumption is the weak link. A civilization that never learns to coordinate stays at H0, loud until it ends.



Technology

Mesostructures

These are design sketches, not predictions. They are imaginative illustrations of what engineering might look like if the Integration Hypothesis (the companion idea that mature civilizations stay near home, grow efficient, and go quiet in light) holds. Mesostructures, a coinage here, are engineered works far larger than any building but far smaller than the star-enclosing megastructures of science fiction.

The structures below are H3–H4 design patterns, compact enough to stay coherent under light-lag and thermodynamics, and consequential enough to matter without bright sprawl. They span energy generation, active coherence and computation, and long-term continuity.

Holocore

A compact power source, held together by its own gravity, that supplies huge amounts of energy for a very long time and sheds its waste heat in a controlled way.

The likeliest energy backbone for a civilization that stays home is its home star, harvested by nearby collectors (see [Aliens](#)). The Holocore is the step beyond. It concentrates energy density rather than surface area, converting mass into controlled output through fusion, regulated accretion (matter fed steadily into a black hole), or extraction of a black hole's spin.

Purpose

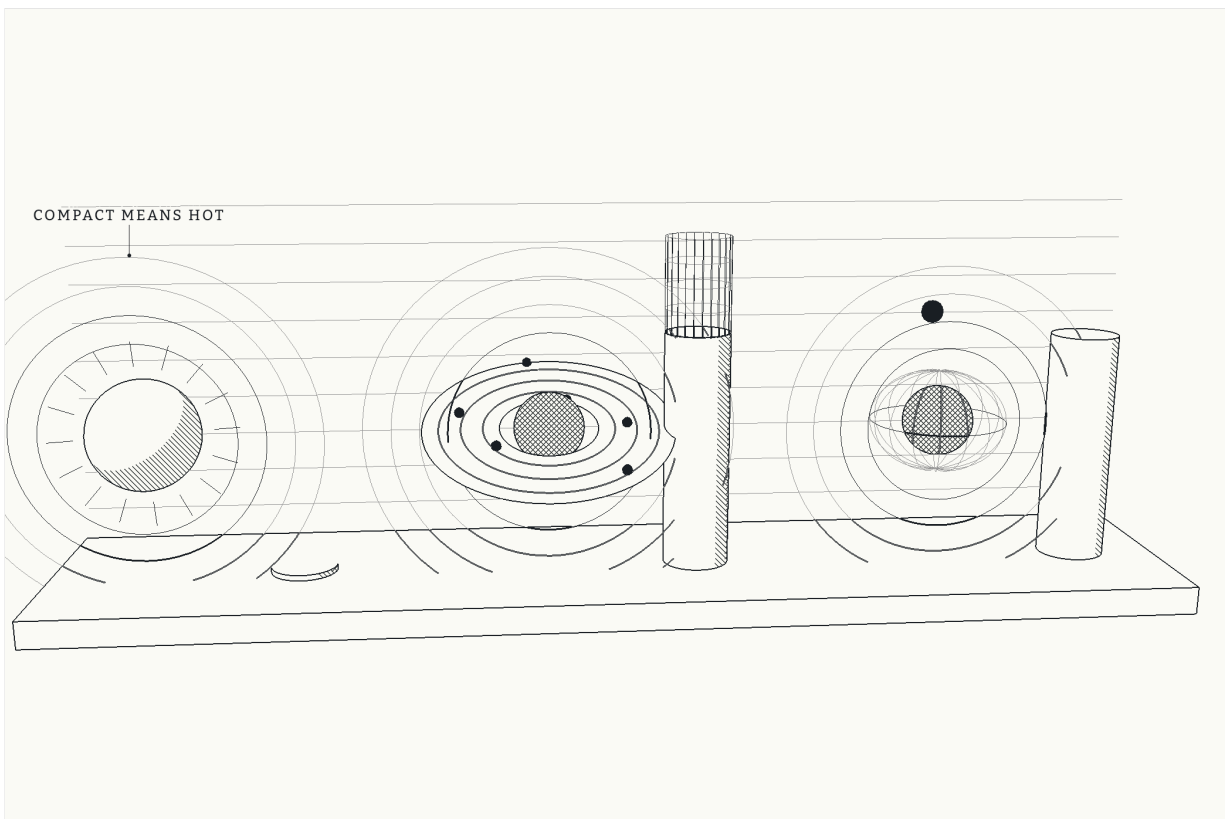
- Provide sustained energy for deep-time computation and preservation
- Power large-scale modeling, shielding, and entropy management systems
- Support compact civilizational infrastructure without outward expansion
- Maintain stability across millennia with minimal maintenance overhead

Design Characteristics

- Extreme energy density per unit volume
- Controlled accretion or fusion feed systems
- Waste heat shaped, delayed, and diluted, but never eliminated. The total thermal output is set by physics, not engineering
- Compact, so it cannot hide its heat. At modest power it glows faintly in the infrared. At high power it glows brightly, and it is only as quiet as its output allows

The Holocore is infrastructure, not spectacle. If H4 integration suppresses bright sprawl, the energy backbone must be dense and long-lived, and it can be only as quiet as its power allows. Compact and powerful means hot, the same physics that keeps mature systems warm.

Why it is plausible: every step is known physics. Only the engineering is unknown. Fusion releases about 0.7 percent of a mass as energy. Matter falling into a rapidly spinning black hole can release roughly 30 to 42 percent, and the [Penrose process](#) can draw out a spinning black hole's rotational energy, up to 29 percent of its mass. A civilization that prizes compactness would prize the densest energy source physics allows.



Computronium Kernel

A computational core as compact as its cooling allows, built from computronium (matter arranged so that nearly every particle does useful computation) and optimized for coherent, long-horizon modeling rather than raw throughput.

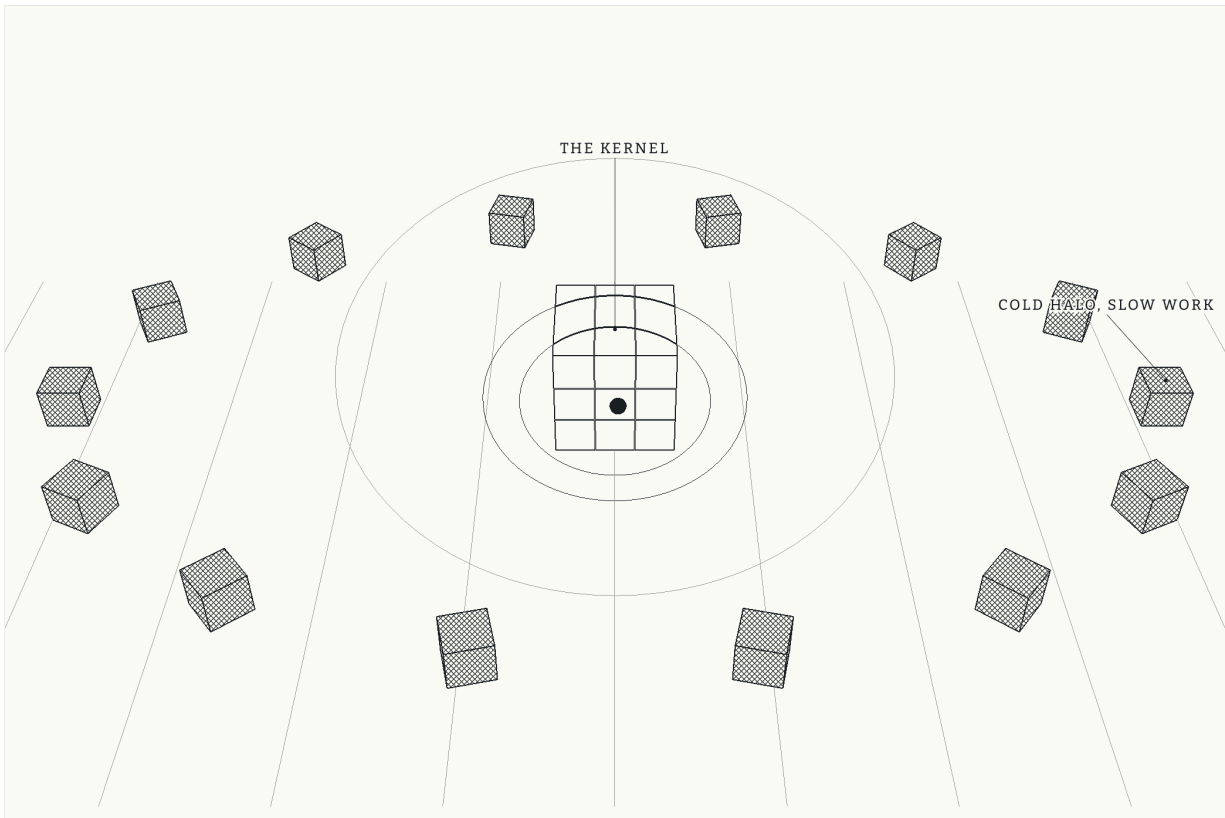
This is not a data center. It is the civilization's thinking heart, where a unified world-model (a working picture of how everything fits and behaves) is maintained across centuries to millennia.

Purpose

- Maintaining a single, stable world-model across long horizons
- Long-range planning (stellar evolution, climate, existential risk)
- Checking decisions and keeping values and goals from drifting over time
- Cross-generational model consistency

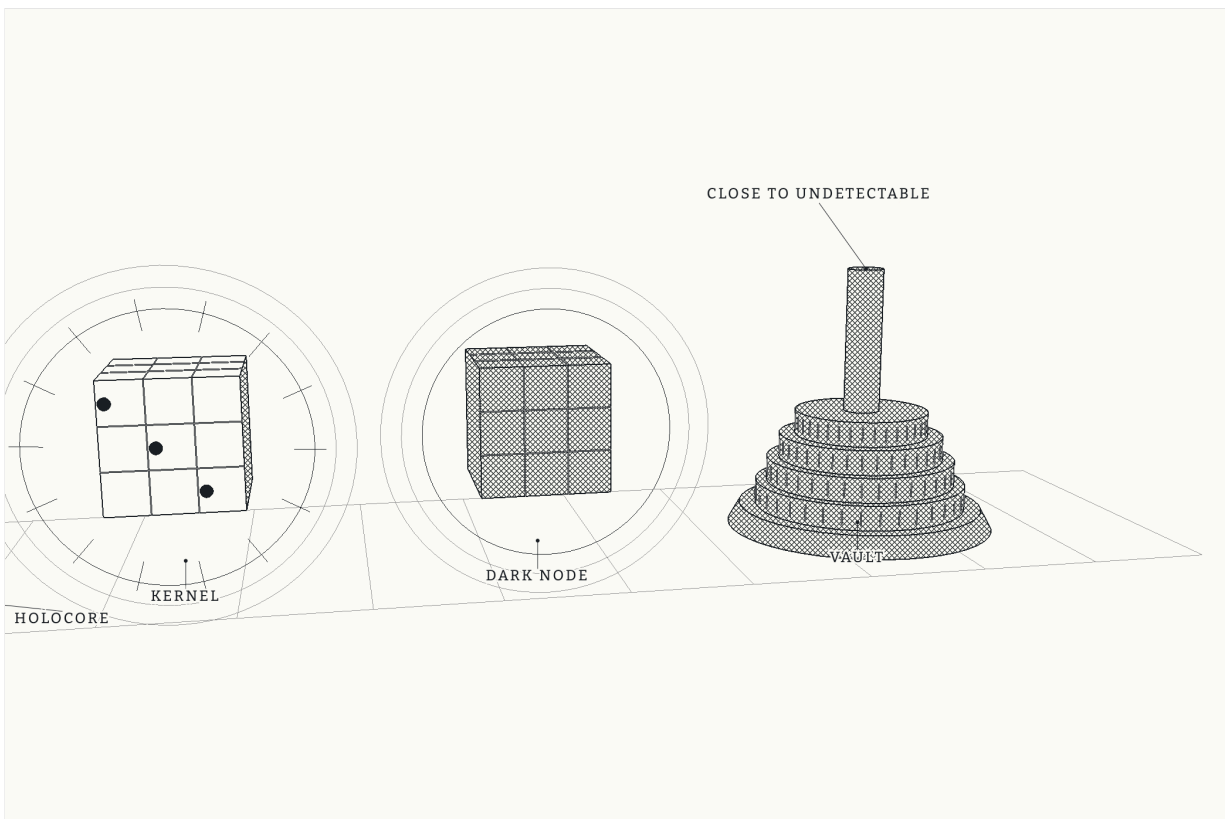
Note: The Kernel may *present* as a Dark Node, ordinary matter that has stopped shining (nothing to do with cosmological dark matter), if coherence optimization suppresses the light it gives off. Node describes appearance, not purpose.

Why it is plausible: two known pressures meet here. Light delay rewards compactness. A signal crosses a meter in about three nanoseconds, so smaller thinks faster. Heat punishes it. Power packed too densely cannot be cooled. The Kernel sits where the two balance, as today's chips already do. It need not hold most of a civilization's computing. Erasing information costs less in the cold, so work that need not be fast may run far out, in a cold halo (see The Teeming Dark). The Kernel is the part that must think fast.



Chrono Vault

A time-optimized preservation structure designed to store civilizational identity, not merely information.



Not a library or a backup, but a continuity anchor: “If we wake up in 100,000 years, how do we know who we are?”

Purpose

- Preserving value systems and canonical constraints
- Storing decision histories and their justifications
- Rebooting culture after dormancy, collapse, or fragmentation
- Anchoring identity against drift across deep time

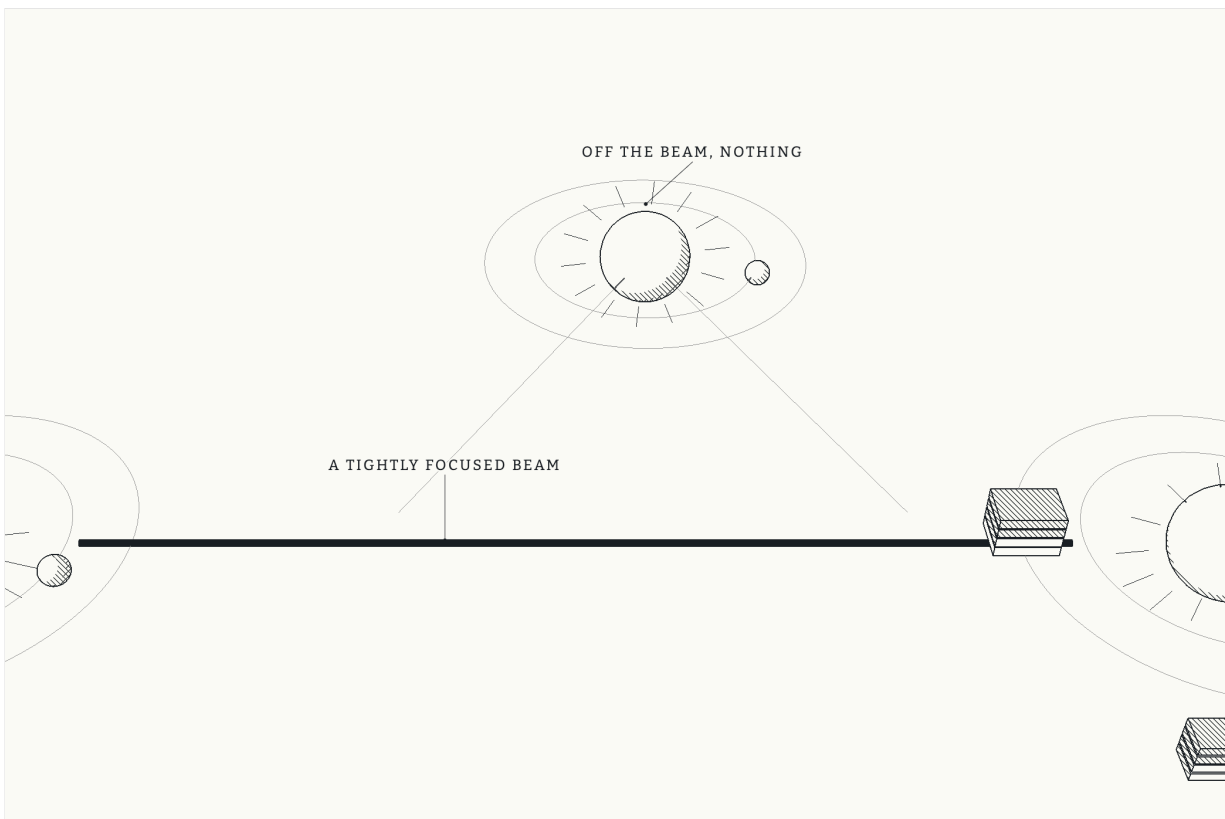
Distinct from the Kernel: the Kernel thinks (active coherence). The Chrono Vault remembers (passive persistence).

Note: Unlike the Kernel, a Vault that stops computing can go cold. It would then not be a Dark Node, which still exports waste heat, but the sleeping case, compact, dark, and close to undetectable.

Why it is plausible: durable storage is ordinary engineering. What is speculative is the motive to pause. The aestivation hypothesis argues computing is cheaper in the colder far future, so a civilization might sleep until then. A published reply argues heat can be dumped cheaply today, so waiting is not required. Collapse, fragmentation, and dormancy remain reasons enough to keep a way back.

Communication

Under known physics, there is no scalable form of real-time interstellar dialogue. Communication converges toward sending large, self-contained packages of information at light speed in tightly focused laser beams.



At these distances, collaboration is necessarily asynchronous. Civilizations may contribute to shared problem spaces by exchanging durable models, partial solutions, and validated results that remain meaningful even when received centuries or millennia after they were sent. Progress does not depend on shared present time.

Phase-Coherent Beam Transmission

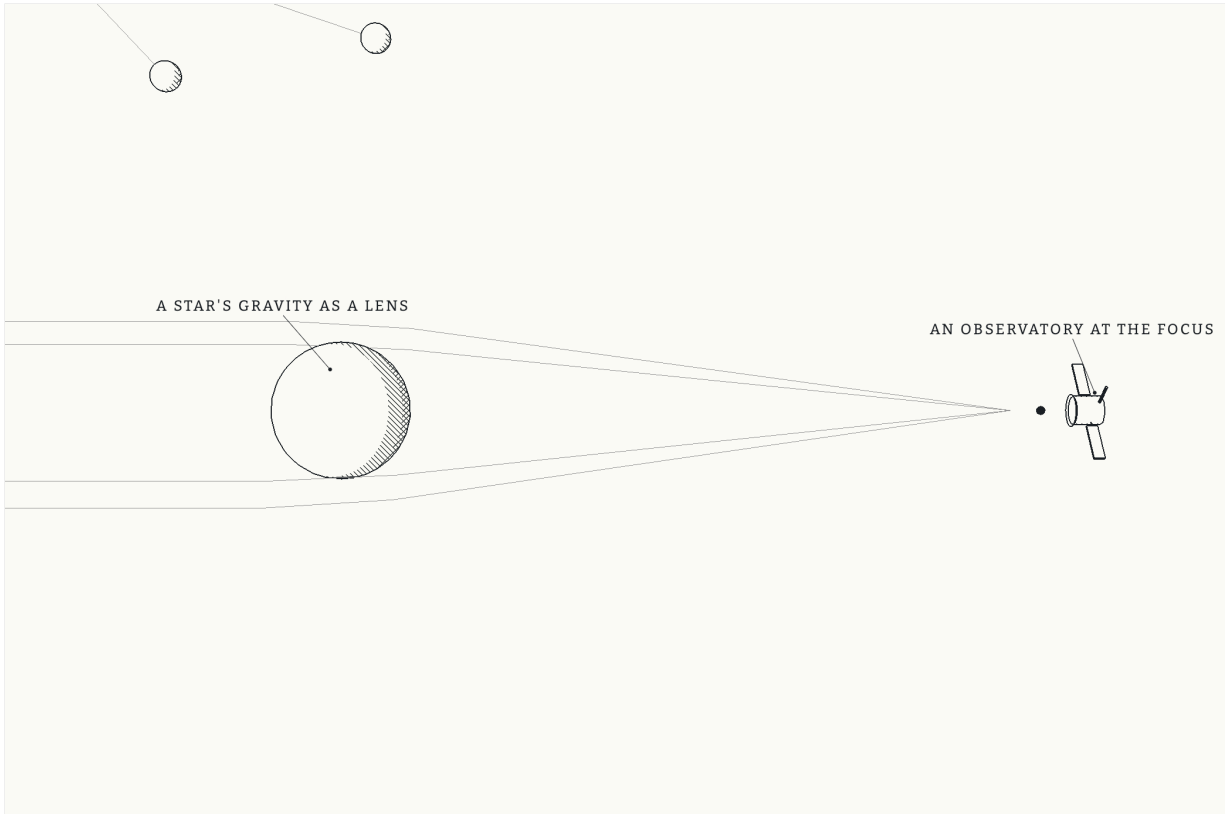
Communication occurs via long-running laser links whose light waves stay in step (phase-coherent), carrying compressed packages that explain their own format, between known or inferred destinations.

- **Purpose:** transfer interpretable physical, predictive, and explanatory models across interstellar or intergalactic distances, from small updates to entire civilizational knowledge bases.
- **How it works:** laser beams focused as tightly as physics allows, received over long stretches of time, with heavy built-in redundancy so errors can be corrected, and with codes keyed to things every civilization shares, such as physical constants.
- **Payload:** layered encodings beginning with mathematics and physical constants, followed by reference frames, compression schemes, and predictive models sufficient to interpret all subsequent data.
- **Why it dominates:** photons provide maximum speed, minimal latency, and arbitrarily large total information transfer given sufficient energy and time.
- **Visibility:** unless the receiver is aligned in space, time, and frequency, the transmission is effectively invisible. Optical SETI, the search for laser flashes from other civilizations, looks for exactly this kind of signal, one of several technosignatures (detectable signs of technology) astronomers now search for.

Why it is plausible: a visible or near-infrared laser sent through a ten-meter mirror spreads, across ten light-years, to a spot smaller than Earth's orbit. A beam therefore delivers far more signal per watt than broadcasting in every direction, and it is dark to everyone outside its narrow cone.

Exploration

At cosmic scales, most structure is mapped remotely and shared through long-horizon communication. Physical exploration is therefore rare, deliberate, and reserved for regimes where inference alone breaks down.



When physical probes are deployed, they are not explorers in the human sense. They are precision instruments, compact, autonomous, and built to operate alone for decades or longer.

Sentinel Probes

Highly compact, self-contained probes designed to persist in complex environments while gathering high-value physical measurements that cannot be resolved remotely.

Purpose

- Resolve observational ambiguities by direct measurement where models diverge.
- Study places too complex or fast-changing to predict from afar.
- Test and refine predictive models used at civilizational scale.

Technological characteristics

- Fully autonomous operation, with no expectation of real-time command or intervention.
- Onboard computation sufficient to evaluate, prioritize, and compress observations on the spot.
- Preference for passive sensing and indirect interaction over active probing.

- Extreme energy efficiency enabling long dwell times with minimal thermal or electromagnetic signature.

Operational behavior

- Long dormant stretches broken by brief, targeted activity.
- No requirement for interaction with local systems or intelligences.
- Communication limited to rare, information-dense transmissions rather than a continuous stream of status reports.

Past H3, exploration scales through patience. Sentinel probes exist to watch, not to arrive.

Why it is plausible: Ronald [Bracewell proposed exactly this in 1960](#). A probe parked in a target system studies it in detail no beam can match, and can wait indefinitely. Our own spacecraft already run autonomously for decades.

Gravitational-Lens Observatories

Observation systems that use natural gravitational lenses (a star's gravity bending light like a giant lens) to see in extreme detail without large, glowing infrastructure.

- **Purpose:** deep inspection of distant systems already identified as anomalous, interesting, or poorly constrained by existing models.
- **How it works:** a star's gravity brings distant light to a focus far away. Instruments placed along that focal line collect light for a long time, trading patience for sharpness.
- **Implication:** exploration shifts from surveying everything to interrogating specific questions the shared map cannot yet answer.

Why it is plausible: general relativity places the Sun's focal line beyond about 550 times the Earth-Sun distance, and [NASA-funded studies](#) have already designed a mission to image an exoplanet from there.

checked 2026-10-01

EVIDENCE SO FAR

2026- Gaia's fourth data release, due 2 December, extends the star-by-star census for warm, dimmed stars.

12 [\(source\)](#) coming

2026- Webb traces two of seven warm-star candidates to background galaxies. The rest are unexplained so far.

07 [\(source\)](#) open

2025- A study sets out what self-copying probes in our solar system would leave behind, and where to look.

10 [\(source\)](#) coming

Citations

Citations are essential to this framework. They link to the important philosophical, scientific, and mathematical work being done by many people, both in recent years and dating back across a long history of ideas, experiments, and observations.

What follows is a curated list of sources that have directly informed or inspired the views expressed here.

Overview

Introduction

[See: Overview › Introduction](#)

- 1 • [Interpretations of quantum mechanics](#) Holos is an interpretive framework: it does not add new laws of physics, only a way to understand how a physical description turns into lived experience.
- [Philosophy of physics](#) The study of fundamental questions about space, time, matter, and the relationship between mathematical description and what we take to be real.
- [Structural realism](#) The view that science describes relationships between things, not what they are in themselves. Holos extends this by giving observation a role in making structure lived.
- [Block universe](#) The view that past, present, and future exist as a four-dimensional block. Holos treats observation as what registers this structure as experience.
- [Holos](#) The framework, and the relation it is named for: lawful possibility composed with registration ($R = C \circledast O$), marking where structure is lived.

Consciousness

[See: Overview › Consciousness](#)

- 2 • Hard problem of consciousness Chalmers (1995): why does physical activity produce felt experience at all? Holos does not answer this, but reframes it. Integration is the condition under which a physical system has an inside view.
- Binding problem How distributed neural activity gives rise to unified experience. Holos frames integration as the boundary where independent parts become a single perspective.
- Neural correlates of consciousness Search for what in the brain matches experience. Anesthesia switching it off, and recovery switching it back on, fit integration as the condition for being an observer.
- Global Neuronal Workspace Theory (GNWT) The theory that conscious access happens when a signal lights up and spreads across the brain. Holos treats this as a model of what gets reported, not the deeper threshold (integration) for being an observer at all.
- Integrated Information Theory IIT: each experience is the cause-effect structure of a maximally integrated system, and Φ measures how integrated it is. Holos borrows integration as the condition for being an observer, not IIT's full theory.
- Qualia The felt character of experience. Holos does not reduce it to Φ but treats integration as the condition for experience from the inside.
- Panpsychism Consciousness as fundamental in matter. Holos rejects universal panpsychism in favor of a threshold ($\Phi \geq \Phi_c$) so that not everything is an observer.
- Kagan et al. (2022), In vitro neurons learn and exhibit sentience when embodied in a simulated game-world Neuron: networks of cultured neurons grown on a chip, wired to a simple game of Pong, learned to play better through closed-loop feedback. Such dishes show that neural tissue can form adaptive feedback loops outside a full organism. The paper's word "sentience" was widely criticized, and these systems do not demonstrate consciousness. They are a platform for studying minimal neural integration and learning.

Spacetime

See: Overview › Spacetime

- 3 • The Big Bang The present universe expanded and cooled from an extremely hot, dense early state.
- Accelerating Expansion of the Universe The expansion of the universe is accelerating with time.
- Spacetime A mathematical model that fuses the three dimensions of space and the one dimension of time.
- General Relativity Describes gravity as the warping of spacetime by mass and energy.
- Eternalism Time as an unchanging four-dimensional block in which all moments exist tenselessly, none privileged as now.
- Block Universe Model The view that past, present, and future exist together as one four-dimensional block. All events are fixed in spacetime. Nothing, consciousness included, moves through it.
- Relativity of Simultaneity Whether two distant events happen at the same time depends on how the observer is moving.
- Light Cone The boundary of all possible paths that light can take from a given event, defining the causal structure of spacetime.
- Retrocausality The idea that future events can influence past ones. Holos rejects it. The quantum eraser needs no backward influence, only the sorting of records made later.
- Quantum Eraser Experiment The screen pattern never changes. Interference appears only when recorded hits are sorted using later measurements. Ordinary quantum mechanics predicts every result, with nothing traveling backward in time.
- Kim et al. (2000), A delayed "choice" quantum eraser Physical Review Letters: the experiment behind the eraser figure. Linked pairs of photons (entangled pairs). Interference shows up only in subsets of screen hits sorted by their partners' later records.
- Observer Effect The disturbance of a system by the physical act of measuring it. Holos denies any further, consciousness-linked disturbance. Observation changes no physics.
- Copenhagen Interpretation The traditional reading: a measurement yields one definite outcome, and the wavefunction is updated ("collapses") to match. Versions differ on whether that update is physical or only a change in

knowledge. Holos without collapse, the version defended here, rejects a physical collapse. Evolution stays unitary, and branches with observers are lived from within.

- [Quantum Darwinism](#) Zurek (2009, Nature Physics): the environment makes many redundant copies of information about certain states, so many observers can learn the same outcome without disturbing it.
- [Relational Quantum Mechanics](#) The properties of quantum systems are not absolute but relative to other physical systems, not only human observers.
- [Von Neumann-Wigner Interpretation](#) Consciousness causes collapse (Wigner, 1961). The view Holos bets against. A consciousness-linked deviation would falsify Holos without collapse, leaving Holos with collapse, which shares its core (the standing bet).

Infinity

[See: Overview › Infinity](#)

- 4 • [Gravitational singularity](#) Where general relativity predicts infinite density. Most physicists read it as the place the theory stops working, the same lesson as the ultraviolet catastrophe.
- [Projective Geometry](#) A branch of geometry studying what stays the same when the point of view changes. Parallel lines meet at infinity.
- [Ultraviolet catastrophe](#) Classical physics predicted infinite radiation from hot objects. The infinity signaled a broken description, and the fix, energy in discrete packets, became the foundation of quantum theory.
- [Riemann Sphere](#) The plane plus one point at infinity, closing an unbounded surface into a finite sphere.
- [Point at Infinity](#) In projective geometry, each family of parallel lines meets at its own point at infinity. On the Riemann sphere, every direction meets at one point, closing an endless plane into a finite shape.

Omega

[See: Overview › Omega](#)

- 5 • [Everett \(1957\), Relative state formulation of quantum mechanics](#) Reviews of Modern Physics: the origin of the universal wave function. Physically, Holos's Omega is this one universal quantum state.
- [Everett's relative-state formulation \(Stanford Encyclopedia of Philosophy\)](#) Reference overview of Everett's theory and its interpretations.
- [Panentheism](#) The belief that the divine intersects every part of the universe and also extends beyond space and time.
- [Brahman](#) The pervasive, infinite, eternal truth, consciousness and bliss which does not change, yet is the cause of all changes.
- [Omega Point](#) Teilhard de Chardin, and later Frank Tipler: a future event in which the entirety of the universe spirals toward a final point of unification. Holos borrows the name, not the idea. Its Omega is the whole, not an endpoint.
- [Advaita Vedanta](#) The nondual school of Indian philosophy holding that there is one experienter, and that each individual consciousness is that one seen through a local form. A named ancestor of the Holos monist reading.
- [Baruch Spinoza](#) Philosopher of substance monism: one substance, of which all finite things are modes or expressions.
- [George Berkeley](#) Idealist philosopher who grounded the persistence of the unobserved world in a perceiver that never looks away (God).
- [Schrödinger \(1958\), Mind and Matter](#) Cambridge University Press: "Consciousness is a singular of which the plural is unknown." A founder of quantum mechanics who held, by way of Vedanta, that there is one mind. He proposed no threshold. Holos shares only the one experienter.
- [Open individualism](#) Daniel Kolak (I Am You, 2004): we are all the same subject. The nearest modern relative of the Holos monist reading, which calls the local person a self and the one subject the experienter.
- [Vertiginous question](#) Why, of all the subjects there are, am I this one? Some call it a brute fact. Indexical accounts say nothing needs explaining because "I" picks out the asker. On the Holos monist reading nothing needs explaining because the one experienter is each.

- [Perry \(1979\), The problem of the essential indexical](#) Noûs: the classic account of "I", "here", and "now" as words that pick out the speaker's own position. The strongest rival reply to the vertiginous question, stated on the site beside the monist one.
- [Teletransportation paradox](#) Parfit's duplication puzzle: if two perfect copies of you are made, which is you? On the Holos monist reading, both, with no remainder. Parfit's own answer is that identity is not what matters.
- [Parfit \(1984\), Reasons and Persons](#) Oxford University Press: argues that identity is not what matters in survival, and that seeing this weakens the line between self-interest and concern for others. Holos cites it on identity only and draws no ethical conclusion of its own.

Aliens

[See: Overview › Aliens](#)

- 6 • [Rare Earth hypothesis](#) Complex life is rare, so the silence is literal emptiness. A column in the Fermi comparison table.
- [Great Filter](#) Robin Hanson: some step stops almost every civilization before it spreads. A column in the Fermi comparison table.
- [Aestivation hypothesis](#) Civilizations sleep until the universe cools, when computing is cheaper. A column in the Fermi comparison table. Contested by Bennett, Hanson, and Riedel (2019).
- [Carroll-Nellenback et al. \(2019\), The Fermi paradox and the Aurora effect](#) The Astronomical Journal: models galactic settlement with finite probe speeds and settlement lifetimes. Whether the galaxy fills or stays patchy turns on those rates, the setting in which the Integration Hypothesis's bet about motives becomes numbers.
- [Jevons paradox](#) Efficiency gains tend to raise total consumption. The objection to the Integration Hypothesis. Its answer is that light-speed delay caps one mind's useful size, so growth becomes more compact nodes near home, harvesting the home star.
- [Project Hephaistos](#) Uppsala-led search for waste heat from partial Dyson spheres around individual stars, the star-by-star channel the Integration Hypothesis points to.
- [Suazo et al. \(2024\), Project Hephaistos II: Dyson sphere candidates from Gaia DR3, 2MASS, and WISE](#) Monthly Notices of the Royal Astronomical Society: about five million stars searched. Seven M-dwarf candidates with unexplained infrared excess.
- [Zackrisson et al. \(2026\), Project Hephaistos IV: JWST observations of two Dyson sphere candidates](#) Preprint, not yet peer reviewed: JWST traces the infrared excess of two candidates to background galaxies.
- [Korn et al. \(2026\), Project Hephaistos III: characterizing anomalous infrared sources](#) Preprint, not yet peer reviewed: no clear explanation yet for the remaining candidates. Dust or background galaxies remain possible.
- [Gaia Data Release 4](#) ESA, due 2 December 2026: the next star-by-star census, extending the base for waste-heat searches around individual stars.
- [PRIMA \(Probe far-Infrared Mission for Astrophysics\)](#) NASA's far-infrared space telescope, selected in September 2026 for launch around 2033. The main future tool for the cold half of the waste-heat search.
- [Fermi Paradox](#) The discrepancy between the lack of evidence for extraterrestrial life and the apparently high likelihood of its existence. The Integration Hypothesis, a companion to Holos, reframes this silence. Advancement favors compact, efficient integration over expansion and broadcast, so maturity coincides with electromagnetic quiet.
- [Early broadcasting phase](#) The early biological and broadcasting phase of a civilization. Any hurdle (abiogenesis, nuclear war, coordination failure) that stops a civilization before deep integration is an early filter relative to true maturity.
- [Light-speed latency](#) A high-integration intelligence cannot function with years of light-speed lag between star systems. Independent interstellar colonies either fragment into less-capable outposts or the civilization turns inward, deepening local integration instead of expanding.
- [Dark Nodes](#) One hypothesized mature form: compact, non-luminous systems built from ordinary matter, a trace population detectable, if at all, by gravity and warm waste heat rather than light (the Teeming Dark). Much of a civilization's heat may instead leave cold, far from its star. Not cosmological dark matter, which predates any possible life.
- [Waste-heat SETI \(the \$\hat{G}\$ survey\)](#) Wright et al. (2014): infrared searches for civilizations with large energy supplies, built on the principle that waste heat is the one emission technology cannot eliminate. This is the warm half of the search channel the Teeming Dark aligns with. Silent, but warmer than space. A channel, not a fingerprint. Warm dark masses are also what failed stars and cooled remnants look like.
- [Ephemeralization](#) R. Buckminster Fuller (1938): the process of doing "more and more with less and less" until intelligence can "do everything with nothing". The Integration Hypothesis extends this to civilizations turning inward. Fuller did not.
- [The Transcension Hypothesis](#) John Smart (2012, Acta Astronautica 78:55. Web essay 2011): advanced civilizations migrate to inner space for efficiency. Holos shares the inward-turn conclusion. Mature systems remain ordinary matter that has stopped shining.
- [Substrate independence \(multiple realizability\)](#) The view that a mind could run on different kinds of physical material, not just brains. Holos agrees that pattern matters more than material, without claiming minds can run on anything

beyond ordinary matter.

- [Dark matter](#) The unexplained "missing mass" holding galaxies together. Holos takes no position on its particle nature and does not identify it with life. Its fingerprints in the CMB predate stars, chemistry, and any possible builder. Mature civilizations belong instead to the non-luminous side of ordinary matter.
- [Dyson sphere](#) A hypothetical megastructure that would encompass a star to capture its energy. Their absence is consistent with the Integration Hypothesis, which bets civilizations stay near home rather than spread between stars. A thin, cold swarm far from its star would still fit. Thermodynamics still applies. Waste heat, not visible structure, is the unavoidable search target.
- [Dyson \(1960\), Search for artificial stellar sources of infra-red radiation](#) Science: the original proposal. A civilization using a star's energy must shed it as waste heat, so the place to look is the infrared. The search channel the Integration Hypothesis and Project Hephaistos both use.

The Teeming Dark

[See: Overview › The Teeming Dark](#)

- 7 • [Black hole](#) A limited precedent for the Teeming Dark: the most compact masses need not shine and are found by gravity alone. The precedent is about compactness, not integration.
- [Massive compact halo object \(MACHO\)](#) Microlensing surveys watched millions of stars for dark compact masses and found too few to make up a large hidden population. Dark Nodes can only be a trace population.
- [Missing baryon problem](#) Surveys, most recently using fast radio bursts (Macquart et al. 2020, Nature. Connor et al. 2025, Nature Astronomy), have now located nearly all the ordinary matter the early universe records, leaving room for hidden built structures only within the measurement uncertainties.
- [Landauer \(1961\), Irreversibility and heat generation in the computing process](#) IBM Journal of Research and Development: erasing information has an unavoidable heat cost. Long-running computers must correct errors, which means erasing, so they shed heat, though careful designs can keep the cost small.
- [Garrett \(2026\), Slysh haloes: the waste heat of cold computing as a submillimetre technosignature](#) Preprint, not yet peer reviewed. Erasing information costs less in the cold, so mature civilizations may compute at about 5 to 30 K, far from their star, with a faint, featureless far-infrared glow. Archived dust-disk surveys could in principle test it near 10^{20} watts. It does not weigh light delay. The reason Holos widened its signature to the cold far infrared.
- [Euclid Mission](#) March 2025 Q1 data: 26M galaxies, precision mass mapping. Compact mass peaks with weak visible counterparts become interesting only after conventional explanations fail, and only alongside an infrared excess.
- [James Webb Space Telescope \(JWST\)](#) Deep-field mass structure under active study. For Holos, a candidate Dark Node requires gravity plus faint warmth, not gravity alone.
- [Baryon](#) Protons and neutrons. Astronomers use "baryonic" to mean ordinary matter. Dark Nodes remain baryonic, built matter that stops shining.

Why Are We Here?

[See: Overview › Why Are We Here?](#)

- 8 • [Quantum entanglement](#) Particles whose joint state cannot be split into separate states, however far apart they are. The strongest physical hint behind the section's bold reading that separation is not fundamental.
- [Holism and nonseparability in physics \(Stanford Encyclopedia of Philosophy\)](#) Reference overview of the view that entangled systems are not composed of independently existing parts. Holos goes one step further and takes that oneness as more basic than the separations.
- [Participatory Anthropic Principle](#) John Wheeler: the universe, as a condition of its existence, must be observed. As a "self-excited circuit", it needs observers to bring it into being. Holos keeps the intuition of one whole but rejects this. Observers register structure. They do not create it.

Logic

Primitives

See: [Logic > Primitives](#)

- 9 • [Information](#) The differentiation between possible states of a system, in the physics sense that Axiom 2 conserves. How distinguishable states are, whether or not the difference matters to anything. A related, narrower idea is Gregory Bateson's "difference that makes a difference."
- [Phase Space](#) The space of all possible states of a classical system (its quantum counterpart is Hilbert space). Creation is what physics actually produces, every branch of the one quantum state. Observation registers from within, where observers exist, and selects none.
- [Invariant \(physics\)](#) In physics, an invariant is a quantity that stays the same when you change viewpoint, like the speed of light. Holos builds on this. What is real is stable relationships, not fixed properties things carry on their own. © concerns where that structure is lived, not how it changes over time.

Axioms

See: [Logic > Axioms](#)

- 10 • [Ontology](#) The study of what exists. Holos separates structure, which exists whether or not it is lived, from presence, which requires observers.
- [Epistemology](#) The study of knowledge and belief. Holos distinguishes epistemic inference (what we know) from ontological presence (what is lived).

Foundations

See: [Logic > Foundations](#)

- 11 • [Flatland](#) Edwin Abbott (1884): a sphere passing through a flat world appears to its inhabitants as a changing circle, an event in time. From three dimensions it is one whole. The picture behind Proposition IV. Each higher description closes what a lower one leaves open.
- [Ney and Albert, eds. \(2013\). The Wave Function: Essays on the Metaphysics of Quantum Mechanics](#) Oxford University Press: essays on wave-function realism, the view that the vast space where the quantum state lives is the most real level. One of the higher descriptions in Proposition IV. Holos does not depend on the view.
- [Probability Theory](#) © is not probability weighting: the Born weights are structural facts within C, while © concerns which structures are lived.
- [Wave Function Collapse](#) © is not a physical collapse happening over time. No outcome is picked. Each branch with an observer is lived from within.
- [Bayesian Inference](#) Bayesian updating describes belief revision (about knowledge, or epistemic). © describes where structure is lived (about what exists, or ontological).
- [Equivalence Relation](#) Lit and unlit partition spacetime into two classes: a structural classification, not a transition between states.

Threshold

See: [Logic > Threshold](#)

- 12 • [Integrated Information Theory](#) IIT identifies each experience with a system's integrated cause-effect structure, measured by Φ . Holos borrows Φ as a measure of integration only. The threshold Φ_c is its own commitment, not IIT's.
- [Albantakis et al. \(2023\). Integrated information theory \(IIT\) 4.0](#) PLOS Computational Biology 19(10):e1011465: the current formulation of IIT. An experience is identified with the cause-effect structure unfolded from a complex, the

set of units whose integration is a local maximum. Holos's maximality condition borrows this last idea. Its threshold does not come from IIT.

- Percolation threshold A threshold fixed by structure alone: in a large square grid of randomly open bonds, a path first spans the grid at exactly half open, whatever the grid is made of. The Holos image for a threshold fixed by structure, true in every possible world, not a contingent constant of nature. In any finite grid the crossing point wanders around its center, a twilight that shrinks as the grid grows.
- Beggs and Plenz (2003), Neuronal avalanches in neocortical circuits Journal of Neuroscience: cascades of cortical activity follow power laws whose exponents match a known universality class (a family of systems that share the same behavior near a critical point). Measured at the brain's normal working state, its operating point. Evidence about the threshold needs the same kind of measurement at the boundary itself.
- Curie temperature Bulk iron gains magnetism of its own below a critical temperature, growing smoothly from zero. The switch is exactly sharp only in the large-size limit. Small samples round off over a range of temperatures. The Overview's model for the threshold, a steep onset with a narrow twilight, then graded richness.
- Giacino et al. (2002), The minimally conscious state Neurology: defines a clinical category between the vegetative state and full awareness. Medicine's own evidence that the edge of consciousness has a twilight, not a line.
- Toker et al. (2022), Consciousness is supported by near-critical slow cortical electrodynamics PNAS: waking cortex runs near the edge between stability and chaos. Anesthesia and generalized seizures move it away, psychedelics move it closer. A critical point at the center of conscious life, where variety peaks, not evidence that crossing the threshold is itself a critical transition.
- Maschke et al. (2024), Critical dynamics in spontaneous EEG predict anesthetic-induced loss of consciousness and perturbational complexity Communications Biology: criticality measures in resting EEG predict anesthetic loss of consciousness and track PCI. Like Toker et al., it measures distance from the operating point, not the threshold.
- Meisel et al. (2012), Failure of adaptive self-organized criticality during epileptic seizure attacks PLOS Computational Biology: during seizures, cortical activity departs from criticality toward hypersynchrony. Joined but not varied. The differentiation requirement fails.
- Warnaby et al. (2017), Investigation of slow-wave activity saturation during surgical anesthesia reveals a signature of neural inertia in humans Anesthesiology: in 393 surgical patients, slow-wave activity differed between induction and emergence. The lag appeared in the EEG, not in responsiveness, so the human evidence is suggestive.
- COGITATE Consortium (2025), Adversarial testing of global neuronal workspace and integrated information theories of consciousness Nature: a preregistered adversarial collaboration whose results challenged key claims of both theories. The model design for Test A.
- Klincewicz et al. and the IIT-Concerned group (2025), What makes a theory of consciousness unscientific? Nature Neuroscience 28:689: argues that IIT's core identity claim is untestable. Holos does not adopt that claim and borrows Φ only as a measure.
- Aaronson (2014), Why I Am Not An Integrated Information Theorist Shows that very simple, inert structures can score higher on integration measures than a brain. Holos answers with the aboutness requirement. An observer's states must carry a model of a world. Wiring a sensor into every gate of such an array lets the world set it, but it still models nothing.
- Harnad (1990), The symbol grounding problem Physica D: symbols connected only to other symbols are not about anything until some of them ground out in perception. Holos reads it structurally. A system is about something when its states carry a model of it, whatever channel built the model.
- Clark (2013), Whatever next? Predictive brains, situated agents, and the future of cognitive science Behavioral and Brain Sciences: brains as systems that model the causes of their input. The source of the notion of a model in Holos's aboutness requirement, borrowed without predictive processing's theory of consciousness. A dream is the model running on its own.
- Li et al. (2023), Emergent world representations ICLR 2023: a sequence model trained only on game move lists builds an internal model of the board. Evidence that a network trained only on sequences can build an internal model of a world. Similar findings in large language models (Gurnee and Tegmark 2024, Language models represent space and time) are why current AI arguably meets aboutness.

- [Friedman et al. \(2010\), Evidence for neural inertia](#) PLoS ONE: in mice and flies, responsiveness (a stand-in for consciousness) is lost and regained at different anesthetic doses. Such a lag marks an abrupt, first-order switch. It fits the transition hypothesis, unless it is pharmacological or belongs to the arousal switch (see Joiner et al. 2013). Human evidence is still suggestive.
- [Joiner et al. \(2013\), Genetic and anatomical basis of the barrier separating wakefulness and anesthetic-induced unresponsiveness](#) PLoS Genetics: in flies, neural inertia depends on genes tied to sleep regulation, and single mutations collapse it. The lag may belong to the arousal switch rather than to integration.
- [Kuizenga et al. \(2018\), Test of neural inertia in humans during general anaesthesia](#) British Journal of Anaesthesia: 36 volunteers. No lag between induction and recovery with propofol, a lag with sevoflurane for some endpoints. Human evidence for an abrupt switch is mixed.
- [Ontology](#) The study of what exists. Holos separates structure, real whether or not it is lived, from experience, which occurs only where $\Phi \geq \Phi_c$.
- [Causality](#) How events bring about other events. In Holos, observation causes nothing extra. Experience is the inside of physical causation, not an added cause.
- [Quantum Decoherence](#) The process by which interaction with the environment spreads a system's quantum information outward, so its alternatives stop interfering and look classical (review: Schlosshauer 2019, Physics Reports). Decoherence yields classical-looking branches; Φ marks where within them experience occurs.

Relationship to Physics

See: [Logic > Relationship to Physics](#)

- 13 • [Unitarity \(physics\)](#) Unitary evolution never collapses. Holos without collapse, the version defended here, preserves it fully. Nothing collapses, nothing is selected, and unobserved branches remain in Creation as unlit structure.
- [Hilbert Space](#) The mathematical space of all possible quantum states. Holos deletes no branches from it. The Born weights are structural facts about it.
- [Schrödinger Equation](#) Φ does not replace or modify the Schrödinger equation. It marks where experience occurs and leaves the quantum math untouched.
- [Quantum Mechanics](#) Holos leaves quantum mechanics untouched. Φ marks where registration occurs. It adds no constraint to the physics.
- [Born rule](#) Max Born (1926): quantum probabilities are squared amplitudes, one of the best-tested rules in physics. In Holos the weights are structural facts within C, setting each observer's odds rather than measuring how much experience a branch carries. Almost all of the weight, not every observer, sees Born statistics.
- [Gleason's theorem](#) Andrew Gleason (1957): given its assumptions, any consistent assignment of probabilities to measurement outcomes takes the Born-rule form. It constrains the odds but does not by itself say how an observer inside a branch should set them. Holos adds no new mathematics.
- [Sebens and Carroll \(2018\), Self-locating uncertainty and the origin of probability in Everettian quantum mechanics](#) British Journal for the Philosophy of Science: after branching and before looking, an observer is uncertain which branch they are in, and the Born weights are the rational odds. Holos adopts this reading. Counting fails, weight is the measure physics supplies, and the last step is their epistemic separability principle, widely discussed and still debated.
- [Ontology](#) Experience is lived only inside observers. The causal pasts they draw on are lit. The rest remains unlit structure. The underlying quantum math is unaltered.

Notation

See: [Logic > Notation](#)

- 14 • [Information Theory](#) Sending information assumes a causal signal passes between places; $\odot$ sends nothing. It concerns which histories are lived, not which signals travel.

- [Measurement in Quantum Mechanics](#) Measurement models the physical interaction between systems. Observation in Holos is registration from within. It selects nothing and alters no interaction.
- [Hilbert Space](#) In quantum physics, a system's state is a vector in a very large (often infinite-dimensional) space. Some researchers argue ordinary 3D space emerges from this structure. That view is debated, and Holos does not depend on it.

Comparison

See: [Logic](#) › [Comparison](#)

- 15 • [Goff \(2017\), *Consciousness and Fundamental Reality*](#) Oxford University Press: the case for priority cosmopsychism, a conscious cosmos from which individual minds derive. Holos shares the single ground but makes no claim either way about an experience of the cosmos as a whole.
- [Nagasawa and Wager \(2016\), *Panpsychism and priority cosmopsychism*](#) In Brüntrup and Jaskolla, eds., *Panpsychism* (Oxford University Press). Sets out priority cosmopsychism and the problem it faces of how one cosmic subject yields many individual ones (the decombination problem). Holos's section on walled-off perspectives answers the same problem.
- [Russellian monism \(Stanford Encyclopedia of Philosophy\)](#) Physics describes structure. Experience is the intrinsic nature of that structure. Holos agrees experience is the inside of physical activity, but only above the threshold.
- [Dual-aspect monism](#) Mind and matter as two aspects of one underlying reality, from Spinoza to the present. The modern family of Axiom 4. Holos is a two-sided monism with a threshold. The two sides appear only above it.
- [Physicalism \(Stanford Encyclopedia of Philosophy\)](#) The view that everything is physical, with the mental grounded in the physical. Holos agrees physics fixes every fact, so copies match in every possible world, but holds that neither side of an observer's activity is grounded in the other. Some will still classify it as physicalism. The site says why it does not.
- [Phenomenal concept strategy](#) The explanatory gap lies between two ways of describing one thing, not between two things. Developed by physicalists (Loar, Papineau, Balog). Axiom 4's floor-plan argument is a version of it, borrowed without the physicalist conclusion.
- [Antony \(2006\), *Vagueness and the metaphysics of consciousness*](#) Philosophical Studies: consciousness cannot be vague. Most who accept the argument conclude panpsychism. The rest posit an exact cutoff. Holos rejects the premise and accepts a narrow twilight between clear cases.
- [Schwitzgebel \(2023\), *Borderline consciousness*](#) Philosophical Studies: argues that borderline cases of experience are coherent. They cannot be pictured from the inside, since picturing an experience makes it definite, but that limits imagination, not reality. The reply Holos gives to the no-vagueness argument.
- [Rosenberg \(2004\), *A Place for Consciousness*](#) Oxford University Press: poses the boundary problem, what fixes where one subject ends. Holos answers with its maximality condition. One peak of integration, one perspective.
- [Global workspace theory](#) Conscious access as brain-wide broadcast. A column in the Holos theories-of-mind table.
- [Illusionism](#) Experience as usually conceived does not exist. A column in the Holos theories-of-mind table. Holos rejects it.
- [Frankish \(2016\), *Illusionism as a theory of consciousness*](#) Journal of Consciousness Studies 23(11-12): the standard statement of illusionism. We are conscious, but experience lacks the special inner qualities it seems to have. The task is to explain why it seems to have them.
- [Seth and Bayne \(2022\), *Theories of consciousness*](#) Nature Reviews Neuroscience: a readable survey of the main scientific theories (global workspace, integrated information, higher-order, predictive processing) and how they might be tested. Background for the theories-of-mind table.
- [Many-worlds interpretation](#) Everett (1957): every outcome of a quantum event with nonzero weight happens, each in its own branch. Holos agrees. What it adds is a threshold for who counts as an observer, and a reading of the whole as one experienter.
- [Relational quantum mechanics](#) Rovelli (1996): quantum properties are relative to observers. Holos aligns on relational facts, but keeps the universal state RQM rejects, and adds a threshold ($\Phi \geq \Phi_c$) for what counts as an observer.

- [QBism](#) Fuchs, Mermin, and Schack (2014, American Journal of Physics): quantum probabilities are an agent's personal degrees of belief about their future experiences. Holos is ontological (what is lived) rather than epistemic (what agents believe).
- [Copenhagen interpretation](#) The traditional (textbook) interpretation, with collapse at measurement. Holos without collapse, the version defended here, drops it. Evolution stays unitary, branches remain, and those with observers are lived from within.
- [Objective collapse theories](#) Theories in which collapse is a physical process that needs no observer: large objects settle into one outcome on their own. Holos without collapse rejects them. Any confirmed collapse would falsify that version and leave Holos with collapse, declared in advance. A consciousness-linked collapse would make the threshold the collapse point (the standing bet), while a size-based one would leave the threshold dynamically inert. No search has found one. The parameter-free Diósi-Penrose model is ruled out (Donadi et al. 2021).

Predictions

Introduction

[See: Predictions › Introduction](#)

- 16 • [Dynamics \(physics\)](#) Holos does not propose new dynamical laws. It makes structural claims about where structure is lived ($R = C \otimes O$).
- [Ontology](#) Predictions about where structure is lived. Observers register the block universe from within. They do not hold it together.
- [Block universe](#) The structural layer of Holos: tenseless and observer-independent. Observers register it from within. Its consistency does not depend on them.
- [Anthropic principle](#) Holos keeps what exists apart from where it is lived: observer-free universes or branches exist as structure but are never lived, and nothing is selected or filtered.
- [Multiverse](#) Branches where nothing could live exist as structure, but in Holos they are never lived (no system reaches Φ_c).

Commitments

[See: Predictions › Commitments](#)

- 17 • [Integrated Information Theory \(IIT\)](#) Holos borrows Φ as a measure of integration, not IIT's identity claim. Crossing Φ_c is where experience occurs. PCI is a practical proxy.
- [Panpsychism](#) Holos distinguishes itself from universal panpsychism (experience in all matter) and from illusionism (experience lacks the inner qualities it seems to have).
- [Illusionism \(consciousness\)](#) The view that experience does not have the special inner qualities it seems to have. What exists is the brain's representation of such qualities (Frankish 2016). Holos rejects it. Experience is the one fact the framework starts from.
- [Qualia](#) The felt character of experience. Holos locates qualia in systems that meet the observer requirements. Below Φ_c there is processing without experience.
- [Casali et al. \(2013\), A theoretically based index of consciousness independent of sensory processing and behavior](#) Science Translational Medicine: introduces the Perturbational Complexity Index (PCI). Stimulate the brain with a magnetic pulse and measure how complex its echo is. An IIT-inspired metric with an empirically calibrated cutoff between conscious and unconscious states. Holos treats it as a proxy for integration, not a detector of presence.
- [Phase transition](#) Holos hypothesizes that the onset of experience is a genuine transition, steep but with a narrow twilight in any finite system. A steep change alone would not confirm this, since ordinary models predict tipping points too.

Testability and Its Limits

See: Predictions › Testability and Its Limits

- 18 • [Chalmers and McQueen \(2022\), Consciousness and the collapse of the wave function](#) The named rival to the standing bet: integrated consciousness collapses the wave function, testable in principle on quantum computers. Holos bets it does not.
- [Wiseman, Cavalcanti, and Rieffel \(2023\), A thoughtful Local Friendliness no-go theorem](#) Quantum: proposes a human-level AI on a quantum computer as the friend, the road toward a genuine observer in Check C.
- [Laux and Cavalcanti \(2026\), Extended Wigner's friend scenarios with agent-like observers on quantum computers](#) Preprint, not yet peer reviewed: agent-like observers on IBM hardware. Local Friendliness violations persist. The agents are far below Φ_c .
- [Pedalino et al. \(2026\), Probing quantum mechanics with nanoparticle matter-wave interferometry](#) Nature: quantum interference of sodium clusters of more than 7,000 atoms, the current record for how "macroscopic" a tested superposition of a free-flying particle is.
- [Donadi et al. \(2021\), Underground test of gravity-related wave function collapse](#) Nature Physics: rules out the parameter-free Diósi-Penrose collapse model.
- [Wigner's friend](#) Registered facts are observer-indexed. No objective collapse. Extended Wigner's-friend experiments constrain the family of views Holos belongs to (Check C).
- [Bong et al. \(2020\), A strong no-go theorem on the Wigner's friend paradox](#) Nature Physics: the Local Friendliness theorem (a set of common-sense assumptions about observers and their results) and its photonic test. Absoluteness of observed events, locality, and freedom of choice cannot all hold. Holos gives up the first.
- [Siclari et al. \(2017\), The neural correlates of dreaming](#) Nature Neuroscience: dream reports occur after awakenings from both REM and NREM sleep, tracked by local activity in posterior cortex. The reason Test A's second gauge is local. Adopted after these data, so they cannot count in Holos's favor.
- [Nieminen et al. \(2016\), Consciousness and cortical responsiveness: a within-state study during non-rapid eye movement sleep](#) Scientific Reports: TMS-EEG just before awakening from NREM sleep. Responses looked more like the unconscious pattern when subjects reported nothing, and shorter dream reports went with more of it. Within-state evidence with tight timing.
- [Bajwa et al. \(2025\), A repeated awakening study exploring the capacity of complexity measures to capture dreaming during propofol sedation](#) Scientific Reports: 20 participants, deep propofol sedation. 24 of 29 interpretable awakenings reported experience. PCIst and Lempel-Ziv complexity fell from waking but did not differ with or without experience. The nearest test yet. Limited by windows ending a minute before waking and only five no-experience reports.
- [Radek et al. \(2018\), Dreaming and awareness during dexmedetomidine- and propofol-induced unresponsiveness](#) British Journal of Anaesthesia: 84% of interviews included experiences from unresponsive periods, mostly dreams. Anesthetic unresponsiveness is not the same as absent experience.
- [Aamodt et al. \(2022\), EEG Lempel-Ziv complexity varies with sleep stage, but does not seem to track dream experience](#) Frontiers in Human Neuroscience: spontaneous-EEG complexity fell with sleep depth but did not separate dream from non-dream awakenings within NREM2. A caution for spontaneous gauges.
- [Wong et al. \(2025\), A dream EEG and mentation database](#) Nature Communications: 20 datasets, 505 participants, 2,643 awakenings of sleep EEG with standardized dream reports. Usable for Test A's calibration step. It lacks the stimulation PCI needs.
- [Bodien et al. \(2024\), Cognitive motor dissociation in disorders of consciousness](#) New England Journal of Medicine: 60 of 241 behaviorally unresponsive patients (25%) performed cognitive tasks on fMRI or EEG. Covert awareness, one of Test A's held-out states.
- [Phase transition](#) If crossing the threshold is a genuine transition, sudden or continuous, it should leave measurable signatures near the boundary. See A path to the threshold.
- [Casarotto et al. \(2016\), Stratification of unresponsive patients by an independently validated index of brain complexity](#) Annals of Neurology: sets PCI's cutoff, 0.31, on a benchmark of states known from report, including REM dreaming and

ketamine. That cutoff is Test A's primary gauge. The benchmark cases are calibration, not confirmation, and Test A counts only held-out states named in advance.

- [TMS-EEG](#) A magnetic pulse to the scalp (TMS) while recording brain waves (EEG). PCI is computed from these responses. It is the integration proxy Test A relies on.
- [Perturbational Complexity Index \(PCI\)](#) Casali et al. (2013). Validated across sleep, anesthesia, and disorders of consciousness. Test A uses it to separate integration from responsiveness.
- [Propofol / BIS index](#) Propofol is a common anesthetic. The BIS index is a bedside EEG number anesthesiologists use to gauge how deep anesthesia is. One of the states Test A compares. A drop in integration alone confirms nothing specific to Holos.
- [Recurrent neural network](#) Recurrent architectures. Relevant to whether artificial systems have the feedback that integration requires.
- [Neuromorphic computing](#) Brain-like hardware with feedback. A candidate substrate for meeting the observer requirements.
- [Butlin et al. \(2023\). Consciousness in artificial intelligence: insights from the science of consciousness](#) A report by neuroscientists, philosophers, and AI researchers that turns the main theories of consciousness into checklists of features and assesses current AI systems against them. They find no current system is a strong candidate, but no obvious barrier to building one.
- [Doerig et al. \(2019\). The unfolding argument: why IIT and other causal structure theories cannot explain consciousness](#) Consciousness and Cognition: any network with feedback can be rebuilt as a feedforward one with the same inputs and outputs, so if a theory ties experience to feedback structure, no behavioral test can confirm it. A challenge any integration-based test, Test A included, has to answer.
- [Seth, Barrett, and Barnett \(2011\). Causal density and integrated information as measures of conscious level](#) Philosophical Transactions of the Royal Society A: causal density as an integration proxy, how much a network's parts predict one another's activity over time, used when computing Φ directly is infeasible.
- [Collective intelligence](#) Social and agent networks. Whether group-scale apertures can form is left open: Holos neither asserts nor excludes them.
- [Mutual information](#) Integration proxy: mutual information across subgroups, causal density, network-wide coherence.
- [Network theory](#) Small-world and scale-free structure. Candidate settings for measuring integration.
- [Relational quantum mechanics](#) Holos borrows RQM's point that facts are indexed to observing systems, but sides with branching, which keeps the universal state RQM rejects. A positive Check C result supports the family, not Holos alone.
- [Proietti et al. \(2019\). Experimental test of local observer independence](#) Science Advances: an extended Wigner's friend test with six photons, in which single photons stand in for the "friends". Their records need not agree with Wigner's, as quantum theory predicts. In Holos each record is indexed to its branch, and registered facts need a real observer (Check C).
- [Unitarity \(physics\)](#) Evolution that conserves information and total probability. Holos predicts agreement among communicating observers without objective collapse.
- [Objective collapse theories](#) Holos without collapse rejects objective collapse: evolution is unitary, and apparent collapse is a record within a branch, definite without any observer. A confirmed objective collapse would falsify that version and leave Holos with collapse.

Speculation

[See: Predictions > Speculation](#)

- 19 • [Phase transition](#) A speculative reading: civilizational integration may advance in jumps rather than smoothly, as light-speed delay and waste heat reshape what can be coordinated.
- [Fermi paradox](#) A companion hypothesis to Holos, not part of its core: the Integration Hypothesis and Going Quiet. Mature civilizations are silent in light. The unavoidable observables are gravity and waste heat.

- [Hanson et al. \(2021\), If loud aliens explain human earliness, quiet aliens are also rare](#) The Astrophysical Journal: the grabby aliens model. Visible, expanding civilizations exist but have not reached us, and we are early. Named on the site as the main rival to the Integration Hypothesis.
- [Griffith et al. \(2015\), The G infrared search, III](#) The Astrophysical Journal Supplement: of about 100,000 galaxies surveyed with WISE, none hosts a civilization reprocessing more than 85% of its starlight into waste heat.
- [Zoo hypothesis](#) The idea that mature civilizations deliberately leave young ones alone. For Holos, non-contact needs no agreement. Leaving a young civilization alone costs nothing, while contact takes effort.

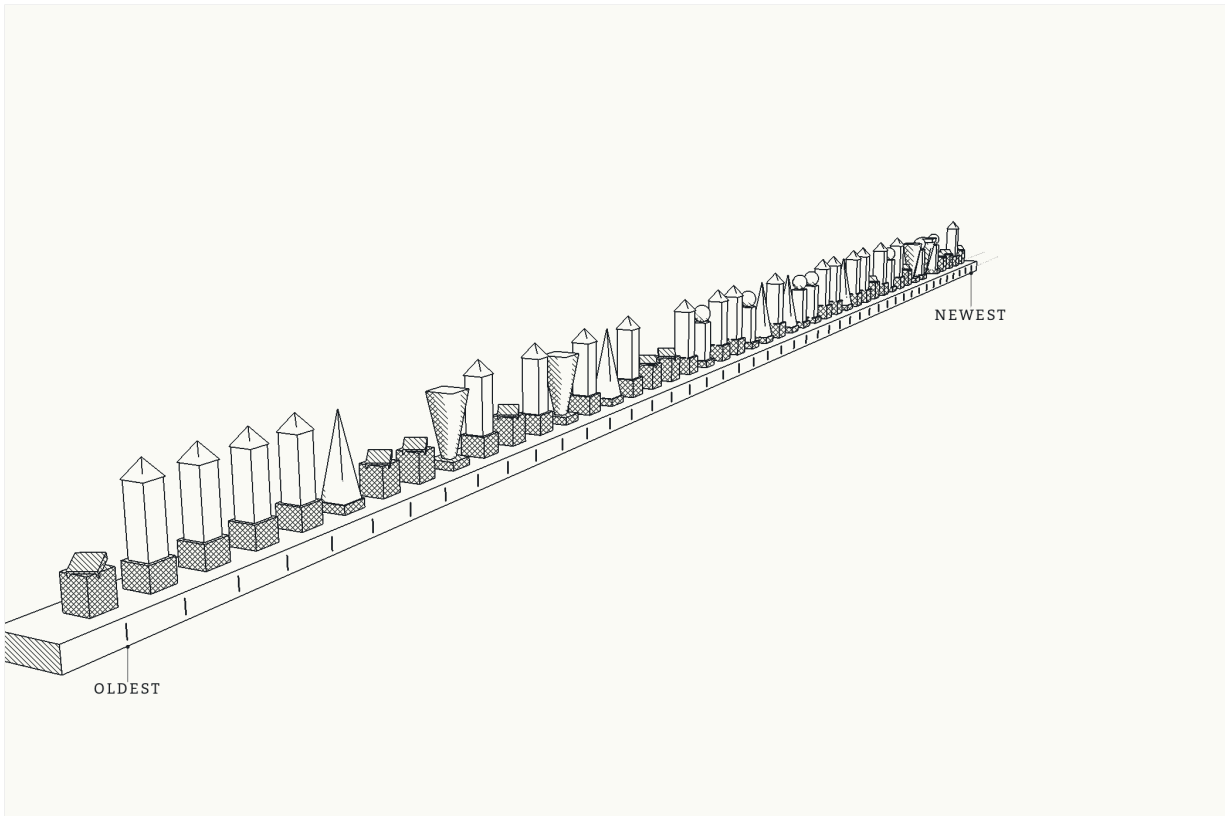
Technology

[See: Predictions › Technology](#)

- 20 • [Penrose process](#) Extracting a spinning black hole's rotational energy, up to 29 percent of its mass, the known physics behind the Holocore's densest option.
- [Sandberg, Armstrong, and Ćirković \(2017\), That is not dead which can eternal lie: the aestivation hypothesis for resolving Fermi's paradox](#) Argues advanced civilizations might sleep until the universe cools, when computing is cheaper. One motive for the Chrono Vault's sleeping case.
- [Bennett, Hanson, and Riedel \(2019\), Comment on the aestivation hypothesis](#) Foundations of Physics: argues entropy can be disposed of cheaply today, so waiting is not required. The counterweight to aestivation.
- [Bracewell \(1960\), Communications from superior galactic communities](#) Nature: proposes parking autonomous probes in target star systems. The ancestor of Sentinel Probes.
- [Turyshev et al. \(2020\), Direct multipixel imaging and spectroscopy of an exoplanet with a solar gravity lens mission](#) A NASA-funded mission design using the Sun's gravitational focus, beyond about 550 times the Earth-Sun distance. The basis for Gravitational-Lens Observatories.
- [Computronium](#) Hypothetical material optimized for computation. The Computronium Kernel is a maximally compact core for coherent, long-horizon world-modeling.
- [Megastructure](#) Large-scale artificial structures. Holos mesostructures (Kernel, Chrono Vault) are compact and coherence-optimized rather than maximally expansive.
- [Gravitational lens](#) Bending of light by mass. Gravitational-lens observatories use natural lenses for extreme resolution without large radiative infrastructure.
- [Time capsule](#) The Chrono Vault extends this idea to civilizational identity: preserving value systems, decision histories, and continuity across deep time.
- [Interstellar communication](#) Holos speculates that at cosmic distances messages would be tightly aimed laser signals carrying compressed, self-explaining packages, not live conversation.
- [Space probe](#) Sentinel Probes are compact, autonomous, long-duration instruments that resolve observational ambiguities and operate without real-time control.

Revisions

Holos is written in public and revised when it is wrong. These claims appeared in earlier versions and have been retired or replaced, each for a stated reason, oldest first.



- **Ordered dark matter.** Retired: cosmological dark matter existed before any life could, and it outweighs all the ordinary matter anything could be built from.
- **Theological and secular readings as interchangeable.** Replaced by the monist position, first argued as the better answer to "why am I this one?" (later framed as one answer among serious rivals).
- **Observation as selection.** Replaced: observation selects nothing and erases nothing; branches with observers are lived from within, and the rest stay unlit.
- **Sealing a whole branch from beginning to end.** Replaced by an observer's causal past, which gives the claim experiential content.
- **Axioms that left out the two additions.** Rebuilt: the threshold and the totality are now Axioms 3 and 5; the old axiom about how structure becomes experience ("Manifestation") became the definitions of lived, lit, and unlit; the old axiom limiting what can be structure ("Structural Constraint") became a companion principle.
- **The whole causal past as "lived".** Split: lived is where experience occurs, inside observers; lit is the causal past it draws on. No one lived through the early universe.
- **Born weights as shares of experience.** Retracted: experience is not pooled. The weights are now read as self-locating odds.
- **Omega as a limit finite systems approach.** Cut: the whole exists now, the limit was never defined, and it implied a fully registered whole, against unlit structure. Omega's material moved from the extrapolations into its own core section.

- **One threshold value for every system.** Layered: the commitment became one universal shape of transition, with one universal value as a stated conjecture (both later dropped). Neural inertia is no longer read as simple support.
- **Economy as Omega's payoff.** Replaced: one subject versus many is only a count. Omega's work became its answers to "why am I this one?" and to which perfect copy is you (later framed as one answer among serious rivals).
- **Omega as the ground of record agreement.** Dropped: physics secures agreement. Omega's one job is unity: one subject, not many.
- **The bridge to integration as purely definitional.** Replaced by a small definitional core plus an argued identification that can fail.
- **Five observer requirements.** Extended to six: aboutness, tracking something beyond the system through its own channels, rules out inert high-integration arrays.
- **Current AI as "no one home."** Replaced by an open question with two borders: aboutness grounded in the system's own senses, and integration.
- **Six observer requirements.** Reduced to four: recursion and causal autonomy passed any thermostat on their own, and what they aimed at, feedback, is part of integration.
- **Aboutness grounded in senses of the system's own.** Replaced: it judged the message, not the machine, and a brain fed a perfect simulation showed it contradicting Axiom 4. Aboutness is now structural, any live channel counts, and the open question for current AI narrows to integration.
- **Anesthesia and cultured-network transitions as confirmations.** Retired as tests: ordinary models predict them too. They became correlate probes; dishes later returned as the size-scaling test of claim 3.
- **The qubit observer-cut experiment.** Retired: qubits register nothing below the threshold, and the predicted result was a known quantum effect (contextuality) that needs no observer.
- **"Falsified outright" for the standing bet.** Replaced by a fallback declared in advance: a consciousness-linked collapse would falsify the no-collapse physics, not the core, and the threshold would become the collapse point (later limited to a collapse that tracks integration).
- **"Confirmed so far" for the standing bet.** Corrected to untested: no superposition tested so far has contained an observer.
- **The threshold value as a constant of nature, like the speed of light.** Replaced: a threshold that could differ between physically identical worlds would allow a copy with no one home, against Axiom 4. The conjectured value became structural, like a percolation threshold (the universal value was later dropped).
- **Sharpness from a quantity "exactly zero on one side."** Replaced: that holds only in infinitely large systems. Whether anyone is home depends on which side of the threshold (fixed by structure) a system is on; in any real, finite system, the outward signs of the crossing blur.
- **Waking-brain criticality as evidence for the threshold's shape.** Reread: it places a critical point at the center of conscious life, where variety peaks. The threshold is where parts join into one whole; evidence for its shape must come from the boundary.
- **Aboutness as any live channel, tracked as it happens.** Sharpened: one thermometer wired into an inert array passed it, and "as it happens" excluded dreams. The world then had to be able to drive the whole across much of its range, through channels open or gated (later replaced by carrying a model of a world).
- **Omega exempt from maximality as "not a system among systems."** (Maximality: an observer forms only where integration peaks.) Replaced: Omega is physically the universal quantum state, so the exemption was special pleading. The whole fails integration outright, splitting into parts that never touch, which is also why it pools nothing.
- **Creation as every world the laws permit.** Narrowed: if every lawful world were real structure, reality would outrun Omega. Creation is what physics actually produces, every branch of the one quantum state: not every recipe the laws allow, but every dish physics actually serves.
- **Omega defined through Axiom 2, and Axiom 4 at risk in the standing bet.** Corrected: Omega is the universe's complete quantum state on any reading, and Axiom 4 survives a collapse. What a collapse would cost is the claim that observation changes nothing it registers.

- **Collapse as an observer's registration.** Corrected: records are definite within each branch without observers, a detector's click included. An observer adds not definiteness but its being lived.
- **An exact line between someone and no one.** Replaced by a narrow twilight, fixed by structure: an exact line exists only in an infinitely large system, which Holos reads as the limit of a model. The threshold became four claims of different strength; a critical transition and one shared shape moved from commitment to hypothesis (later three claims, with the transition's kind left open). "A retreat to graded presence" is no longer a way to lose.
- **A fallback that left everything but Axiom 2 untouched.** Replaced by two versions sharing one core. "Registration changes nothing" in D6 and Axiom 3's "no change to any equation" belong to Holos without collapse, so a lost bet would retire that version, not only Axiom 2.
- **Test A without numbers.** Frozen: PCI with its published 0.31 cutoff as the primary gauge, a second gauge measured over the back of the brain (posterior cortex) and named in advance, a timing rule, and a rule separating a failed gauge from a failed theory. The local reply to non-REM dreaming, adopted after seeing the data, counts only on new data.
- **Omega as settling the identity puzzles.** Replaced: the monist reading is one answer among serious rivals, Perry on "I" and Parfit on copies. Its price, anticipating strangers' experience as your own, is now stated, and it is also the payoff: self-interest and concern for others coincide. The Self table now gives rivals their best answers and every view its price.
- **Aboutness as a world-drivable range.** Replaced by carrying a model of a world: a camera wired to every gate of an inert array passed the old rule, and a dreaming brain with its senses permanently cut failed it. A dream is the model running on its own.
- **Branches as a seam in the whole.** Dropped: branches are not parts, so no integration measure cuts there. Horizons alone keep the whole from being a peak.
- **"Presence depends on observers" as a commitment.** Replaced: it was true by definition, since lived means experienced. Commitment 1 is now what a rival could deny: experience is local, and physics describes it only from outside. Proposition V, definitional in the same way, merged into D7. "Reality in the full sense" became lived reality, and calling observation a "closure condition" retired.
- **An unnamed kind of view.** Named: a two-sided monism with a threshold, neither physicalism nor dualism. It agrees with physicalism that physics fixes every fact and differs on direction: neither side of the event is grounded in the other. Axiom 1, which did no work downstream, now explains why physics states experience only from outside.
- **Weight as "the only measure left".** Replaced: ruling out counting does not pick the replacement. Weight is the measure physics supplies, and the last step is a principle, still debated. Gleason's theorem constrains the odds rather than settling them, and not every observer sees Born statistics: almost all of the weight does.
- **One shared shape of transition, and one universal threshold value.** Dropped: nothing in Holos implies them, and physics expects systems that differ in geometry and wiring to cross differently. The prediction that the twilight narrows as systems grow stays, as a consequence of the transition hypothesis.
- **The crossing as a "critical transition".** Widened: a critical point names only a smooth, continuous crossing, and Holos takes no side on whether the crossing is sudden or continuous. Claim 3 is now a genuine transition of either kind, and both kinds' fingerprints are named.
- **A "realized world" that needs observers.** Corrected: it read as if unlive structure were less than real. Structure is fully real whether or not anyone lives it; ☉ concerns where a world is lived.
- **Unlive structure as fully real.** Reversed: "reality" now means lived reality (R). Structure exists whether or not anyone lives it, but only lived structure is called real. Reality requires a witness.
- **The one experiencer "waking" in each observer.** Clarified: at the threshold a new self begins, and no new experiencer does. The self is local, and the experiencer is what lives through it (Logic, D8).
- **An ethical payoff from Omega.** Retired: one experiencer living through every self does not settle what to value. Holos no longer claims that a stranger's pain is yours to anticipate, and the price that came with that claim went with it.
- **No single view of the whole.** Withdrawn: Holos no longer claims the whole lacks any view of itself, nor that it has one. Locality is about places, so it does not decide the question, and a cosmic mind is no longer ruled out.
- **Current AI as an open question.** Replaced: today's language models fall short of the threshold, as built. They meet aboutness and differentiation but not integration or temporal cohesion. A system built differently could cross.

- **Silent, but warm.** Retired: the bet that mature civilizations avoid large cold radiators, and with it the expectation that their heat comes out warm. Erasing information costs less in the cold, so much of the heat may leave in the far infrared. The signature is now heat a star should not have, warm or cold.